SPECIFICATION PACKAGE

Packing House Operations Automation System

Functional & Process Specification — Package for Management Review

Receiving & Pallet ID registration · pre-cooling · QC/QA gates · parallel production lines · finished goods · cold storage · dispatch

DRAFT — for management review. Not approved. Not for build.

Prepared by	Operations — ERP & SaaS Transformation
Prepared for	Packing House Manager; QC and QA managements; Warehouse Management; Operations management
Evidence base	SOP001 Packhouse SOP Manual (supplied file; revision status inconsistent — see EV-17) · Packing House Operations — Engineering Summary Rev 2.2, 18 Oct 2026 (includes the Packing House Manager's corrections) · Fridge capacity table (image)
Companion files	PH-OAS-SPEC-001_Registers.xlsx (all tables, filterable) · PH-OAS-SPEC-001.docx (editable text) · diagram sources (.drawio, .dot)

C **Confirmed** — stated in a supplied source

P **Proposed** — design proposal; needs approval

NYV **Not Yet Verified** — gap, conflict or assumption

Document control

Revision history

Version	Date	Author	Change
0.1	19 October 2026	Operations — ERP & SaaS Transformation	First complete draft for management review. No decisions applied yet.

Review and approval

This draft is issued for review. Signing below records a review decision on the specification, not approval of any open decision; those are recorded in the Open Decisions Register (Section 13).

Role	Name	Decision (approve / approve with changes / reject)	Signature	Date
Packing House Manager				
QC Manager				
QA Manager				
Warehouse Manager				
Operations (system owner)				
IT				

Evidence labels

Label	Meaning	How it is used
C	Stated in a supplied source document (SOP001 or the Packing House Summary Rev 2.2, which includes the Packing House Manager's corrections). Documented — not independently observed on the floor.	Build as specified once the AS-IS walk-through confirms the floor matches the document.
P	A design proposal made in this specification. Requires approval by the named owner before build.	Build only after the named owner approves; most proposals have a matching change proposal (CP) or open decision (OD).
NYV	An assumption, a gap, or a conflict between sources. Must be validated or decided (see Open Decisions Register) before the dependent requirement is built.	Do not build as fixed behaviour. Implement as configuration that warns or blocks until the open decision is closed.

Identifier conventions

Prefix	Meaning	Section
EV-	Evidence item (fact, conflict or gap from the sources)	4.5
A0–A16	AS-IS process stage	4.2
CP-	Change proposal (TO-BE vs AS-IS)	5.5
ST-	TO-BE stage	5.6
X-	Cross-cutting control	5.7
HO- / D-	Handoff / decision point	5.4
SoD-	Segregation-of-duties rule	6.4
GEN- / TQ- / MB-	Genealogy rule / trace query / mass balance	7
SM- / G / V- / E- / AT-	State model / gate / validation / exception / audit rule	8
IB-	Integration boundary	9.5

Prefix	Meaning	Section
FR- / NFR-	Functional / non-functional requirement	10
B- / H-	Confirmed bottleneck / hypothesis	11
T- / K	Test scenario / KPI	12
OD-	Open decision	13
STD-	Standards traceability row	14

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1 Executive summary

This package specifies an internal Packing House Operations Automation System: a workflow and traceability system that records and controls the packing-house flow from gate arrival and Pallet ID registration, through pre-cooling, QC/QA gates, the parallel production lines, finished-goods identification and cold storage, to dispatch. It coordinates departments with different reporting lines — the Packing House, Data Entry (which reports to Warehouse Management), Quality Control and Quality Assurance (both outside the Packing House), Sales, Logistics and Finance — without moving any decision away from the department that owns it today. The existing ERP keeps the transactions it handles today; no interface is assumed.

36
Evidence items
25 confirmed · 11 gaps/conflicts

16
Change proposals
each with owner and evidence needed

90
Functional requirements
+ 12 non-functional

41
Open decisions
6 critical · 16 high · 15 medium · 4 low

Key findings

1. **The quantity chain is open at both ends of receiving.**
The harvesting tag carries no quantity, and product rejected twice by QA is, per SOP001, never entered in the system. Received = accepted + rejected cannot be proven, so no mass balance can close. Decision OD-01 (register every pallet at the scale; add a farm dispatch note) is the single most important design choice.

2. **The traceability key is not defined.**
SOP001 assigns the lot number at three different points, and complaints must quote a 'box code' that no document defines. Until OD-02 and OD-19 are decided, labels and trace reports cannot be specified to build level.

3. **Acceptance criteria are missing from the supplied documents.**
The SOP refers to an 'Incoming raw material grading schedule' and a 'Product specification' for tolerances; neither is in the supplied file, nor is the organogram. The system can record QA/QC decisions, but cannot check them against tolerances until QA supplies them (OD-05, OD-06).

4. **Three time rules conflict or are ambiguous.**
The 48 h clock (harvest vs order receipt), the 72 h maximum vs 'regrading after long storage', and the logger scope (all loads vs export) give different answers at dispatch (OD-09, OD-10, OD-18).

5. **Quality authority is clear; recording authority is not.**
QC and QA decisions, the PHM's order acceptance, waste sign-off thresholds and the rule that all instructions to PH staff pass through the PHM or Production Engineer are documented. Who records production consumption, finished pallets, fridge moves, loggers and seals is not (OD-12 – OD-14).

6. **Line capacity may be the real constraint, not data.**
At the documented nominal rates, a full Harvest Fridge of cherry tomato (25.6 t) needs about 21 ELIFAB running hours. On peak days the 24-hour grading rule is at risk regardless of software. This is a hypothesis to measure in Phase 0 (H-03) before investing in automation.

Readiness by area

Area	Status	Comment
Process architecture (AS-IS / TO-BE)	Ready for review	AS-IS must be confirmed by a floor walk with PHM, QC, QA and Warehouse.
Roles & permissions	Partly ready	Quality and approval authority confirmed; 12 action-level assignments (mostly who records) not stated in sources (Section 6.3).

Area	Status	Comment
Traceability model	Blocked on decisions	OD-01, OD-02, OD-19 decide the identifiers and the lot key.
Workflow rules	Mostly ready	Time rules OD-09/10; temperature limits OD-08; tolerances OD-05.
Data model & integration	Ready for review	Boundary defined; the 'pack house system' must be identified (OD-03).
Functional requirements	Ready for Phase 1 after Phase 0	Phase 1 FRs depend on OD-01, OD-03, OD-05, OD-24.
Implementation plan	Ready for review	No durations: they depend on the build approach (OD-25).

Recommendation

1. Approve this specification as the basis for design, not for build. Its status is Draft 0.1.
2. Assign an owner to each open decision. Close the six Critical decisions (OD-01, OD-02, OD-03, OD-05, OD-24, OD-25) in Phase 0, together with a signed walk-through of the AS-IS maps.
3. Pilot one crop through Phases 1–3 with a paper parallel run. No ERP, scale or temperature-system integration in Phase 1; decide integrations later from measured re-keying effort.
4. Run the Phase 0 baseline measurements (Section 11) in parallel, so that the system's KPIs have a defensible starting point.

Decisions requested from management

- Accept this document as the design basis (Draft 0.1) and confirm the system owner.
- Name an owner and a date for each of the six Critical open decisions: OD-01 entry point & rejected pallets, OD-02 lot definition, OD-03 'pack house system' identity, OD-05 tolerances, OD-24 infrastructure, OD-25 build approach.
- Authorise Phase 0 (decide & validate — no build), including the floor walk-through of the AS-IS maps and the baseline measurements.

2 Introduction and scope

2.1 Purpose

This specification gives management a basis to decide, and gives a later build team (including Claude Code) a basis to implement, an internal system that controls and traces the packing-house workflow. It separates the documented current state (AS-IS, Section 4) from the proposed controlled state (TO-BE, Sections 5–10). Every material requirement carries a label and a source.

It is a specification, not a design for a specific product. It does not choose software, does not assume integrations, and does not change department authority. Code is out of scope until the Phase 0 decisions are made.

2.2 Scope

In scope

- Order intake interface: accepted order lines and the daily plan (ST-00) — as an interface, not a sales system.
- Gate arrival, incoming vehicle inspection and Incoming Pallet registration (ST-01, ST-02).
- Arrival visual check, pre-cooling, QA raw-material acceptance and the rejection/regrade loop (ST-03 – ST-05).
- Harvest storage, fridge locations, temperatures, FIFO and ageing (ST-06, ST-12, X-07).
- QC-1 release, Job Cards on parallel lines, line-start checks, QC-2 inspection (ST-07 – ST-09).
- Finished-goods pallets, labels, split/merge, waste capture and approval (ST-10, ST-11).
- Allocation, dispatch QC, vehicle checks, loading, logger and seal, packing list, shelf-life samples (ST-13, ST-14, X-04).
- Regrade/rework, holds and nonconformity, CAPA, customer deviations, emergency overrides (ST-15, X-01 – X-03, X-06).
- Complaints, trace and recall scope, daily stock report and KPIs (ST-16, ST-17).

Out of scope (initial)

- Harvest and sales forecasting (SOP-FC) beyond recording accepted orders.
- Cleaning, hygiene, staff entry/exit and pest-control checklists (SOP-CL, SOP-SE) — candidate later module.
- Staff scheduling, HR, maintenance management and finance postings.
- Machine control or data capture from Aweta, ELIFAB or Ingro equipment.
- Farm operations before dispatch, and transport after departure (except logger and seal records).
- Replacement of the ERP or of any existing departmental system.

2.3 Source documents

Section references use these codes. 'Confirmed' means stated in these documents; nothing in this specification has been observed on the floor by the author. The SOP's page numbers are not used because the supplied file's pagination differs from its index.

Code	Document	Section	Notes
SOP	SOP001 — Packhouse SOP Manual (file: English SOP – 2026 Updated.docx)	Whole document	Header pp.1–7: SOP001, Rev 002 – 10/08/2025, draft date 16/06/24. Header pp.8+: Draft version 001, draft date 15/06/24. Revision status inconsistent — see EV-17.
SOP-FC	SOP001	Forecasting Procedure / Weekly Planning	Harvest, sales and logistics forecasts; order acceptance.

Code	Document	Section	Notes
SOP-RC	SOP001	Receiving Procedure	Harvest tag, visual check, QA sample check, system entry, label, rejection loop.
SOP-PC	SOP001	Pre-cooling and Cold Storage	Pre-cooling dwell, Harvest Fridge, Fridges 1–7, FIFO, 48/72 h, storage temperatures, storage groups, packaging storage.
SOP-GR	SOP001	Grading Procedure	Receiving grading, production grading, dispatched product grading, shelf-life samples.
SOP-CC	SOP001	Crate Coding and Washing	Crates, LSS boxes, customer crates.
SOP-PL	SOP001	Production, Labelling and Palletizing	Packaging issue, line checks, grades, label content, palletizing, deviations.
SOP-WS	SOP001	Waste Procedure	Processed waste, long-storage waste, packaging waste, weight loss, approval thresholds.
SOP-DS	SOP001	Despatch Procedure	Order allocation, FIFO, vehicle inspection, dispatch QC, logger, seal, loading window.
SOP-TR	SOP001	Transport Procedure (Warehouse department)	Incoming and outgoing vehicle conditions.
SOP-CP	SOP001	Customer Complaint Procedure	Validity, investigation, recall within 6 h, 24 h resolution.
SOP-ST	SOP001	Staff Shifts	Instruction routing via PHM/PE; data entry clerks own ERP transactions and traceability.
SOP-SR	SOP001	Pack House General Stock Reporting	Daily stock report content.
SOP-CL	SOP001	Cleaning / Inspection	Machine cleaning 1 h per shift; daily inspections by QA and PH managers.
SOP-SE	SOP001	Staff Entry / Exit	Hygiene checklist (out of initial scope).
SOP-PF	SOP001	Process Flow Diagram	10-step flow: receiving → dispatch; lot number assigned after pre-cooling in Harvest Fridge.
SOP-SP	SOP001	Process Flow — Site Plan	Hand-marked site plan (PRO 03, draft 02/07/24). Positions not used as evidence.
SUM-1	Packing House Operations — Engineering Summary, Rev 2.2, 18 Oct 2026	§1 Conceptual Process Flow	Includes Packing House Manager's corrections and SOP001 review.
SUM-2	Packing House Summary Rev 2.2	§2 Equipment, Capacity, Storage & Labor Targets	Lines, throughputs, cold-storage capacity, labor targets.
SUM-3	Packing House Summary Rev 2.2	§3 Operational Process, Checks & Responsibilities	Process stages, quality checkpoints, roles and reporting lines, open items 1–8.
FCT	Fridge capacity table (image supplied with the request)	Three tables: bags; 35 field boxes (cherry 70); Option 3 — 49 field boxes	Pallet positions and kg by crop group and location.

Document	Reason not used
agri-erp-report.pdf (project file)	Market study of agriculture ERP vendors. Contains no packing-house process evidence; not used as a source for this specification.

2.4 Method

- Extracted every operational statement from SOP001, the Summary and the capacity table into the evidence register (Section 4.5), recording conflicts between sources rather than resolving them silently.
- Mapped the AS-IS process with the documented roles and reporting lines, then designed the TO-BE as the same process with records, states and gates added (principle DP-01).
- Labelled each requirement Confirmed, Proposed or Not Yet Verified and linked every unresolved point to an owner in the Open Decisions Register.
- Re-computed all capacity figures from the source tables; derived numbers are marked as such.

- Checked standards references against published structures; clause wording is paraphrased (Section 3).

3 Standards basis and applicability

- Standards are used for specific principles, not as a template. Each use is traced in Section 14 to the requirement it informs, with its type: a formal requirement of a standard (which binds only if the company adopts or is certified to it), a recognised code of practice, an information-security control, or a recommended practice.
- The company's certification status and customer standards are not stated in the supplied documents (the Summary says QA monitors 'the international quality and food-safety standards and certifications applicable'). Applicability is therefore an open decision (OD-22).
- This specification does not identify critical control points, critical limits or legal requirements. Where the system must hold such values, it holds them as QA-owned configuration (OD-08, OD-23).
- Designing to these principles does not make the company compliant or certifiable. Compliance is demonstrated by operating records and audit, not by a specification.
- Clause references are paraphrased from the published standards (ISO 9001:2015, ISO 22000:2018, ISO/IEC 27002:2022, Codex CXC 1-1969 Rev. 2022, CAC/RCP 53-2003). QA should check them against the licensed copies before audit use.

Reference	Principles applied in this design	Type
ISO 9001:2015	Process control, responsibilities, documented information, identification and traceability, customer property, preservation, release, nonconforming output, corrective action	Standard requirement — binds only if adopted/certified
ISO 22000:2018	Traceability system (§8.3), hazard control plan records (§8.5), monitoring equipment (§8.7), potentially unsafe product and recall (§8.9)	Standard requirement — binds only if adopted/certified
Codex CXC 1-1969 (Rev. 2022)	GHP: time/temperature control, records, recall, lot identification, transportation; HACCP principles 4–7 for records	Recognised code of practice
Codex CAC/RCP 53-2003	Fresh produce packing: cold storage monitoring, condensate, incoming material, water, records, recall	Recognised code of practice
ISO/IEC 27001:2022 Annex A / 27002:2022	Segregation of duties, access control and review, privileged access, logging, clock sync, records protection, backup, change management	Information-security control set
ALCOA+	Data integrity of electronic records	Recognised practice
GS1 SSCC	Pallet identification if customers require it	Optional practice
Reg. (EC) 178/2002 Art. 18 (EU) / UK equivalent	One-step-back / one-step-forward traceability expected by EU/UK buyers	Legal for operators in those markets; relevance via export customers to be confirmed

No compliance claim

Nothing in this document certifies or claims compliance with any standard. The full standards-to-requirements traceability matrix, with implementation status per row, is in Section 14.

4 AS-IS baseline (current state)

This section records the current process exactly as the sources describe it. It is the baseline the TO-BE must preserve (DP-01). It has not been verified on the floor; Phase 0 includes a walk-through to confirm or correct it.

4.1 Organisation and reporting lines

Function	Reports to	Documented responsibilities	Label	Source
Receiving / Data Entry	Warehouse Management	Weighing at receiving; records weight and Pallet ID (1 employee). Data entry clerks handle all ERP transactions and traceability.	C	SUM-3; SOP-ST
Quality Control (QC)	QC management (outside the Packing House)	Arrival check, QC-1, QC-2, dispatch approval; approves waste before scaling; inspects customer crates; provides grade-to-customer schedule.	C	SUM-3; SOP-WS; SOP-CC; SOP-PL
Quality Assurance (QA)	QA management (outside the Packing House)	Raw-material acceptance; records dispatch findings; shelf-life samples 10–14 days; compliance with applicable standards and crop-protection/pre-harvest-interval follow-up; complaint investigation with QC.	C	SUM-3; SOP-RC; SOP-GR; SOP-CP
Packing House Manager / Production Engineer	Not stated	All instructions to Packing House staff from other departments, including QC, pass through them. PH management accepts or rejects orders; PHM signs waste reports.	C	SOP-ST; SUM-3; SOP-FC; SOP-WS
Warehouse Dept	Not stated	Transport procedure owner; packaging stock and ERP removal on dispatch; waste sign-off above 1 t; approves truck arrival times with PH management.	C	SOP-TR; SOP-PL; SOP-WS; SOP-FC
Organogram	—	Referenced in the SOP index (p.19) but not present in the supplied file.	NYV	EV-17

4.2 Process stages

ID	Stage	What happens today	Who	Records today	Label	Source
A0	Weekly planning & order acceptance	Harvest forecast per crop per day to Sales and PH by Thu 12:00 (Sat–Fri week, 15 % accuracy level). Sales forecast by Thu 16:00 (Export UK/Abay/GU/EU, Local, Retail standing order, LSS). PH management compares sales vs harvest vs grading %, accepts or rejects orders, and plans staff, hours, grading and packaging. Confirmations: local 24 h, export 48 h ahead.	Farm, Sales, PHM, Warehouse, Logistics	Forecasts (format not stated); white board	C	SOP-FC
A1	Farm dispatch & transport	Pallets leave the farm with a harvesting tag (field, variety, time, date, supervisor). No quantity on the tag. Last harvest received by 17:00. External supplier loads arrive in supplier vehicles.	Farm harvest supervisor; supplier	Harvesting tag	C	SOP-RC; SUM-3; SOP-TR
A2	Gate arrival & vehicle check	Vehicle checked against transport conditions. Non-compliant vehicle: product not accepted until pre-approved by PHM, QA and top management; condition noted on the raw material receiving report.	Receiving / QA	Raw material receiving report	C	SOP-TR
A3	Arrival visual check	QC/QA visually check every pallet: covered, palletised, acceptable crates, insects, disease, temperature, appearance, foreign matter, freshness, spec conformance, GR3/waste in crates. Excessive waste → rejected, returned to farm or waste area.	QC / QA	Product intake report (SOP-PF)	C	SOP-RC; SOP-GR
A4	Weighing & Pallet ID	Data Entry (1 employee) records weight and Pallet ID at 3 gates / 3 scales. (The SOP places system entry after QA acceptance — conflict EV-04.)	Data Entry Clerk (Warehouse Mgmt)	Weight record; Pallet ID	NYV	SUM-1; SUM-3; SOP-RC
A5	Pre-cooling	Product held 15–30 min max. (by receiving temperature). Capacity 16 pallets.	Packing House	None stated	C	SOP-PC; SUM-2

ID	Stage	What happens today	Who	Records today	Label	Source
A6	QA raw-material acceptance	QA samples per section, reports temperature, disease, deformities, shape, size, foreign matter, over-ripeness, with photos/videos; accept or reject. If accepted: entered into the 'pack house system' to receive lot number and label (section/block, date/time, weight, pallet no.). If rejected: engineer regrades → re-scaled → QC/QA recheck → second rejection to waste, not entered into the system.	QA (QC/QA sign report); Engineer; Data Entry	Sample QC report (signed); receiving label	C	SOP-RC; SOP-GR
A7	Harvest storage & issue	Harvest Fridge (64 pallets) issues to production by FIFO. Not graded the same day → Fridges 5–7. Graded within 24 h of receiving.	Packing House; Store keeper (FIFO issue)	Fridge temperature records (manual + computerised)	C	SOP-PC; SOP-PL; SUM-2
A8	QC-1 initial QC	Quality and grading check in the Sorting Area; approved product released to production.	QC	Not stated	NYV	SUM-1; SUM-3
A9	Production on parallel lines	Packaging requested from the store controller and collected daily; line supervisors issue packaging per plan/order; QC and line leader inspect product before packing; packing to spec; grades Premium/GR1/GR2/waste; QC provides grade-to-customer schedule; spec deviations need written customer approval. Routes: Figure 4.2.	PHM / PE / Line leaders / packers; QC	Working plan / sales order	C	SOP-PL; SUM-1
A10	QC-2 online & final QC	Random visual checks on the lines; final inspection of finished product before weighing.	QC	Not stated	C	SOP-GR; SUM-3
A11	FG weighing, labelling, palletising	Finished product weighed and labelled (weight, lot, grade or customer, …); Pallet ID per pallet; palletised to customer requirement; non-order GR1/GR2 on wooden pallets in black boxes.	Packing House; Data Entry	FG label; Pallet ID	C	SOP-PL; SUM-3
A12	Waste handling	QC-approved label → scaled → entered in ERP → waste area → farm waste area at start of day shift. Signed waste report (thresholds EV-15). Long-storage waste and weight loss recorded separately.	PH; QC; Data Entry; Finance; Warehouse; QA; Top mgmt	Waste report (printed, signed); ERP entry	C	SOP-WS
A13	FG cold storage	Fridges 1–4 with traceability label (crop, date, grade, customer, section). FIFO. Temperatures manual + computerised. 48 h / 72 h limits.	Packing House	Temperature records	C	SOP-PC
A14	Dispatch & loading	Orders on white board; stock allocated FIFO; shortage → issue from Harvest Fridge for grading. Vehicle inspection; shipping plan and pack list printed; QC inspects with photos/videos; product temperature checked; QA records findings; QC signs and stamps label. Reject → back to fridge, replaced by next FIFO pallet, regraded. Logger, seal (no. on packing list), no loading if not lockable; 09:00–01:00. Shelf-life sample per product/customer/load.	PH; QC; QA; Logistics; Data Entry	Shipping plan; pack list; QC report; logger; seal no.	C	SOP-DS; SOP-GR
A15	Complaints & recall	Via Sales → PH, QC, QA. Validity rules; investigation by QA and QC confirmed with PHM; recall within 6 h if health risk; resolution within 24 h.	Sales; QA; QC; PHM	Customer complaint report; trend analysis	C	SOP-CP
A16	Daily stock reporting	Per crop: daily harvest, stock by customer and grade, waste, season-to-date harvest and waste; crop colour separately; harvest vs supplier crop separately. Sent daily to all parties.	Packing House / Data Entry (not stated)	Daily stock report	C	SOP-SR

4.3 Process maps

AS-IS (A) — Farm dispatch to release for production

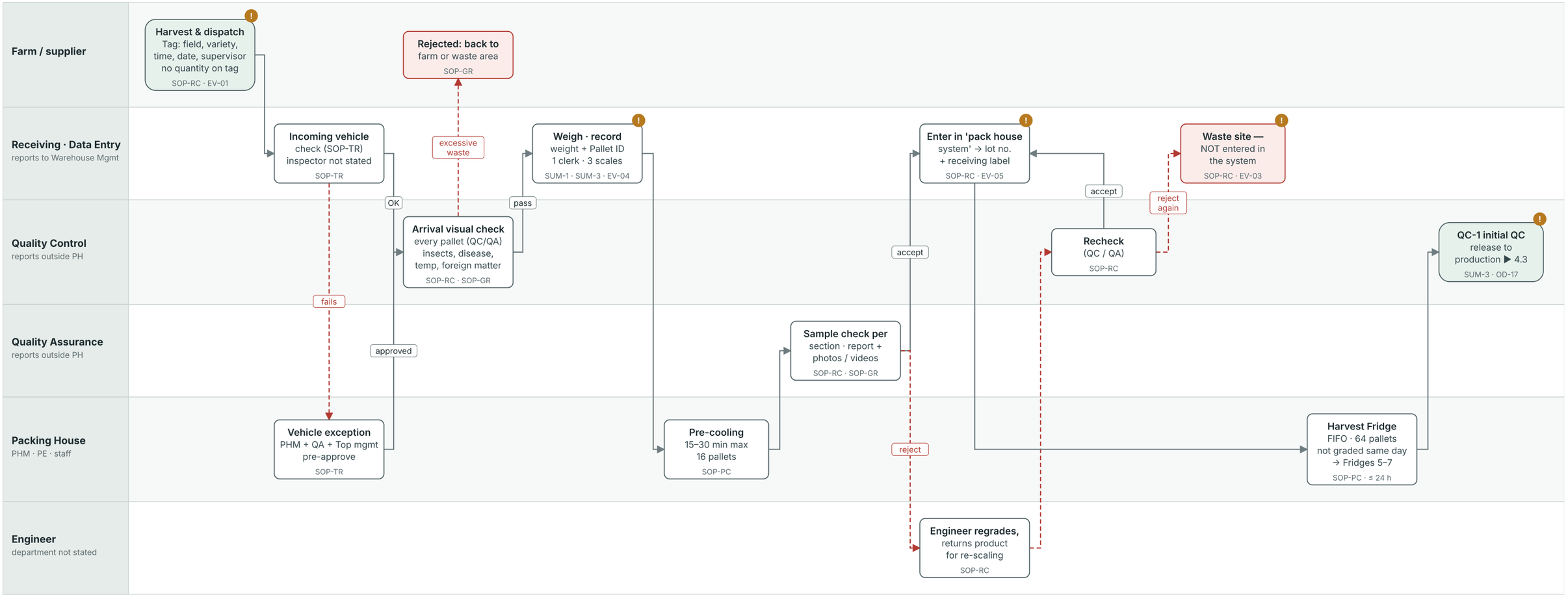


Figure 4.1 — AS-IS (A): farm dispatch to release for production. Amber markers flag documented conflicts or gaps.

Lanes follow documented reporting lines. The engineer's department is not stated (OD-12). The weighing step reflects the Summary; SOP001 places system entry after QA acceptance (EV-04).

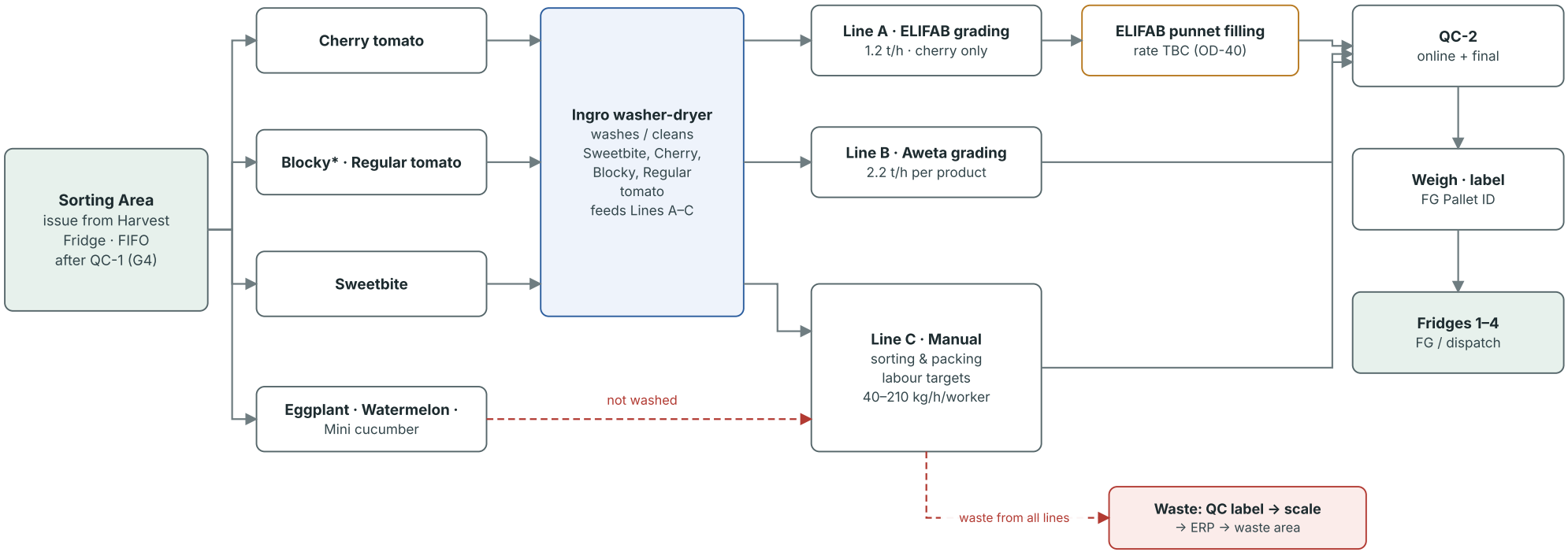
Legend

- Activity (lane = performing department)
- Decision point
- Rejection / waste outcome
- Proposed change (TO-BE)
- System record / state / rule
- Conflict or gap in the sources
- Exception path (dashed red)

Key observations

- The tag carries no quantity, so farm dispatch cannot be reconciled with the weights recorded at the scale (EV-01).
- Two documents disagree on when a pallet first enters the system: at the scale (Summary) or after QA acceptance (SOP001) (EV-04).
- Product rejected twice goes to the waste site and is not entered in the system, so rejected quantities are invisible (EV-03).
- The lot number is assigned at different points in different parts of the SOP (EV-05).
- Three different departments act on every pallet before it reaches production: Data Entry (Warehouse Mgmt), QC/QA (outside PH) and the Packing House. Every hand-off is informal today (HO-01 – HO-07).

Crop routes through production — parallel lines (AS-IS = TO-BE)



Lines A, B and C can operate simultaneously (SUM-1). 1 h per shift is reserved for machine cleaning (SOP-CL). Throughputs are running rates, not holding capacity.

* Crop identity of "Blocky" (pepper or tomato type) and the washer crop list are open (EV-29, OD-30). Routes per SUM-1 Rev 2.2 (manager-corrected).

Figure 4.2 — Crop routes through the parallel production lines (unchanged in the TO-BE).

AS-IS (B) — Production to truck departure

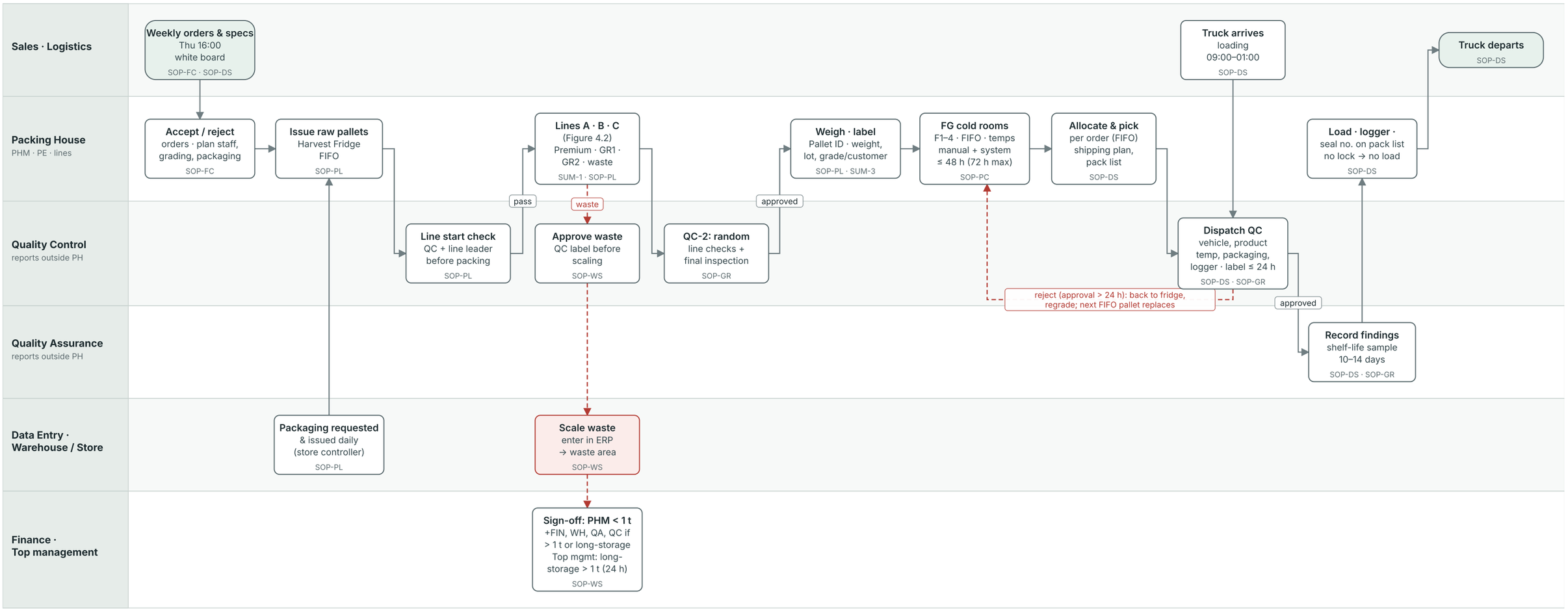


Figure 4.3 — AS-IS (B): production to truck departure.

Waste sign-off thresholds and the 24 h top-management rule per SOP-WS. Dispatch rejection only for pallets whose QC approval is older than 24 h (SOP-DS).

Legend

- Activity (lane = performing department)
- Decision point
- Rejection / waste outcome
- Proposed change (TO-BE)
- System record / state / rule
- Conflict or gap in the sources
- Exception path (dashed red)

Key observations

- Orders reach the floor on a white board; there is no record linking an order to the pallets loaded against it (EV-32).
- The link from incoming pallets to finished pallets depends on the traceability team's manual recording; the team's identity is not stated (EV-19).
- Waste needs QC approval before scaling and up to five signatures (or top-management approval) after; this is a likely approval-latency point (B-06, H-05).
- The 24 h QC-approval rule decides whether dispatch QC may reject a pallet; it is hard to check by eye at loading (EV-13).
- Logger scope differs between SOP sections: all loads vs export loads (EV-23).

4.4 Capacity and time-rule baseline

Item	Documented value	Label	Source
Gates / scales at receiving	3 gates, 3 scales; 1 data-entry employee	C	SUM-1
Pre-cooling	16 pallets; 6–12 t depending on crop; dwell 15–30 min max	C	SUM-2; SOP-PC
Harvest Fridge	64 pallets; 22.4–44.8 t by crop; FIFO to production	C	FCT; SUM-2
Fridges 1–7	60 pallets each (420); 21–42 t per room; F1–4 finished goods/dispatch, F5–7 harvest goods not graded the same day	C	FCT; SOP-PC
Field boxes per pallet	35 (cherry 70); Option 3 = 49 boxes is an option, not current practice	C	FCT
Washer-dryer (Ingro)	Washes Sweetbite, Cherry, Blocky, Regular tomato; feeds Lines A–C; throughput not stated	C	SUM-2
Line A — ELIFAB	Grading 1.2 t/h (cherry only); punnet filling rate TBC	C NYV	SUM-2
Line B — Aweta	2.2 t/h per product (Blocky, Regular tomato)	C	SUM-2
Line C — manual	Labour targets 40–210 kg/h/worker by category (9 categories)	C	SUM-2
Cleaning	1 h per shift for machine cleaning and preparation; 1 day per week deep clean	C	SOP-CL
Time rules	Last harvest 17:00; grade ≤ 24 h from receiving; load ≤ 48 h from harvest (72 h absolute max for a specific order); QC approval ≤ 24 h at dispatch; loading 09:00–01:00; LSS trucks 11:00 and 19:00	C	SOP-RC; SOP-PC; SOP-GR; SOP-DS

4.5 Evidence register — facts, conflicts and gaps

Every design decision in Sections 5–10 traces to at least one item here. Items labelled Not Yet Verified are conflicts, gaps or data-quality findings and each has an open decision.

ID	Statement	Type	Label	Source	Design consequence
EV-01	Every arriving pallet carries a harvesting tag stating field, variety, time, date and supervisor. The tag carries no quantity.	Fact / gap	C	SOP-RC; SUM-3 key point	Farm dispatch cannot be reconciled with received weight. Drives CP-02 (dispatch note) and OD-01.
EV-02	Receiving has 3 gates and 3 scales; one data-entry employee records weight and Pallet ID. Data Entry reports to Warehouse Management.	Fact	C	SUM-1; SUM-3	Receiving registration is a Warehouse-line role, not Packing House. Possible queue — hypothesis H-01.
EV-03	SOP001 says product is entered into the 'pack house system' only after QA acceptance; product rejected twice is not entered at all.	Fact / control gap	C	SOP-RC	Rejected quantities leave no system record. Drives CP-01 and OD-01.
EV-04	The Summary records Pallet ID and weight at receiving, before pre-cooling and QA acceptance.	Conflict with EV-03	NYV	SUM-3 open item 6	Point of system entry must be decided (OD-01).
EV-05	Lot-number assignment is described at three different points: on QA acceptance (SOP-RC), after pre-cooling in the Harvest Fridge (SOP-PF), and 'during production' (SOP-RC).	Conflict	NYV	SOP-RC; SOP-PF; SUM-3 open item 8	Lot definition and assignment point must be decided (OD-02).
EV-06	Pre-cooling: 16 pallets; dwell 15–30 min maximum depending on receiving temperature. The SOP flow diagram states 30 min.	Fact / gap	C	SOP-PC; SOP-PF; SUM-2	No rule links receiving temperature to dwell time, so the system cannot enforce it until defined (OD-07).
EV-07	Arrival visual check by QC/QA on every pallet; QA sample check per section after pre-cooling, documented with photos/videos; accept / reject.	Fact	C	SOP-RC; SOP-GR; SUM-3	Two distinct incoming gates (G2, G3).
EV-08	Acceptance tolerances are in the 'Incoming raw material grading schedule' (SOP index p.7). This schedule is not present in the supplied SOP file. The SOP also refers to a 'Product specification' for tolerance per defect, which was not supplied.	Gap	NYV	SOP-RC; SOP-GR; SOP index	Acceptance criteria cannot be configured. The system will hold them as QA-owned master data (OD-05, OD-06).

ID	Statement	Type	Label	Source	Design consequence
EV-09	QA rejection: notify the engineer; engineer regrades and returns product for re-scaling; QC/QA recheck; second rejection → waste.	Fact	C	SOP-RC	Rework loop with a maximum of one recheck (G3, E-04). The engineer's department is not stated (OD-12).
EV-10	Harvest Fridge: 64 pallets; issue to production by FIFO. Product not graded the same day moves to Fridges 5–7. Grade within 24 h of receiving.	Fact	C	SOP-PC; SUM-2	24-hour grading timer from receiving (V-06).
EV-11	QC-1 (initial QC) in the Sorting Area: quality and grading check; approved product released to production. Wording to be confirmed against the SOP.	Fact (wording open)	NYV	SUM-1; SUM-3 open item 4	Gate G4 is specified but its scope must be confirmed (OD-17).
EV-12	Before packing starts, the Quality Controller and the line leader visually inspect product. QC does random checks on the line and at final product.	Fact	C	SOP-PL; SOP-GR; SUM-3	Gates G5 (line start) and G6 (QC-2).
EV-13	Full pallets cannot be rejected after completion at the line unless the pallet has been in the fridge for more than 24 h before dispatch. A pallet approved less than 24 h before loading cannot be rejected at dispatch. Every dispatched pallet needs a QC approval label not older than 24 h.	Fact	C	SOP-GR; SOP-DS	QC approval is time-stamped with a computed 24 h validity (V-11).
EV-14	Loading within 48 h of harvest; up to 72 h absolute maximum only for a specific order; more than 72 h is 'strictly not allowed'. The Packing House Manager's note says 'Order Received — Max. 48 Hours'. Unsold graded product waits a maximum of 48 h, then goes to DC.	Fact / ambiguity	NYV	SOP-PC; SOP-PL; SUM-3 open item 7	The 48 h clock start is ambiguous (OD-09). The 72 h limit and 'regrading after long storage' conflict (OD-10).
EV-15	Waste: bagged, QC-approved label before scaling, scaled during the shift, entered in the ERP, moved to the waste area. Waste report signed by the PHM if under 1 t; over 1 t also by Finance, Warehouse, QA and QC. Long-storage waste is recorded separately and signed off by Finance/Warehouse, QC/QA and the PHM; above 1 t it also needs top-management e-mail approval, and if no approval arrives within 24 h the waste is removed anyway.	Fact	C	SOP-WS	Approval routing by weight and category (V-16, V-17, E-20). The basis of the 1 t threshold (per report, per shift or per day) is not stated (OD-11).
EV-16	Instructions from other departments to Packing House staff, including QC, pass through the Packing House Manager or the Production Engineer.	Fact	C	SOP-ST; SUM-3	The system must not assign tasks directly from QC to PH staff (CP-15).
EV-17	SOP001 headers carry two revision states (Rev 002 – 10/08/2025 vs Draft version 001, 15/06/24). The SOP index lists a Packhouse Organogram (p.19) that is not present in the supplied file.	Document-control gap	NYV	SOP	The current controlled revision and the organisation chart must be confirmed (OD-36).
EV-18	QC and QA report to their own managements outside the Packing House. QA and QC are separate functions.	Fact	C	SUM-3	Quality decisions are recorded only by QC/QA users. The PHM cannot approve quality gates.
EV-19	Data entry clerks handle all ERP transactions and traceability. A 'traceability team' keeps traceability during production.	Fact / ambiguity	NYV	SOP-ST; SOP-PC	Whether the traceability team is the data-entry clerks is not stated (OD-13).
EV-20	Fridge temperatures are recorded manually at regular intervals and in a computerised record. The interval is not stated.	Fact / gap	NYV	SOP-PC	Reading interval (OD-32). The computerised system is not identified (IB-03).
EV-21	Storage groups kept separate: tomato & cherry; mini cucumber; pepper & sweet bite; eggplant; mango; plus finished goods/dispatch and harvest. Allocations may change with need.	Fact	C	SOP-PC	Fridge storage-group assignment is configurable, with a placement check (V-07).
EV-22	Documented storage temperatures: tomato 10–12 °C; eggplant 10–12 °C; mini cucumber 10–12 °C; sweet pepper 8–10 °C; hot pepper 8–10 °C; capsicum 8–10 °C; plum tomato 10–12 °C; onion 0–5 °C; mango room temperature 2 days then 15 °C; chestnut pumpkin 10–15 or 20–25 °C.	Fact	C	SOP-PC	Used as documented reference ranges. Alarm thresholds, tolerances and any critical-limit status are not documented (OD-08).

ID	Statement	Type	Label	Source	Design consequence
EV-23	Dispatch: vehicle inspected (temperature, cleanliness, smell, contamination) before loading; product temperature checked; QC inspects to customer specification with photos/videos; a temperature logger is added to all loads (SOP-DS) / all loaded export trucks (SOP-GR); seal number on packing list; no loading if the truck cannot be locked; loading 09:00–01:00; LSS trucks 11:00 and 19:00.	Fact / conflict	C	SOP-DS; SOP-GR; SOP-TR	Dispatch gate G8. Logger scope conflict (OD-18).
EV-24	Shelf-life sample of every product, per customer, per load, kept by QA for 10–14 days.	Fact	C	SOP-GR	Shelf-life sample record linked to the load (FR-DS-09).
EV-25	Complaints: received via Sales; valid only with nature, % out of specification, photos/videos, delivery date, box codes and lot number; must arrive the same day as delivery (more than 24 h → invalid); recall within 6 h for health-risk incidents; investigation and resolution within 24 h; returns within 24 h.	Fact	C	SOP-CP	Complaint module timers (V-18). 'Box code' is not defined anywhere (OD-19).
EV-26	Finished label content: weight, lot number, supplier name, product description (grade if not packed for a specific customer), customer name and address if order-specific, customer extras, origin for certain orders. The traceability label adds crop, date, grade, customer and section number.	Fact	C	SOP-PL; SOP-PC	FG label data set (Section 7.3).
EV-27	Harvest crop and supplier crop are recorded separately per supplier. Labels carry the supplier name. Incoming supplier vehicles are subject to the transport procedure.	Fact	C	SOP-SR; SOP-PL; SOP-TR	Receiving must support own-farm and external-supplier sources (OD-31).
EV-28	Production lines: Ingro washer-dryer feeds Lines A–C; Line A ELIFAB grading 1.2 t/h (cherry only) and ELIFAB punnet filling (rate TBC); Line B Aweta 2.2 t/h per product (Blocky, Regular tomato); Line C manual sorting & packing (washed Sweetbite; unwashed eggplant, watermelon, mini cucumber). 1 h per shift for machine cleaning.	Fact	C	SUM-1; SUM-2; SOP-CL	Line and route master data (Figure 4.2).
EV-29	The Summary groups 'Blocky' with Regular tomato on Line B (Aweta), but cites Aweta's pepper page [R2], and 'blocky' is commonly a pepper type. The SOP stores pepper & sweet bite (8–10 °C) separately from tomato (10–12 °C). The SOP site-plan note lists the washing line crops as 'Pepper/Sweet bite'.	Ambiguity	NYV	SUM-1; SUM-2; SOP-PC; SOP-SP; SUM-3 open item 5	Crop identity affects storage-group and temperature rules (OD-30).
EV-30	Fridge capacity: Harvest Fridge 64 pallets; Fridges 1–7 60 pallets each; pre-cooling 16 pallets. Field boxes per pallet 35 (cherry 70). Option 3 (49 boxes per pallet) is shown as an option.	Fact	C	FCT; SUM-2	Location capacity master data (V-08). Option 3 is not current practice (OD-29).
EV-31	In the capacity table, the column headed 'Kg per pallet' holds totals per location (e.g. Fridge 1 tomato 42,000 = 60 × 700 kg). Option 3 figures are not a uniform 49/35 scaling: implied kg per pallet rises ×1.36 (tomato), ×1.43 (pepper), ×1.33 (mini/eggplant), ×1.50 (onion), against ×1.40 for box count.	Data-quality finding	NYV	FCT (recomputed)	Nominal kg per pallet must be confirmed before use as a plausibility check (OD-29).
EV-32	Orders are prepared from the forecast and displayed on a white board. PH management accepts or rejects orders from Sales.	Fact	C	SOP-DS; SOP-FC	No recorded order hand-off today. Order source and format must be decided (OD-37).
EV-33	Packaging: requested from the store controller and collected daily; only 24 h of packaging is kept inside the packhouse; packaging used is removed from stock in the ERP by the Warehouse Dept on dispatch. LSS box inventory is kept by the PH data entry clerk.	Fact	C	SOP-PL; SOP-PC; SOP-CC	Packaging consumption stays an ERP/Warehouse transaction (IB-01).
EV-34	Vehicle non-compliance at incoming: product not accepted until pre-approved by the PHM, the QA team and top management; condition noted on the raw material receiving report.	Fact	C	SOP-TR	Three-party exception approval (E-02).
EV-35	Customer crates delivered for supply are inspected by the Quality controller; if dirty, PH staff wash them or they are returned to the supplier.	Fact	C	SOP-CC	Customer property control (FR-ST-07).

ID	Statement	Type	Label	Source	Design consequence
EV-36	The SOP documents no control of wash water for the washer-dryer.	Gap (observation)	NYV	SOP (absence); SUM-2	Referred to QA for hazard analysis (OD-23). No requirement is invented here.

5 TO-BE process architecture

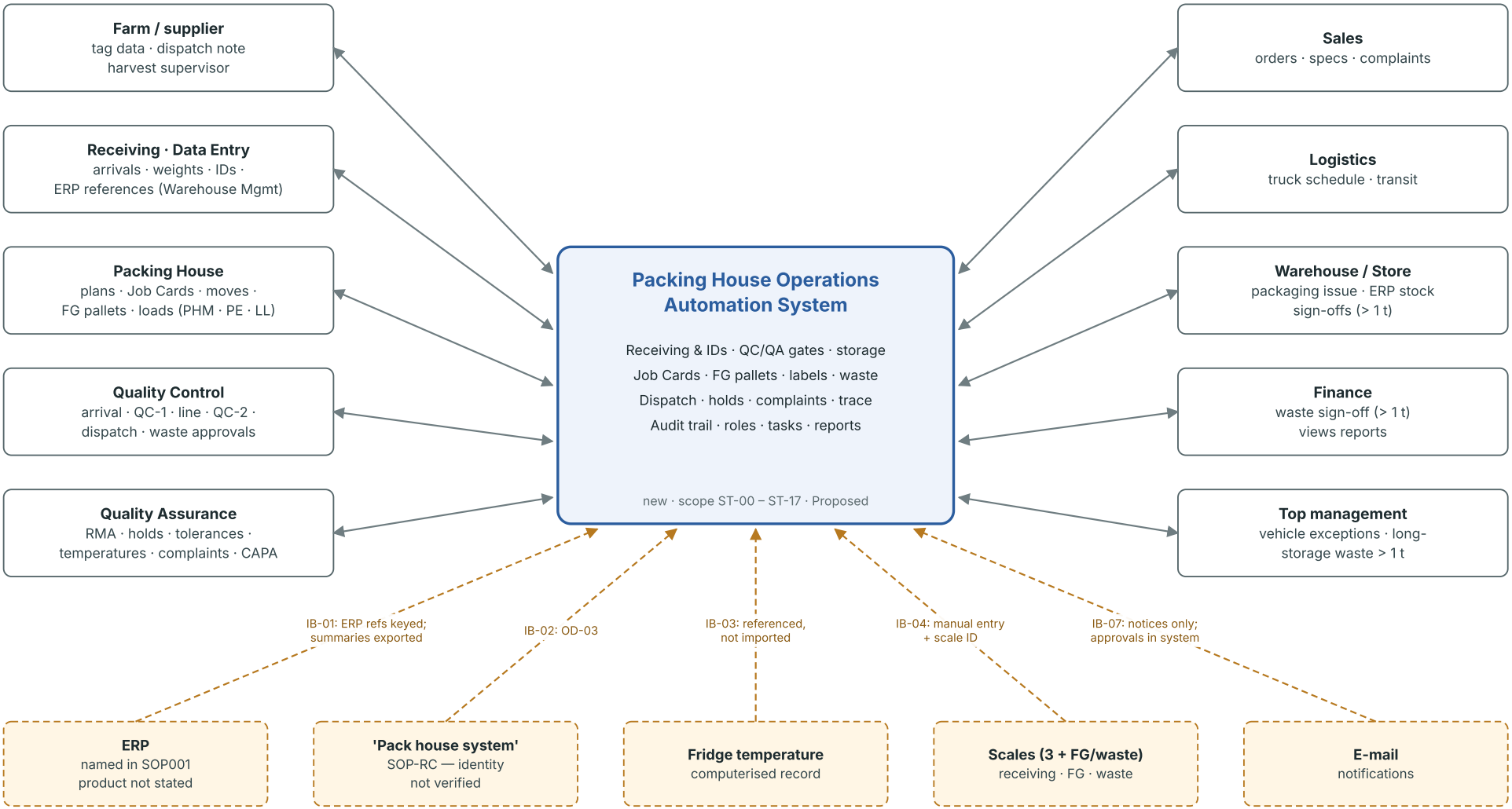
The TO-BE keeps the AS-IS sequence, crop routes and department authority, and adds identity, records, states and gates around them. Changes to the AS-IS are listed as change proposals (Section 5.5) with rationale, risk, evidence and an approval owner.

5.1 Design principles

ID	Principle	Meaning
DP-01	Preserve the AS-IS	The system records and enforces what SOP001 and the Summary already require. Sequence, crop routes and department authority change only where an approved change proposal (CP) says so.
DP-02	Record once, at source	The role that observes an event records it, or a data-entry clerk records it on that role's behalf with both identities stored. No re-keying between steps.
DP-03	Identity before movement	No pallet moves between stages without a scannable ID.
DP-04	Gates are states	A QC/QA decision changes the item's state; the next stage accepts only items in the required state.
DP-05	Exceptions are records	Holds, overrides, deviations and late entries each have an owner, a reason and a closure.
DP-06	Configuration, not code	Time limits, thresholds, tolerances, checklists and label templates are data with owners and effective dates.
DP-07	Existing systems stay	The ERP keeps its transactions; the computerised fridge record stays; no API or integration is assumed.
DP-08	Minimum approvals	Build only the approvals the SOP requires or an approved CP adds. Routine recording needs no approval.
DP-09	Degrade safely	Paper fallback with pre-numbered forms and flagged back-entry when the system or a device is unavailable.

5.2 System context

System context — departments, the new system and existing systems



Existing systems (dashed) — kept as they are; no API or integration assumed. Departments keep their own decisions; the system records them.

Figure 5.1 — System context. Each department keeps its decisions; the system records them. Existing systems are not replaced and no interface is assumed.

5.3 End-to-end TO-BE flow

TO-BE overview — stages, gates and key records

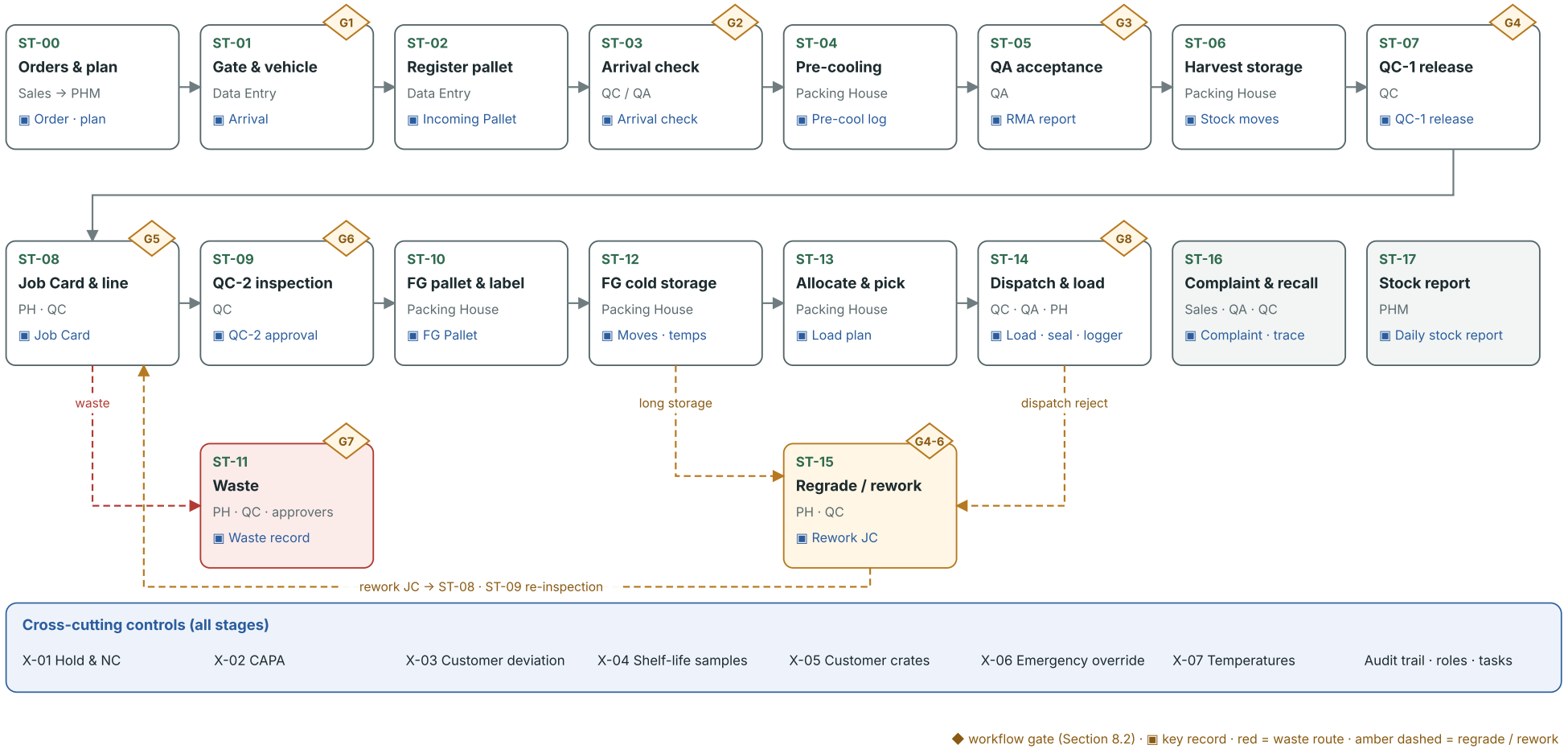


Figure 5.2 — TO-BE overview: stages, gates and key records.

TO-BE (A) — Arrival to issue for production

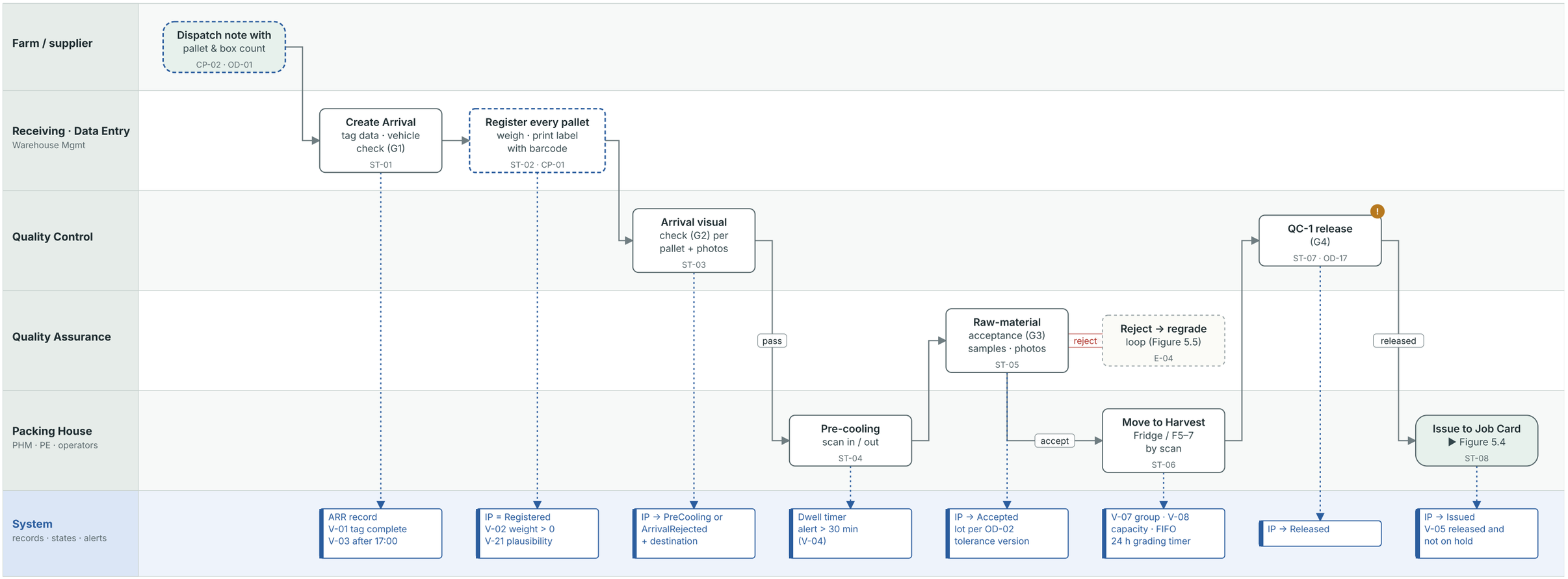


Figure 5.3 — TO-BE (A): arrival to issue for production, with system records and states. Blue dashed borders mark proposed changes.

Registering every pallet at the scale (CP-01) and the dispatch note (CP-02) depend on OD-01. QC-1 scope depends on OD-17.

Legend

- Activity (lane = performing department)
- Decision point
- Rejection / waste outcome
- Proposed change (TO-BE)
- System record / state / rule
- Conflict or gap in the sources
- Exception path (dashed red)

What changes from the AS-IS

- Same sequence and the same departments as Figure 4.1; the system adds IDs, states and gates (DP-01, DP-04).
- Every pallet is registered at the scale with status 'Registered'; acceptance is a QA state change, not a data-entry event (CP-01, pending OD-01).
- Each QC/QA decision is recorded by QC/QA users only and moves the pallet's state; Data Entry cannot accept a pallet (SoD-01).
- Timers make the SOP time rules visible: pre-cooling dwell (V-04) and grading within 24 h of receiving (V-06).
- Rejections follow Figure 5.5 and leave a record; nothing disappears from the quantity chain.

TO-BE (B) — Job Card to truck departure

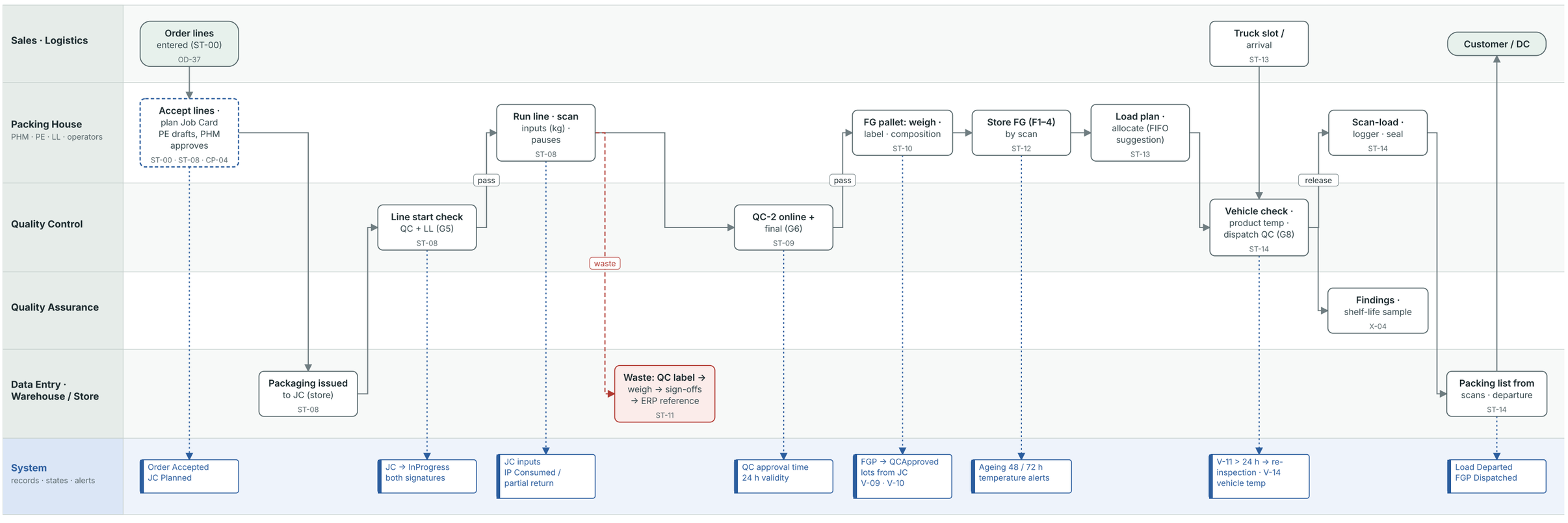


Figure 5.4 — TO-BE (B): Job Card to truck departure.

Instructions arising from QC findings go to the PHM/PE queue, never directly to PH staff (V-30, SOP-ST).

Legend

- Activity (lane = performing department)
- Decision point
- Rejection / waste outcome
- Proposed change (TO-BE)
- System record / state / rule
- Conflict or gap in the sources
- Exception path (dashed red)

What changes from the AS-IS

- A Job Card links the pallets consumed to the finished pallets and waste produced, giving a mass balance per run (CP-04, MB-01).
- QC-2 approval is time-stamped; at loading the system computes whether the 24 h validity has passed and asks for re-inspection (V-11).
- Waste is approved by QC before weighing and routed for sign-off by weight and category (V-16, V-17); the ERP posting stays with Data Entry and its reference is stored (IB-01).
- The packing list is generated from scanned pallets, so it equals what was loaded (CP-10).
- QC findings that need staff action go to the PHM/PE queue (CP-15, V-30).

5.4 Handoffs, decision points and exception paths

Handoffs between departments

ID	From	To	Handed over	System mechanism	Label
HO-01	Farm / supplier	Receiving (DEC)	Load with harvesting tag (+ dispatch note if CP-02)	Arrival record; tag fields mandatory	C P
HO-02	Receiving (DEC)	QC / QA	Registered pallets awaiting arrival check	Task queue 'Arrival check due' per arrival	P
HO-03	QC / QA	Packing House	Passed pallets to pre-cooling	IP state PreCooling; pre-cooling board	C P
HO-04	Packing House	QA	Pre-cooled pallets awaiting raw-material acceptance	Task queue 'RMA due' per section and day	P
HO-05	QA	Engineer	Rejected product for regrade	Regrade task created on rejection	C P
HO-06	Engineer	Receiving (DEC) → QC / QA	Regraded product for re-scaling and recheck	Re-weigh task, then recheck task	C P
HO-07	QA	Packing House	Accepted pallets for storage	IP state Accepted; storage move task	C P
HO-08	Packing House	QC	Pallets requested for production (QC-1)	QC-1 request task	NYV
HO-09	Warehouse store	Packing House (LL)	Packaging for the day's plan	Packaging issue recorded on JC	C
HO-10	QC	PHM / PE	Findings needing staff action	Task to PHM/PE queue only (V-30)	C
HO-11	Packing House	QC	Output ready for final inspection	QC-2 task per JC	C P
HO-12	Packing House	QC → approvers → Data Entry	Waste for approval, sign-off and ERP posting	Waste record routing (V-16, V-17)	C
HO-13	Sales	PHM	Order lines and customer specifications	Order lines 'Pending acceptance'	C
HO-14	PHM	Sales / Logistics	Order acceptance; shortages; load readiness	Notifications with timestamps	C P
HO-15	Logistics	Packing House	Truck schedule and ETA (one day before)	Schedule entry on load	C
HO-16	Packing House	QC	Picked load ready for dispatch inspection	Dispatch QC task per load	C P
HO-17	QC	QA	Dispatch findings; shelf-life sample	QA dispatch record and sample per load	C
HO-18	Packing House / DEC	Warehouse (ERP)	Packaging consumed per load; waste per day	Exportable summaries; ERP reference returned	C P
HO-19	Sales	QA / QC / PHM	Customer complaint	Complaint record with timers	C
HO-20	QA	Farm	Farm-related complaint (residues, MRLs) in writing	Letter generated from the complaint record	C

Decision points

ID	Decision	Owner	Inputs	Outcomes	Label	Source
D-01	Accept or reject an order line	PHM	Sales forecast vs harvest forecast vs grading %	Accepted / Rejected	C	SOP-FC
D-02	Accept a non-compliant incoming vehicle	PHM + QA + Top mgmt	Vehicle inspection	Accepted with exception / Not accepted	C	SOP-TR
D-03	Arrival visual check	QC / QA	Checklist per pallet	To pre-cooling / Rejected to farm or waste area	C	SOP-RC
D-04	Raw-material acceptance	QA	Samples, tolerance set	Accepted / Rejected → regrade	C	SOP-RC

ID	Decision	Owner	Inputs	Outcomes	Label	Source
D-05	Recheck after regrade	QC / QA	Recheck sample	Accepted / Waste	C	SOP-RC
D-06	Fridge allocation (storage group)	PHM	Stock by group, capacity	Fridge group assignment	C	SOP-PC
D-07	QC-1 release	QC	Quality and grading check	Released / Hold	NYV	SUM-3
D-08	Line start	QC + Line Leader	Product vs requirements	Start / Correct and re-check	C	SOP-PL
D-09	Customer deviation	Customer (written)	Deviation request	Approved / Not approved	C	SOP-PL
D-10	QC-2 pass	QC	Line and final inspection	Approved / Re-sort	C	SOP-GR
D-11	Waste approval and sign-off	QC; PHM; + FIN/WHM/QA/QC above 1 t and for all long-storage waste; TOP long-storage above 1 t	Weight, category, photos	Approved / Refused	C	SOP-WS
D-12	Unsold product after 48 h	PHM / Sales	Age, orders	Dispatch to DC	C	SOP-PL
D-13	Dispatch vehicle acceptance	QC	Temperature, cleanliness, smell, contamination, lockable	Load / wait / reject vehicle	C	SOP-DS
D-14	Dispatch pallet release	QC	QC approval age; product temperature; packaging	Load / cool / re-inspect / reject	C	SOP-DS; SOP-GR
D-15	Rejected pallet at dispatch	QC + PHM (customer optional)	Rejection reason	Resort / replace / customer concession	C	SOP-GR
D-16	Complaint validity	QA / QC	Completeness; 24 h window	Valid / Invalid	C	SOP-CP
D-17	Recall	Not stated (OD-22)	Health risk	Recall within 6 h	C NYV	SOP-CP
D-18	Hold disposition	QCM / QAM	Assessment	Release / regrade / rework / waste / return / concession	P	X-01

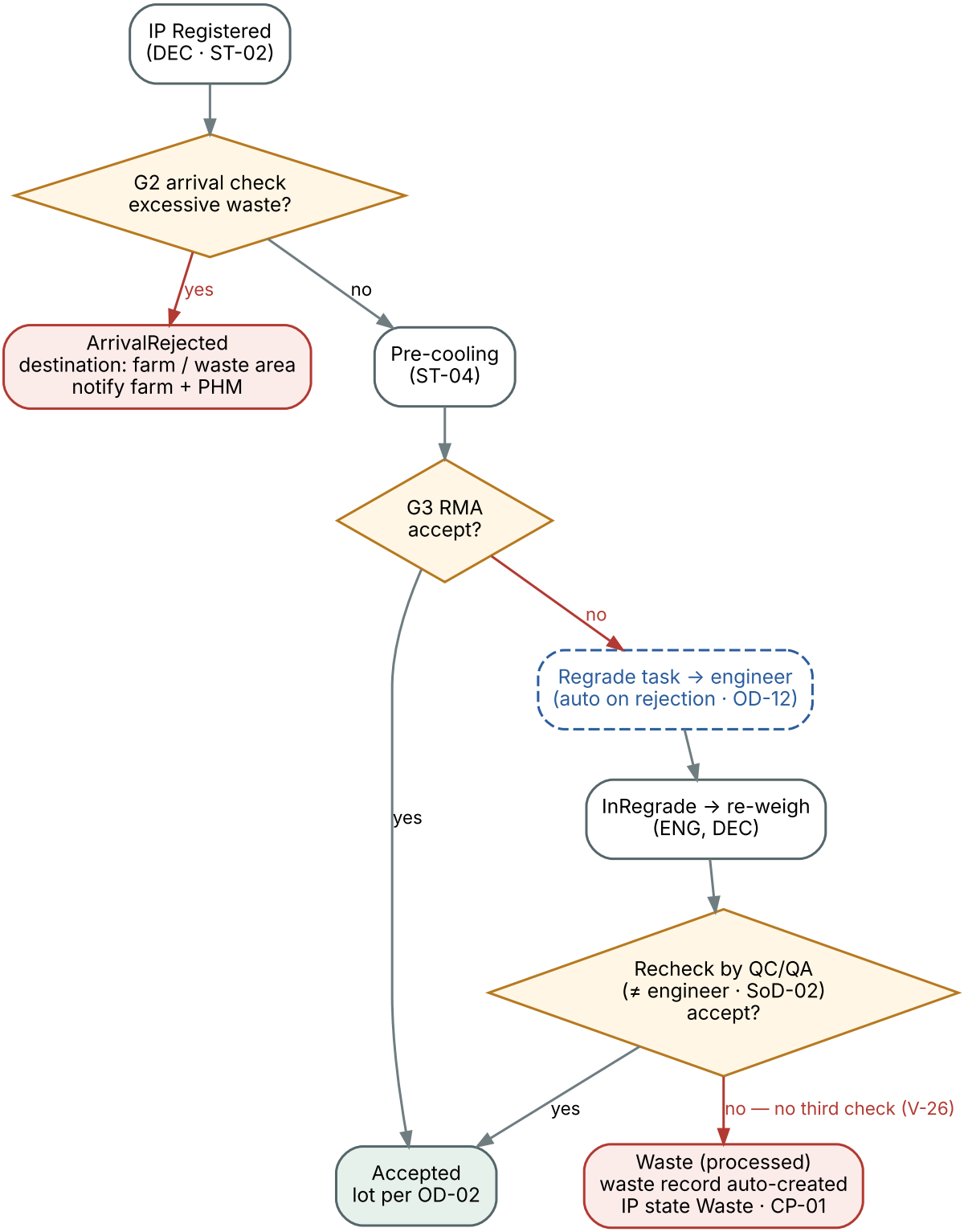


Figure 5.5 — Exception path — incoming rejection and regrade (E-03, E-04). Red = rejection path; blue dashed = system-created task.

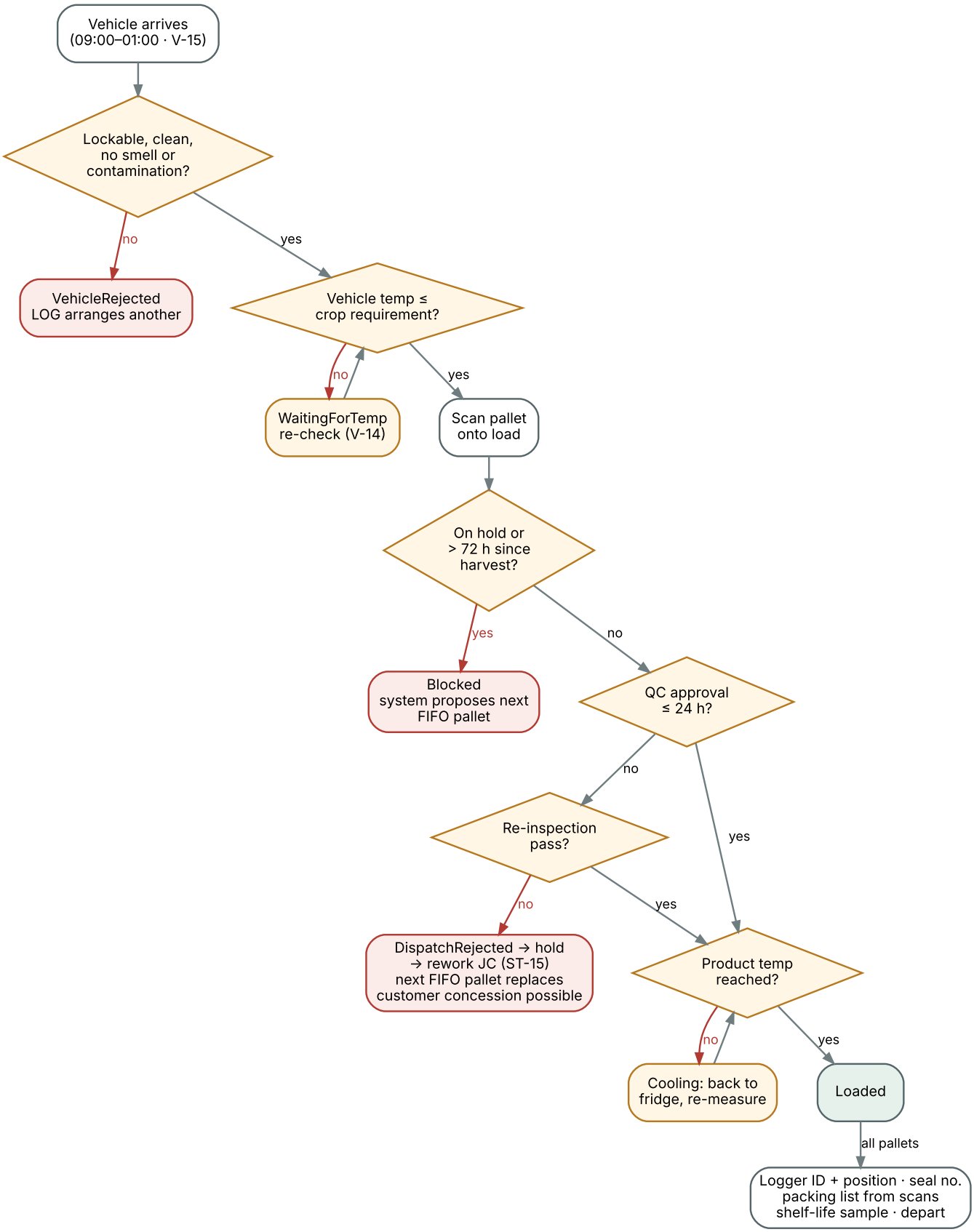


Figure 5.6 — Exception path — dispatch (E-14 to E-18). Every block or rejection leaves the pallet traceable and proposes the next FIFO pallet.

5.5 Change proposals (TO-BE vs AS-IS)

Each proposal states why it is needed, what it costs operationally, and what evidence must be collected before approval. None is approved.

ID	Change	Rationale	Benefit	Risk	Evidence required	Approval owner	OD
CP-01	Register every arriving pallet at the scale with status 'Registered', including pallets later rejected. 'Accepted into stock' becomes a status set by the QA decision, not the moment of first entry.	Rejected product currently leaves no system record (EV-03), so received = accepted + rejected cannot be proven.	Complete quantity reconciliation from farm to waste; evidence of rejection rates per section.	Contradicts the current SOP wording; adds records at the busiest point (receiving).	Count of rejected pallets per week (currently unrecorded); receiving time per pallet.	PHM + Warehouse Manager + QA Manager	OD-01
CP-02	Farm/supplier dispatch note with quantity (pallets, boxes) per load, captured at the gate and compared with the received count.	The tag has no quantity (EV-01; Summary key point).	Detects in-transit loss and counting errors; enables farm-to-packhouse reconciliation.	Needs Farm Operations to issue the note; adds a step at farm loading.	Sample of loads comparing dispatched vs received counts.	Farm Operations Manager + PHM	OD-01
CP-03	One defined lot-number assignment point and lot definition, configured in the system.	Three conflicting assignment points (EV-05).	One unambiguous traceability key on every label.	Changing the label format affects customers and existing records.	Customer label requirements; current label samples.	QA Manager + PHM	OD-02
CP-04	Production Order / Job Card per line run, recording input pallets consumed (kg), outputs by grade/customer, waste and residue.	Today the link from incoming pallets to finished pallets depends on the traceability team's manual recording (EV-19).	Mass balance per run; one-step-back / one-step-forward genealogy.	Extra recording on the line; needs a clear owner.	Pilot timing study: recording minutes per run.	PHM	OD-15
CP-05	Electronic QC/QA gate records (arrival check, raw-material acceptance, QC-1, line start, QC-2, dispatch) with mandatory photo/video attachments.	SOP requires photos/videos on all QA/QC reports; paper reports cannot block a pallet's next movement.	Gates become enforceable; evidence retrievable by lot.	Device availability on the floor; photo storage volume.	Device and network survey; average photos per report.	QC Manager + QA Manager	OD-24
CP-06	Electronic QC approval with computed 24 h validity in place of (or alongside) the signed and stamped label.	The 24 h rule (EV-13) is hard to verify by eye at loading.	Automatic re-inspection prompts; fewer disputes at dispatch.	Customers or auditors may still require a physical stamped label.	Customer and certification label requirements.	QC Manager	OD-20
CP-07	System FIFO suggestion and age alerts (24 h grading; 48 h / 72 h harvest-to-loading).	FIFO and age limits are manual today (SOP-PC).	Fewer overdue pallets; less long-storage waste.	Alerts ignored if too frequent.	Baseline age distribution of stock at dispatch.	PHM	OD-09
CP-08	Waste records with QC approval before weight entry and automatic sign-off routing by category and threshold.	Multi-party paper sign-off (EV-15).	Faster, traceable approvals; ERP posting reference captured.	Electronic sign-off may not be accepted by Finance in place of a wet signature.	Current sign-off lead time.	PHM + Finance + QC Manager + QA Manager	OD-11
CP-09	Pallet location recorded per fridge (and optional position) at each move.	Location is not recorded at pallet level today (no source states it).	Faster picking; trace and recall locate stock immediately.	Requires scanning at every move.	Pilot: moves per pallet; scan compliance.	PHM + Warehouse Manager	OD-14

ID	Change	Rationale	Benefit	Risk	Evidence required	Approval owner	OD
CP-10	Dispatch load record: order, pallets scanned, vehicle inspection, product temperatures, QC release, logger ID and position, seal number, packing list generated from scans.	Pack list is printed separately from the loaded pallets today (SOP-DS).	Pack list equals what was loaded; complete load evidence.	Loading speed if scanning is slow.	Loading time per truck (baseline).	PHM + QC Manager + Logistics	OD-18
CP-11	Hold / Nonconformity record with disposition and optional CAPA link.	Rejections and regrades are handled by discussion today (SOP-DS).	Status visible on every affected pallet; closure evidence.	Overhead if used for trivial issues.	Monthly count of rejections and regrades.	QA Manager	OD-16
CP-12	Trace report (backward/forward) and mock-recall test from any ID.	6 h recall requirement (EV-25) depends on manual record search today.	Recall scope in minutes; demonstrable traceability effectiveness.	Only as good as the recorded links.	Mock-recall test results before and after.	QA Manager	OD-21
CP-13	System-printed barcode labels and scanning at moves.	Manual transcription of IDs is error-prone.	Fewer ID errors; faster recording.	Hardware cost; printer failure becomes a stoppage risk.	Label volumes per hour; existing printer inventory.	PHM + IT	OD-24
CP-14	Optional GS1 SSCC as the FG Pallet ID.	Some retail/export customers require SSCC-labelled pallets (not confirmed for these customers).	Customer-ready logistic labels.	GS1 membership and label redesign.	Customer label requirements.	Sales + QA Manager	OD-20
CP-15	QC/QA findings that need PH staff action create a task for the PHM/PE queue, never directly for PH staff.	SOP-ST instruction routing (EV-16).	Preserves the reporting line; records response time.	Possible delay if the PHM/PE is unavailable.	QC finding → action time (H-08).	PHM + QC Manager	OD-16
CP-16	Manual fridge temperature readings entered in the system with the reading time; computerised record referenced, not replaced.	Interval undefined; two parallel records (EV-20).	Temperature history per fridge linked to stored pallets.	Double entry until the computerised record can be imported.	Identity and export capability of the computerised record.	QA Manager + PHM	OD-32

5.6 Stage specifications

For each TO-BE stage: owner, records and creators, permitted actions, data and its authoritative source, execute/verify/approve split, start/complete/hold/reject/reopen conditions, records and audit, exceptions, and the applicable standard or control principle. Inline labels show the status of each statement.

ST-00 Order intake & daily production plan (interface)

AS-IS: A0

Process owner / responsible department	Sales owns orders C; PH management accepts or rejects orders C (SOP-FC). Plan owner: PHM C.
Records created & creator	Customer Order (Sales or PH planner on Sales' behalf NYV); Order Acceptance decision (PHM C); Daily Production Plan (PHM/PE P).
Permitted actions	SAL: create/submit order. PHM: accept/reject order lines, create plan. PE: draft plan. Others: view.
Required data & authoritative source	Customer, order line (product, grade, pack spec, quantity, required date, export/local/retail/LSS), customer specification reference (Sales/QC C SOP-PL), forecast reference. Source: Sales C; the format of today's orders is not known NYV (OD-37).
Execute · Verify · Approve	Execute: SAL. Verify: PE checks against forecast and grading % C. Approve: PHM accepts/rejects C.
Start condition	Order received from Sales.
Completion condition	Each order line is Accepted or Rejected with a reason; accepted lines are available for allocation.
Hold condition	Not applicable.
Reject condition	PHM rejects the line (capacity, stock or packaging) C; reason mandatory P.
Reopen condition	SAL may amend an accepted order; the amendment returns the line to 'Pending acceptance' P.
Records, timestamps, links, audit	Order ID; acceptance timestamp and user; link to dispatch loads and FG pallets.
Exceptions & escalation	Shortage of stock, packaging or trucks must be communicated 24 h (local) / 48 h (export) ahead C → notification to SAL and LOG P.
Standard / control principle	ISO 9001 §8.2.3 review of requirements for products.

ST-01 Gate arrival & incoming vehicle inspection

◆ G1 AS-IS: A1–A2

Process owner / responsible department	Transport procedure belongs to the Warehouse department C (SOP-TR). Gate recording: Receiving/Data Entry (Warehouse Mgmt) P.
Records created & creator	Arrival (load) record — DEC P; Incoming Vehicle Inspection — inspector role not stated NYV; Vehicle Exception Approval — PHM + QA + Top management C.
Permitted actions	DEC: create arrival, capture tag and dispatch-note data. Inspector: record vehicle condition. PHM, QAM/QAO, TOP: approve a non-compliant vehicle exception.
Required data & authoritative source	Arrival time (server), vehicle plate, driver, source (own farm block/section or supplier) C, harvesting tag fields: field, variety, harvest time, date, supervisor C, dispatch-note quantity (pallets/boxes) P (CP-02), vehicle condition checks per SOP-TR C.
Execute · Verify · Approve	Execute: DEC (arrival), inspector (vehicle). Verify: — . Approve: only for exceptions (PHM + QA + Top management) C.
Start condition	Vehicle at the gate.
Completion condition	Arrival record submitted with complete tag data, and vehicle inspection Pass (or exception approved).
Hold condition	Vehicle Fail → arrival 'On Hold — vehicle' until the three-party approval or rejection C.
Reject condition	Exception refused → load not accepted; disposition recorded (returned to farm/supplier) P.
Reopen condition	DEC may correct tag data before G3 with a reason; after G3 only via correction record approved by QAO P.

Records, timestamps, links, audit	Arrival ID → incoming pallets; tag photo P; vehicle inspection linked to the arrival; all timestamps server-generated.
Exceptions & escalation	E-01 incomplete tag; E-02 non-compliant vehicle; late arrival after 17:00 flagged with reason P (17:00 rule C).
Standard / control principle	Codex CXC 1-1969 Ch.1 §9 Transportation; CXC 53-2003 §5.3 incoming material; ISO 9001 §8.4 externally provided products (supplier loads).

ST-02 Weighing & Incoming Pallet registration

AS-IS: A4 (+A6 entry)

Process owner / responsible department	Data Entry / Receiving (Warehouse Management) C (SUM-3).
Records created & creator	Incoming Pallet (IP) — DEC C; receiving label printed C with barcode P.
Permitted actions	DEC: create, edit before submit, submit, print or reprint label (reprint logged P). Others: view.
Required data & authoritative source	IP ID (system), Arrival ID, crop, variety, block/section or supplier, harvest date/time (from tag), box count (35; cherry 70 C), gross/tare/net kg or net kg (weighing method NYV), scale ID, registration timestamp, label number per crop/section/block/day C (SOP-RC).
Execute · Verify · Approve	Execute: DEC. Verify: system plausibility check against nominal kg per pallet (non-blocking) P (OD-29). Approve: none.
Start condition	Arrival record exists (ST-01 complete or held).
Completion condition	IP submitted with all mandatory fields; status 'Registered'; label attached.
Hold condition	Inherited from arrival hold.
Reject condition	Not at this stage (registration records facts, not decisions) P.
Reopen condition	Weight correction before QA decision: DEC with reason. After QA decision: correction record verified by QAO P.
Records, timestamps, links, audit	IP ↔ Arrival; scale ID and weighing timestamp; label print log.
Exceptions & escalation	Scale failure → use alternate scale and record scale ID P; system down → paper fallback E-22.
Standard / control principle	ISO 9001 §8.5.2 identification; ISO 22000 §8.3 traceability (incoming material identified); ISO 9001 §7.1.5 measuring resources (scale calibration status recorded) P.

Registering before QA acceptance is CP-01 and depends on OD-01. Under the current SOP wording, registration happens only after acceptance.

ST-03 Arrival visual check (every pallet)

◆ G2 AS-IS: A3

Process owner / responsible department	QC / QA team C (SOP-RC).
Records created & creator	Arrival Visual Check per IP (or per arrival with per-pallet results P) — QCI or QAO C.
Permitted actions	QCI/QAO: create, submit Pass/Reject. PHM: view. DEC: view.
Required data & authoritative source	Checklist per SOP-RC C: covered on tractor; on pallets in acceptable crates; insects/insect damage; disease; temperature; general appearance; foreign contamination; freshness; conformance to raw-material spec; GR3/waste in crates. Result, remarks, photos C.
Execute · Verify · Approve	Execute: QCI/QAO. Verify: — . Approve: decision is the inspector's C.
Start condition	IP registered (or arrival registered if CP-01 is not approved).
Completion condition	Each IP has Pass or Reject.
Hold condition	Inspector may place 'Hold — pending QA' if unsure P.
Reject condition	Excessive visible waste → rejected; returned to farm or waste area C (SOP-GR); destination mandatory P.

Reopen condition	QCM/QAM only, with reason, before the pallet is moved P.
Records, timestamps, links, audit	Check ↔ IP; inspector; timestamp; photos.
Exceptions & escalation	E-03 arrival rejection.
Standard / control principle	ISO 9001 §8.4.2 verification of externally provided product; CXC 53-2003 §5.3.

ST-04 Pre-cooling

AS-IS: A5

Process owner / responsible department	Packing House C.
Records created & creator	Pre-cooling log (entry/exit per IP) — PHO/DEC P.
Permitted actions	PHO/DEC: record entry and exit (scan). PHM: view, acknowledge overrun.
Required data & authoritative source	IP, receiving product temperature C (checked at arrival), entry and exit time, target dwell from rule table NYV (OD-07).
Execute · Verify · Approve	Execute: PHO. Verify: system dwell check against 15–30 min max C. Approve: — .
Start condition	G2 Pass.
Completion condition	Exit recorded.
Hold condition	—
Reject condition	—
Reopen condition	Time correction by PHM with reason P.
Records, timestamps, links, audit	Entry/exit timestamps; pre-cooling room ID; dwell computed.
Exceptions & escalation	E-05 dwell > 30 min → alert to PHM P; capacity 16 pallets C — queue warning P.
Standard / control principle	Codex CXC 1-1969 Ch.1 §7.2 time and temperature control (applicability of a limit to be set by QA).

ST-05 QA raw-material acceptance (sample per section)

◆ G3 AS-IS: A6

Process owner / responsible department	QA C (SOP-RC: 'QA will … reject or accept').
Records created & creator	Raw-Material Acceptance (RMA) report per section/source and day C — QAO; signed by QA/QC team C; Regrade Request (on rejection) — system on QAO decision P; Recheck report — QCI/QAO C.
Permitted actions	QAO: create, submit, accept, reject. QCI: co-sign / recheck C. ENG: record regrade completion P. DEC: re-weigh after regrade C. PHM: view, notified.
Required data & authoritative source	Sampled IPs, sample size NYV, product temperature, disease, deformities, shape, size, foreign matter/insects, over-ripeness/softness, photos/videos C; tolerance set version NYV (OD-05); acceptance basis 'mix GR1/GR2, minimum to 0 % waste' C.
Execute · Verify · Approve	Execute: QAO. Verify: QC/QA co-signature C. Approve: QAO decision C.
Start condition	Pre-cooling exit (or first IP of the section ready).
Completion condition	Accepted → IP status 'Accepted' (raw stock); lot assigned if OD-02 places it here.
Hold condition	QAO may hold the section's pallets pending a retest P.
Reject condition	Rejected → notify engineer C; Regrade Request; re-scaled C; QC/QA recheck C; second rejection → Waste (processed) C, record kept (CP-01 P).

Reopen condition	QAM only, with reason, before any pallet is issued to production P.
Records, timestamps, links, audit	RMA ↔ all IPs of the section and day; tolerance-set version; photos; regrade and recheck chain.
Exceptions & escalation	E-04 rejection loop; missing tolerance data → RMA cannot be submitted as 'Accepted against spec' until OD-05 closes P.
Standard / control principle	ISO 9001 §8.4.2, §8.6; ISO 22000 §8.9.4 handling of potentially unsafe product (hold until evaluated); CXC 53-2003 §5.3.

ST-06 Harvest storage (Harvest Fridge / Fridges 5–7) & FIFO issue

AS-IS: A7

Process owner / responsible department	Packing House C; stock custody of fridge moves not stated NYV (OD-14).
Records created & creator	Stock Move (location change per IP) — PHO/DEC P; Temperature Reading per fridge — role not stated NYV.
Permitted actions	PHO/DEC: move (scan from/to). PHM: assign fridge storage group, override placement warning with reason P. QAO/PHM: record temperature NYV.
Required data & authoritative source	IP, from/to location, timestamp, fridge storage group C, capacity in pallets C.
Execute · Verify · Approve	Execute: PHO. Verify: system checks group, capacity and FIFO P. Approve: — .
Start condition	IP Accepted.
Completion condition	IP issued to production (ST-08) or regraded/wasted.
Hold condition	QA/QC hold blocks issue P.
Reject condition	— (quality decisions belong to QC/QA stages).
Reopen condition	—
Records, timestamps, links, audit	Full location history; dwell time per location; temperature history of each location during the stay.
Exceptions & escalation	E-06 not graded within 24 h of receiving C; E-07 group conflict or full fridge; E-08 temperature outside documented range.
Standard / control principle	ISO 9001 §8.5.4 preservation; CXC 53-2003 §5.2.2.4 cold storage controlled and monitored; §5.2.2.3 condensate (SOP: not under evaporators).

ST-07 QC-1 initial QC — release to production

◆ G4 AS-IS: A8

Process owner / responsible department	Quality Control C (SUM-3).
Records created & creator	QC-1 Release per IP or per issue batch P — QCI.
Permitted actions	QCI: release, reject (to hold). PHM/PE: view; request QC-1.
Required data & authoritative source	IPs, quality and grading check result C, grade estimate % P, remarks, photos P.
Execute · Verify · Approve	Execute: QCI. Verify: — . Approve: QCI decision C.
Start condition	IP Accepted and requested for production.
Completion condition	IP status 'Released to production'.
Hold condition	QCI holds → Hold record (X-01).
Reject condition	Through Hold disposition (regrade/waste) P.
Reopen condition	QCM with reason before issue P.

Records, timestamps, links, audit	Release ↔ IP; inspector; time.
Exceptions & escalation	Scope overlap with ST-05 to be resolved (OD-17).
Standard / control principle	ISO 9001 §8.5.1 controlled conditions; §8.6 release.

ST-08 Production order (Job Card) & line execution

◆ G5 AS-IS: A9

Process owner / responsible department	Packing House — PHM/PE C; line execution — Line Leader C.
Records created & creator	Job Card (JC) — PE/PHM P (OD-15); JC Input consumption (IP, kg) — LL/TRC P; Packaging Issue — STK C (SOP-PL); Line Start Check — QCI + LL C; Line Stop events P.
Permitted actions	PE: create/plan JC. PHM: approve/cancel JC. LL: start, pause, record inputs, complete. QCI: line start check Pass/Fail. STK: issue packaging. TRC/DEC: record inputs NYV.
Required data & authoritative source	Line (A ELIFAB / B Aweta / C manual) C, route incl. washer-dryer or not C, crop, order line(s) or grade target, customer specification version, inputs (IP, kg or boxes), start/stop times, cleaning time (1 h/shift C), crew count P.
Execute · Verify · Approve	Execute: LL. Verify: QCI + LL line start check C. Approve: PHM approves the plan P.
Start condition	JC Planned; inputs released (G4); packaging issued; line start check Pass.
Completion condition	All outputs (FG pallets, waste, residue returned) recorded; mass balance computed.
Hold condition	QCI finding → Hold on JC outputs; instruction to staff via PHM/PE task C (CP-15).
Reject condition	Line start check Fail → JC cannot start until a re-check Passes P.
Reopen condition	PHM, with reason, within a window to be set (OD-15) P.
Records, timestamps, links, audit	JC ↔ IPs consumed (kg) ↔ FG pallets ↔ waste records; line, time window, crew.
Exceptions & escalation	E-09 line start Fail; E-10 QC-2 non-conformance; E-11 packaging shortage; E-12 spec deviation needs written customer approval C; E-13 mass-balance variance.
Standard / control principle	ISO 9001 §8.5.1; ISO 22000 §8.3 (relation of received lots to end products; rework).

ST-09 QC-2 online & final inspection

◆ G6 AS-IS: A10

Process owner / responsible department	Quality Control C.
Records created & creator	Online check (random, per JC) and Final Inspection (before weighing) — QCI C.
Permitted actions	QCI: record Pass/Fail and remarks; raise Hold. LL: view.
Required data & authoritative source	JC, sample, defects observed, grade conformance vs customer specification C, photos C.
Execute · Verify · Approve	Execute: QCI. Verify: — . Approve: QCI Pass is the QC approval used for the FG pallet's 24 h validity P (CP-06).
Start condition	JC In Progress.
Completion condition	Final inspection Pass for the output batch/pallet.
Hold condition	Fail → affected output on hold; re-sort instruction via PHM/PE C.
Reject condition	Fail before palletising returns product to the line for re-sort C. Completed full pallets cannot be rejected later except under the >24 h rule C.
Reopen condition	QCM with reason P.

Records, timestamps, links, audit	Inspection ↔ JC ↔ FG pallet(s).
Exceptions & escalation	E-10.
Standard / control principle	ISO 9001 §8.6 release of products; §8.7 nonconforming outputs.

ST-10 FG weighing, labelling & palletising (FG Pallet ID)

AS-IS: A11

Process owner / responsible department	Packing House C; recording role: DEC/TRC NYV.
Records created & creator	FG Pallet with composition (lots/IPs, kg) — PHO/DEC P; FG label printed C.
Permitted actions	PHO/DEC: create, weigh, print. LL: confirm pallet complete. PHM: split/merge approval P.
Required data & authoritative source	FG Pallet ID, JC, product/pack spec, grade (if not for a customer) or customer and order line C, box count, net kg, lot number(s) C, harvest date(s), section(s) C, supplier name C, origin when required C, QC-2 reference.
Execute · Verify · Approve	Execute: PHO/DEC. Verify: system checks label fields (V-09/V-10). Approve: — .
Start condition	QC-2 Pass for the output.
Completion condition	FG Pallet 'QC Approved', label attached, moved to cold room.
Hold condition	Hold inherited from JC or QC.
Reject condition	— (see ST-09).
Reopen condition	Relabel/correction: DEC with reason, old label voided P.
Records, timestamps, links, audit	FG Pallet ↔ JC ↔ IP/lots; label print log; weight and scale ID.
Exceptions & escalation	E-23 printer/scanner failure → pre-printed ID labels and back-entry P.
Standard / control principle	ISO 9001 §8.5.2; ISO 22000 §8.3; CXC 1-1969 Ch.1 §8.1 lot identification.

ST-11 Waste capture, approval & disposal

◆ G7 AS-IS: A12

Process owner / responsible department	Packing House (PHM signs report) C; QC approves waste C; ERP entry by data entry C.
Records created & creator	Waste Record (category: processed / long-storage / packaging & labels / weight loss C) — PHO/DEC P; QC Waste Approval (label) — QCI C; Waste Sign-off — PHM (<1 t) or PHM+FIN+WHM+QA+QC (>1 t, and all long-storage waste) C; Top-management approval for long-storage >1 t C; ERP posting reference — DEC P.
Permitted actions	PHO/DEC: create, weigh after QC approval. QCI: approve/reject waste label. Sign-off roles: approve. TOP: approve long-storage >1 t. DEC: enter ERP reference.
Required data & authoritative source	Category, source (JC, IP, FG pallet, storage), crop, kg, bags, photos (long-storage C), QC approval, sign-offs, disposal time to farm waste area C.
Execute · Verify · Approve	Execute: PHO. Verify: QCI approval before scaling C. Approve: sign-off by threshold C.
Start condition	Waste generated and bagged.
Completion condition	Signed off, ERP reference recorded, removal to farm waste area recorded.
Hold condition	—
Reject condition	QCI refuses the waste label (e.g. saleable product) → product returned to the JC P.
Reopen condition	PHM with reason before ERP posting P.

Records, timestamps, links, audit	Waste ↔ source record; weight; approvals with timestamps; ERP reference.
Exceptions & escalation	E-20 long-storage >1 t with no top-management approval in 24 h → removal recorded as 'removed under 24 h rule' with report and photos C.
Standard / control principle	ISO 9001 §8.7; ISO 22000 §8.9; ISO 27002 §5.3 segregation (maker ≠ approver).

ST-12 FG cold storage (Fridges 1–4), temperature & ageing

AS-IS: A13

Process owner / responsible department	Packing House C; temperature records — not stated NYV.
Records created & creator	Stock Move per FG pallet P; Temperature Reading C; Ageing alerts (system) P.
Permitted actions	PHO/DEC: move. PHM: assign fridge groups. QAO/PHM: record temperatures NYV.
Required data & authoritative source	Location, timestamps, fridge group, documented temperature range per crop C.
Execute · Verify · Approve	Execute: PHO. Verify: system FIFO/group/capacity checks P. Approve: — .
Start condition	FG pallet QC Approved.
Completion condition	Pallet allocated and picked for dispatch, or regraded/wasted.
Hold condition	QA/QC hold → not allocatable P.
Reject condition	—
Reopen condition	—
Records, timestamps, links, audit	Location history; temperature exposure; age since harvest and since QC approval.
Exceptions & escalation	E-08 temperature; E-14 age >48 h without order → DC decision C; >72 h → loading blocked C (OD-10).
Standard / control principle	ISO 9001 §8.5.4; CXC 53-2003 §5.2.2.4.

ST-13 Order allocation & picking

AS-IS: A14 (part)

Process owner / responsible department	Packing House C.
Records created & creator	Load (shipment) plan — PHM/PE P; Allocation of FG pallets to order lines P; Pick confirmation (scan) — PHO P.
Permitted actions	PE/PHM: create load, allocate. PHO: pick. LOG: view schedule.
Required data & authoritative source	Order lines, FIFO-suggested pallets P, truck schedule (LOG C), loading slot.
Execute · Verify · Approve	Execute: PHO. Verify: system matches pallet spec to order line P. Approve: PHM confirms load plan P.
Start condition	Accepted order line due; truck scheduled.
Completion condition	All pallets picked and staged.
Hold condition	Allocated pallet placed on hold → system proposes the next FIFO pallet P.
Reject condition	—
Reopen condition	PE/PHM may re-allocate before loading P.
Records, timestamps, links, audit	Allocation history per pallet.

Exceptions & escalation	Shortage → issue harvested product from Harvest Fridge for grading C (creates a JC).
Standard / control principle	ISO 9001 §8.5.1, §8.5.4 (FIFO).

ST-14 Dispatch QC, loading, sealing & departure

◆ G8 AS-IS: A14

Process owner / responsible department	Packing House (loading) C; QC (dispatch approval) C; QA (records findings, shelf-life sample) C; Logistics (transport) C.
Records created & creator	Dispatch Vehicle Inspection — QCI C; Product temperature check — QC/QA C; Dispatch QC (re-inspection if approval >24 h) — QCI C; QA dispatch findings — QAO C; Logger record (ID, pallet and box position) — role NYV; Seal record — role NYV; Packing List — generated P; Shelf-life sample — QAO C.
Permitted actions	QCI: pass/fail vehicle, release or reject pallets. QAO: record findings, sample. PHO: scan loaded pallets. DEC: finalise packing list, seal number NYV. PHM: approve override of loading window P.
Required data & authoritative source	Vehicle temperature vs crop requirement C, cleanliness, smell, contamination C; product temperature C; packaging (primary/secondary/tertiary) C; photos/videos per shipment C; logger ID and pallet/box position C; seal number C; lockable vehicle C.
Execute · Verify · Approve	Execute: PHO (loading). Verify: QCI checks. Approve: QCI final approve/reject per order C.
Start condition	Vehicle arrived and load picked.
Completion condition	All pallets loaded with valid QC approval; logger and seal recorded; packing list issued; departure time recorded.
Hold condition	Vehicle temperature above requirement → loading on hold until reached C; product temperature not reached → pallet returned to fridge C.
Reject condition	QC rejects a pallet (only if approval >24 h C) → back to fridge, regrade (ST-15), replaced by next FIFO pallet C. Vehicle not lockable → no loading C.
Reopen condition	Packing list correction before departure only P.
Records, timestamps, links, audit	Load ↔ order ↔ FG pallets ↔ lots; vehicle and temperatures; seal; logger position; photos; shelf-life sample.
Exceptions & escalation	E-15, E-16, E-17, E-18, E-19; customer concession for a rejected pallet C (X-03).
Standard / control principle	ISO 9001 §8.6 release; CXC 1-1969 Ch.1 §9 Transportation; ISO 22000 §8.3 (first stage of distribution).

ST-15 Regrade / rework (storage or dispatch rejection)

◆ G4–G6 AS-IS: A9/A14

Process owner / responsible department	Packing House C; QC re-inspects C.
Records created & creator	Rework Job Card (type 'Regrade') — PE/PHM P; new FG pallets; long-storage waste C.
Permitted actions	As ST-08.
Required data & authoritative source	Source FG pallet(s) or IPs; reason (long storage / dispatch reject); original lots and harvest dates preserved P.
Execute · Verify · Approve	As ST-08/ST-09.
Start condition	Hold disposition = Regrade.
Completion condition	Rework JC closed; source pallets consumed.
Hold condition	As ST-08.
Reject condition	As ST-09.
Reopen condition	As ST-08.

Records, timestamps, links, audit	Rework genealogy: old pallet → rework JC → new pallets; waste coded 'long storage' C.
Exceptions & escalation	Regrade beyond 72 h after harvest not permitted for loading until OD-10 is decided NYV.
Standard / control principle	ISO 22000 §8.3(b) reworking; ISO 9001 §8.7.

ST-16 Customer complaint, trace & recall

AS-IS: A15

Process owner / responsible department	QA and QC investigate, confirmed with PHM C; intake via Sales C.
Records created & creator	Complaint — SAL C; Validity check — QAO P; Investigation — QAO/QCI C; Trace report (system) P; Recall decision NYV (authority not stated); Response — SAL C; CAPA P.
Permitted actions	SAL: create. QAO/QCI: assess validity, investigate. PHM: confirm. QAM: open CAPA. Recall authority: OD-16/OD-22 NYV.
Required data & authoritative source	Nature, % out of spec, photos/videos, delivery date, box codes, lot number C; lot, supply date, section, quantity supplied, sample quantity C.
Execute · Verify · Approve	Execute: QAO/QCI. Verify: PHM confirms C. Approve: response agreed with customer (credits) C; approver not stated NYV.
Start condition	Complaint received the same day as delivery C.
Completion condition	Investigated and resolved within 24 h of the complaint C; response sent; CAPA linked if valid C.
Hold condition	Related stock in the packhouse placed on hold P.
Reject condition	Invalid: incomplete information or received more than 24 h after delivery C.
Reopen condition	QAM with reason P.
Records, timestamps, links, audit	Complaint ↔ load ↔ FG pallets ↔ lots ↔ IPs ↔ RMA ↔ sections; shelf-life sample.
Exceptions & escalation	E-21 recall within 6 h for a health-risk incident C; farm-related complaints (residues/MRLs) communicated to the farm in writing C.
Standard / control principle	ISO 9001 §10.2; ISO 22000 §8.9.5 withdrawal/recall; CXC 1-1969 Ch.1 §7.5 recall procedures.

ST-17 Daily stock report & reconciliation

AS-IS: A16

Process owner / responsible department	Packing House C; preparer not stated NYV.
Records created & creator	Daily Stock Report (generated) P; Weekly weight-loss adjustment — role NYV.
Permitted actions	System generates; PHM reviews and issues P; FIN views.
Required data & authoritative source	Per crop: daily harvest, stock by customer and grade, waste, season-to-date harvest and waste; crop colour; own vs supplier crop C; weight-loss % per crop C (value owner OD-28).
Execute · Verify · Approve	Execute: system. Verify: PHM. Approve: — .
Start condition	End of day (cut-off time NYV).
Completion condition	Report issued to distribution list C.
Hold condition	—
Reject condition	—
Reopen condition	Re-issue with version number P.

Records, timestamps, links, audit	Report version; underlying transactions; reconciliation variance.
Exceptions & escalation	Unreconciled variance above tolerance → investigation task to PHM P.
Standard / control principle	ISO 9001 §9.1.3 analysis and evaluation.

5.7 Cross-cutting controls

ID	Control	Specification	Standard / principle
X-01	Hold & nonconformity	Any QC/QA user may place a Hold on an IP, JC output, FG pallet or load. A held item cannot move to the next state. Release or disposition only by QC/QA (QCM/QAM for food-safety holds) P. Disposition options: release, regrade, rework, waste, return to farm/supplier, customer concession C (SOP-DS, SOP-RC).	ISO 9001 §8.7; ISO 22000 §8.9.4
X-02	Corrective action (CAPA)	Opened from a valid complaint C (SOP-CP: immediate corrective action), a repeated nonconformity, or an audit finding P. Root cause, action, owner, due date, effectiveness check P.	ISO 9001 §10.2; ISO 22000 §10.1
X-03	Customer deviation / concession	Temporary deviation request sent to the customer in writing before producing out of specification C (SOP-PL); customer agreement to load a rejected pallet C (SOP-GR). The record stores the written approval as an attachment P.	ISO 9001 §8.7.1 (concession)
X-04	Shelf-life samples	One sample per product, per customer, per load, kept by QA for 10–14 days; removed early if mould appears, recorded and communicated C.	ISO 22000 §8.9.5 (supports investigation)
X-05	Customer-owned crates & LSS boxes	Customer crates inspected by QC; wash or return C. LSS box inventory by the PH data entry clerk C.	ISO 9001 §8.5.3 property of customers
X-06	Emergency override	Time-boxed bypass of a blocking rule with justification, notification to the gate owner and mandatory retrospective review P. Authority and excluded gates: OD-16.	ISO 27002 §5.3, §8.2, §8.15
X-07	Temperature records	Fridge, pre-cooling, vehicle and product temperatures recorded with time, location and user; out-of-range readings raise an alert to PHM and QA P against documented ranges C; limits and tolerances OD-08.	CXC 53-2003 §5.2.2.4; CXC 1-1969 Ch.1 §7.2

6 Roles and permissions

Authority stays with the department that holds it today. QC and QA decisions are recorded only by QC and QA users; the PHM cannot approve quality gates; instructions to PH staff pass through the PHM or Production Engineer. Where the sources are silent, the assignment is labelled and an open decision names who must decide.

6.1 Role catalogue

Code	Role	Department	Reporting line	Label	Source	System use
PHM	Packing House Manager	Packing House	Not stated in sources	C	SUM-3; SOP	Full PH operational authority; order acceptance; waste sign-off; instruction channel
PE	Production Engineer	Packing House	Not stated (assumed PHM)	C	SOP-ST; SUM-3	Instruction channel; production planning
LL	Line Supervisor / Line Leader	Packing House	Not stated (assumed PHM/PE)	C	SOP-PL	Line execution, packaging hand-out, line-start check with QC
PHO	Packing House Operator (weigher, palletizer, forklift)	Packing House	Line Leader	NYV	SOP-PL (staff)	Scan/record only, if given access (OD-24)
TRC	Traceability team member	Not stated	Not stated	NYV	SOP-PC	Production traceability recording (OD-13)
DEC	Data Entry Clerk (Receiving / PH)	Warehouse Management	Warehouse Management	C	SUM-3; SOP-ST	Receiving registration, ERP transactions, LSS box inventory
WHM	Warehouse Manager	Warehouse	Not stated	C	SUM-3; SOP-TR; SOP-WS	Packaging stock, ERP postings, waste sign-off (>1 t), arrival time approval
STK	Store Controller / Store Keeper	Warehouse (assumed)	Not stated	NYV	SOP-PL	Packaging issue (department assumed — OD-14)
QCI	QC Inspector / Quality Controller	Quality Control	QC management (outside PH)	C	SUM-3; SOP-GR	Arrival check, QC-1, line checks, QC-2, dispatch QC, waste label, crate inspection
QCM	QC Manager	Quality Control	Not stated	C	SUM-3	QC disposition authority, QC master data approval
QAO	QA Officer	Quality Assurance	QA management (outside PH)	C	SUM-3; SOP-RC	Raw-material acceptance, dispatch findings, shelf-life samples
QAM	QA Manager	Quality Assurance	Not stated	C	SUM-3	Holds, NC/CAPA, complaints, tolerances, temperature parameters
ENG	Engineer (regrades rejected incoming product)	Not stated (farm or PH)	Not stated	NYV	SOP-RC	Regrade of rejected incoming product (OD-12)
FRM	Farm Harvest Supervisor	Farm Operations	Not stated	C	SOP-RC (tag)	Tag data; dispatch note if CP-02 approved
SAL	Sales	Sales	Not stated	C	SOP-FC; SOP-CP	Orders, customer specs, complaint intake
LOG	Logistics	Logistics	Not stated	C	SOP-FC; SOP-TR	Truck schedules, transit times
FIN	Finance	Finance	Not stated	C	SOP-WS	Waste sign-off (>1 t), packaging waste sign-off
TOP	Top Management	Executive	—	C	SOP-WS; SOP-TR	Long-storage waste >1 t approval; vehicle non-compliance approval

Code	Role	Department	Reporting line	Label	Source	System use
ADM	System Administrator	IT	Not stated	P	Proposed	User/role administration only; no operational transactions
AUD	Auditor (read-only)	Internal audit / certification	—	P	Proposed	Read-only access to records and audit trail

6.2 Responsibility matrix — who creates, executes, verifies, approves and closes

Record / task	Stage	Creates	Executes	Verifies	Approves	Closes
Customer order line	ST-00	SAL C	—	PE P	PHM accept/reject C	System on dispatch P
Daily production plan	ST-00	PE P	LL P	—	PHM P	PHM P
Arrival (load) record	ST-01	DEC P	DEC P	—	—	System when all pallets registered P
Incoming vehicle inspection	ST-01	Inspector — role not stated NYV	Inspector NYV	—	Exception only: PHM + QA + Top mgmt C	System P
Incoming Pallet registration (weight, ID, label)	ST-02	DEC C	DEC C	System plausibility check P	—	System on full consumption P
Arrival visual check	ST-03	QCI / QAO C	QCI / QAO C	—	Inspector's decision C	Inspector C
Pre-cooling log	ST-04	PHO / DEC P	PHO P	System dwell check P	—	System P
Raw-material acceptance report	ST-05	QAO C	QAO C	QC/QA co-signature C	QAO C	QAO C
Regrade of rejected incoming product	ST-05	System on QA rejection P	ENG C; DEC re-weighs C	QCI / QAO recheck C	QAO C	QAO C
Stock move (fridge location)	ST-06/12	PHO / DEC P	PHO P	System group/capacity/FIFO check P	PHM override of warnings P	—
Fridge temperature reading	X-07	Not stated NYV	Not stated NYV	QAO P	—	—
QC-1 release	ST-07	QCI C	QCI C	—	QCI C	QCI C
Job Card (production order)	ST-08	PE / PHM P	LL C	QCI + LL line-start check C	PHM P	PHM after mass balance P
Packaging issue	ST-08	Requester not stated NYV	STK C	LL hands out per plan C	—	WHM removes from ERP stock on dispatch C
QC-2 online / final inspection	ST-09	QCI C	QCI C	—	QCI C	QCI C
FG Pallet (weight, label, composition)	ST-10	PHO / DEC / TRC NYV	PHO P	LL confirms complete P	QC-2 pass = QC approval P	System on dispatch P
FG pallet split / merge	ST-10	PHO P	PHO P	System compatibility rules P	PHM P	System P
Waste record	ST-11	PHO / DEC P	PHO P	QCI approval label before scaling C	PHM (<1 t) C; PHM+FIN+WHM+QA+QC (>1 t or long-storage) C; TOP long-storage >1 t C	PHM after ERP reference P
Hold / nonconformity	X-01	QCI / QAO / PHM P	Per disposition P	QCI / QAO P	QCM / QAM P	QAM P
Load plan & allocation	ST-13	PE / PHM P	PHO P	System spec match P	PHM P	System on departure P

Record / task	Stage	Creates	Executes	Verifies	Approves	Closes
Dispatch vehicle inspection	ST-14	QCI C	QCI C	—	QCI C	QCI C
Dispatch QC release / reject	ST-14	QCI C	QCI C	QAO records findings C	QCI C; rejected pallet decided with PHM C	QCI C
Logger and seal record	ST-14	Not stated NYV	Not stated NYV	QCI checks logger C	—	On departure P
Packing list	ST-14	System from scans P	DEC NYV	LOG NYV	—	On departure P
Shelf-life sample	X-04	QAO C	QAO C	—	—	QAO after 10–14 days C
Customer complaint	ST-16	SAL C	QAO / QCI C	PHM confirms C	Response approver not stated NYV	QAM P
Recall initiation	ST-16	Not stated NYV	Not stated NYV	—	Not stated NYV	Not stated NYV
CAPA	X-02	QAM / QCM P	Assigned owner P	QAM effectiveness check P	QAM P	QAM P
Customer deviation request	X-03	Not stated NYV	—	—	Customer (written) C	Requester P
Daily stock report	ST-17	System P	—	PHM P	—	PHM issues P
Quality master data (tolerances, temperatures, specs)	M01	QAO / QCI P	—	—	QAM / QCM P	—
Operational master data (crops, lines, locations)	M01	ADM P	—	—	PHM P	—
User and role assignment	M02	ADM P	ADM P	—	Department head of the user P	—
Emergency override	X-06	PHM / QCM / QAM P	Invoker P	Gate owner retrospective review P	Second authoriser where available P	Gate owner P

6.3 Action-level permission matrix

● Confirmed authority — the source assigns this action or decision to the role (system recording of it is part of the TO-BE). ○ Proposed — needs approval by the owner of the record type. **?** Not yet verified — source silent or ambiguous; see the open decision in the note. Blank = no permission. View rights are proposed for all listed roles.

Record / object	Action	PHM	PE	LL	PHO	TRC	DEC	WHM	STK	QCI	QCM	QAO	QAM	ENG	SAL	LOG	FIN	TOP	ADM	AUD	Note
Customer order	View	○	○	○			○	○		○	○	○	○		○	○				○	
	Create / Edit / Submit														●						Sales sends orders (SOP-FC). PH planner keying on Sales' behalf: OD-37.
	Approve (accept) / Reject	●																			PH management accepts or rejects orders (SOP-FC).
	Cancel / Reopen (amend)	○													○						Amendment returns line to Pending acceptance.
Daily plan / Job Card	View	○	○	○			○	○	○	○	○	○	○							○	
	Create / Edit / Submit	○	○																		OD-15.
	Approve	○																			
	Execute (start / pause / complete)			○																	
	Record inputs consumed			○		?	?														Traceability team identity: OD-13.
	Cancel / Reopen / Close	○																			Reopen window: OD-15.
Arrival record	Create / Edit / Submit						○														
	Record incoming vehicle inspection						?					?									SOP-TR names no inspector.
	Approve vehicle exception (all three required)	●											●					●			SOP-TR: PHM, QA team and top management. QA signing level: OD-38.
Incoming Pallet	View	○	○	○			○	○		○	○	○	○	○						○	
	Create / Submit (register weight, ID)						●														SUM-3.
	Edit weight before QA decision						○														Reason mandatory.
	Correct after QA decision (maker / checker)						○					○									DEC creates the correction; QAO verifies.
	Print / reprint label						○														Reprints logged.
	Cancel (void registered in error)						○	○													DEC requests; WHM approves.
Arrival visual check	Create / Submit (Pass / Reject)									●		●									SOP-RC: QC/QA team.
	Reopen										○		○								
Pre-cooling log	Record entry / exit				○		○														
	Correct times	○																			
Raw-material acceptance	Create / Submit											●									SOP-RC.

Record / object	Action	PHM	PE	LL	PHO	TRC	DEC	WHM	STK	QCI	QCM	QAO	QAM	ENG	SAL	LOG	FIN	TOP	ADM	AUD	Note
	Verify (co-sign)									●			○								SOP-GR: report signed by QA/QC team.
	Approve (accept) / Reject											●									
	Recheck after regrade									●		●									
	Reopen												○								
Incoming regrade	Execute / mark complete													●							Engineer's department: OD-12.
	Re-weigh						●														SOP-RC: returned for re-scaling.
Stock move	Execute (scan from / to)				○		○		?												SOP-PL: store keeper issues product FIFO — custody: OD-14.
	Assign fridge storage group	○																			Allocations may change with need (SOP-PC).
	Override placement warning	○																			Reason mandatory; logged.
Temperature reading	Create				?							?									OD-32.
	Acknowledge excursion	○											○								
QC-1 release	Release / Hold									●											SUM-3.
	Reopen										○										
Packaging issue	Request		?	?																	SOP-PL does not name the requester.
	Issue								●												SOP-PL.
	Remove from ERP stock							●													SOP-PL: Warehouse Dept on dispatch.
Line start check	Create / Submit (joint)			●						●											SOP-PL: Quality controller and line leader.
QC-2 inspection	Create / Submit (Pass / Fail)									●											SOP-GR.
	Reopen										○										
FG Pallet	View	○	○	○			○	○		○	○	○	○		○	○				○	
	Create / weigh / print label				○	?	?														Recording role: OD-13.
	Confirm complete			○																	
	Split / merge — execute				○																Compatibility rules V-22.
	Split / merge — approve	○																			
	Relabel / correct						○														Old label voided; reason mandatory.
Hold / NC	Raise (Hold)	○								○		○									PHM may raise operational holds.
	Decide disposition — quality										○		○								
	Decide disposition — food safety												○								Which holds are food-safety: OD-23.
	Release										○		○								PHM cannot release a quality hold (SoD-05).
	Close												○								
Waste record	Create / weigh				?		○														Scaling role not stated.

Record / object	Action	PHM	PE	LL	PHO	TRC	DEC	WHM	STK	QCI	QCM	QAO	QAM	ENG	SAL	LOG	FIN	TOP	ADM	AUD	Note
	Verify — QC approval label									●											SOP-WS: before scaling and loading.
	Approve — sign-off < 1 t	●																			Threshold basis: OD-11.
	Approve — sign-off > 1 t or long-storage (all required)	●						●			?		?				●				SOP: 'QA and QC team' — signing level: OD-11.
	Approve — long-storage > 1 t																	●			If no approval in 24 h: removal recorded under the 24 h rule (E-20).
	Record ERP posting reference						●														SOP-ST: data entry clerks handle ERP transactions.
	Reopen (before ERP posting)	○																			
Load	Create / allocate pallets	○	○																		
	Pick / scan loaded pallets				○																
	Vehicle inspection Pass / Fail									●											SOP-GR dispatched product grading report.
	Product temperature check									●		●									SOP-DS: QC / QA.
	Release / reject pallet (dispatch QC)									●											Reject only if QC approval > 24 h (SOP-DS).
	Decide rejected pallet: resort or replace	●								●											SOP-GR: discussed with QC and PHM.
	Record QA dispatch findings											●									SOP-DS.
	Record logger ID & position / seal number				?		?														OD-18.
	Finalise packing list / record departure						?														
	Approve loading-window override	○																			OD-39.
Shelf-life sample	Create / dispose											●									SOP-GR.
Complaint	Create														●						SOP-CP.
	Validity decision									○		○									
	Investigate									●		●									SOP-CP.
	Confirm investigation	●																			SOP-CP.
	Initiate recall												?					?			Authority not stated: OD-22.
	Close												○								
CAPA	Open / approve closure										○		○								
Customer deviation	Create request / record approval														?						SOP-PL names no requester.
Daily stock report	Issue	○																			
	View	○	○	○			○	○		○	○	○	○		○		○	○		○	
Master data — quality parameters	Edit (maker)									○		○									Tolerances, temperatures, specs.

Record / object	Action	PHM	PE	LL	PHO	TRC	DEC	WHM	STK	QCI	QCM	QAO	QAM	ENG	SAL	LOG	FIN	TOP	ADM	AUD	Note
	Approve (checker)										○		○								Effective-dated versions.
Master data — operational	Edit (maker)																		○		Crops, lines, routes, locations, capacities.
	Approve (checker)	○																			
Master data — customers & specs	Edit / approve										○				○						SAL maintains; QCM approves spec content.
Users & roles	Administer																		○		On approved request from the department head.
	Approve access request / periodic review	○						○			○		○								Each for own staff. OD-41.
Emergency override	Invoke	○									○		○								Scope by gate owner: OD-16.
	Retrospective review	○									○		○								Gate owner reviews; invoker cannot review own override.
Audit trail	View (read-only)												○						○	○	No role can edit or delete audit entries.

6.4 Segregation of duties

ID	Rule	Constraint	Label	Basis
SoD-01	The person who registers a pallet's weight cannot record the QA acceptance decision for that pallet.	DEC ≠ QAO	C	Separate departments today (SUM-3); the system enforces it.
SoD-02	Regrade of rejected incoming product and the recheck must be by different people.	ENG ≠ QCI/QAO	C	SOP-RC.
SoD-03	Waste maker cannot approve the QC waste label or sign off the same waste record.	PHO/DEC ≠ QCI ≠ sign-off	P	ISO 27002 §5.3.
SoD-04	Dispatch release is performed only by QC, never by the Packing House role that packed or loaded the pallet.	QCI ≠ PHO/LL	C	QC is outside the PH reporting line (SUM-3).
SoD-05	A quality hold can be released only by QC/QA (QCM/QAM). PH roles may raise but not release.	QCM/QAM only	P	ISO 9001 §8.7; ISO 22000 §8.9.4.
SoD-06	System administrators cannot perform operational transactions or approvals and cannot alter audit records. Admin actions are logged and reviewed.	ADM ∩ ops = ∅	P	ISO 27002 §8.2, §8.15.
SoD-07	Quality parameters (tolerances, temperatures, specifications) follow maker-checker: the editor cannot approve the same version.	Maker ≠ checker	P	ISO 27002 §5.3, §8.32.
SoD-08	An emergency override cannot be reviewed by the person who invoked it.	Invoker ≠ reviewer	P	ISO 27002 §5.3.
SoD-09	After a record is verified or approved, its author cannot edit it. Changes go through a correction record with reason and approval.	No self-edit after approval	P	ALCOA+ data-integrity practice; ISO 9001 §7.5.3.
SoD-10	Where one person legitimately holds two roles (small team, night shift), the system records which role was used and flags any SoD conflict for review rather than silently allowing it.	Conflict flag	P	Practical exception handling; needs PHM/QAM approval (OD-41).

6.5 Emergency override (break-glass)

- Trigger: a blocking rule (gate, validation or time window) prevents an operationally urgent action, or the system is unavailable. P
- Who may invoke: the gate owner's delegated authority — PHM for Packing House rules; QCM for QC gates; QAM for QA gates and food-safety holds. Final list: OD-16. P
- What can never be overridden (proposal for decision): release of a pallet under a food-safety hold; loading beyond 72 h after harvest (SOP 'strictly not allowed' C); loading into a vehicle that cannot be locked C; QC approval label on a pallet that has not passed QC-2. P
- Mandatory data: rule overridden, records affected, reason category and free-text justification, expected duration, invoker re-authentication. P
- System actions: create an Override record linked to every affected record; mark affected records 'Overridden'; notify the gate owner immediately; add to the daily exception report. P
- Time box: the override expires automatically at the end of the stated duration (maximum to be set — OD-16). P
- Retrospective review: the gate owner reviews within a window to be set (OD-16), records Accept / Corrective action required, and may open a CAPA. The invoker cannot review their own override (SoD-08). P
- Monitoring: override count per gate per week is a KPI (K11); repeated overrides on one rule trigger a rule review. P
- Downtime: when the system is down, pre-numbered paper forms are used; back-entry within a set window with the original event time and a 'late entry' flag (E-22). P

6.6 Access governance

- Unique named accounts for every user; no shared logins, including on shared floor tablets (fast badge or PIN switching). P — ISO 27002 §5.16, §8.5
- Roles granted on a request approved by the user's own department head (QC staff by QC management, QA staff by QA management, Data Entry by Warehouse Management). P — §5.18

- Least privilege: default role grants view of own-stage records only; cross-department views are added explicitly. P — §8.3
- Joiner / mover / leaver: access changed on role change and removed on exit (timing to be set — OD-41). P
- Periodic access review by each department head (frequency to be set — OD-41); the review result is recorded in the system. P — §5.18
- Approval actions require re-authentication (PIN) to give an electronic signature its meaning (approved / verified, by whom, when). P
- All server timestamps come from a synchronised clock; device clocks are not trusted for event time. P — §8.17
- Audit log is append-only and retained for the record-retention period (OD-21). P — §8.15, §5.33

7 Job card and traceability model

Traceability is built from three things: a unique ID on every physical unit, an event record each time a unit changes state or location, and a link record every time units are combined, split or transformed. The 'Job Card' is the production order that links incoming pallets to finished pallets.

7.1 Identifiers

Code	Identifier	Identifies	Issued when / by	Proposed format	Label	Notes
ARR	Arrival ID	One vehicle load arriving at the gate	Gate arrival — DEC (system-generated)	ARR-YYMMDD-NNNN	P	Links all incoming pallets of one load and the dispatch note (CP-02).
DN	Dispatch note number	Farm or supplier dispatch document for a load	Issued by the farm/supplier; captured at the gate	As issued by the farm/supplier	P	Not in use today (EV-01). OD-01.
IP	Incoming Pallet ID	One field pallet as received	Weighing — DEC	Internal unique key + printed number per crop / section / block / day (SOP-RC), e.g. CT-261019-B2S05-007	C P	The concept and the per-crop/section/block/day numbering are confirmed (SOP-RC); the format is proposed. The printed number must be unique system-wide, hence the date in it.
LOT	Lot number	Raw-material lot (proposal: crop × source block/section or supplier × harvest date)	Assignment point per OD-02	L-<crop>-<YYMMDD>-<source>	NYV	Three conflicting assignment points (EV-05). This is the key printed on FG labels and asked for in complaints (SOP-CP).
JC	Job Card / Production Order	One line run for one product/grade target or order line	Planning — PE/PHM	JC-YYMMDD-<line>-NN	P	New record (CP-04).
FGP	Finished-Goods Pallet ID	One finished pallet	FG palletising — PHO/DEC	FGP-YYMMDD-NNNNN, or GS1 SSCC (18 digits) if adopted	C P	A Pallet ID per finished pallet is confirmed (SUM-3); the format is proposed; SSCC is optional (CP-14, OD-20).
BOX	Box / case code	One box or punnet case	Not defined	Not defined	NYV	Complaints must quote 'box codes' (SOP-CP), but no source defines them (OD-19).
QCA	QC approval reference	QC-2 or dispatch QC approval of an FG pallet	QC-2 / dispatch QC — QCI	Approval timestamp + inspector + record ID	P	Electronic version of the stamped QC label; validity 24 h (EV-13).
WR	Waste Record ID	One bag batch / weighing of waste	Waste capture	WR-YYMMDD-NNNN	P	Category: processed, long-storage, packaging, weight loss (SOP-WS).
HLD	Hold / NC ID	One nonconformity affecting one or more items	Hold raised	NC-YYMMDD-NNN	P	
LD	Load ID	One outbound vehicle load	Load planning	LD-YYMMDD-NNN	P	Packing list number = Load ID (proposal).
SEAL	Seal number	Truck seal	Pre-printed on the seal	As printed	C	Recorded on the packing list (SOP-DS).
LOGR	Temperature logger ID	Logger placed in the load	Logger serial	As printed + pallet ID + box position	C	Pallet and box position recorded (SOP-TR).
SLS	Shelf-life sample ID	Sample per product / customer / load	QAO at dispatch	SLS-<LD>-NN	P	Kept 10–14 days (SOP-GR).
CMP	Complaint ID	One customer complaint	Sales intake	CMP-YYMMDD-NN	P	

Why the lot definition matters

If the lot is defined per source and harvest date (OD-02 option A or C), a complaint quoting a lot number resolves directly to a block/section and harvest day, which is what residue and MRL investigations need (SOP-CP). If the lot is a production batch only (option B), every trace must go through Job Card composition, and mixed-lot pallets widen every recall.

7.2 Genealogy rules — splitting, merging, rework, waste

ID	Rule	Label
GEN-01	One Arrival contains 1..n Incoming Pallets. Every IP belongs to exactly one Arrival.	P
GEN-02	Every IP carries one source (block + section, or supplier) and one harvest date/time from the harvesting tag.	C
GEN-03	An IP belongs to exactly one raw-material lot. A lot contains 1..n IPs. (Lot definition per OD-02.)	NYV
GEN-04	A Job Card consumes 1..n IPs. Consumption is recorded per IP in kg (or boxes); an IP may be split across several JCs. Any unconsumed remainder returns to storage under the same IP ID with reduced quantity (partial issue).	P
GEN-05	A Job Card produces 0..n FG Pallets, 0..n Waste Records and an optional residue return. Each FG Pallet stores its composition: the list of lots/IPs and kg contributed.	P
GEN-06	Lot attribution to FG Pallets is proportional to consumption within the JC time window unless the line allows exact segregation; the method is recorded on the JC (exact / time-window / proportional) so the trace report can show its confidence.	P
GEN-07	Split: an FG Pallet split into 2..n children closes the parent; each child inherits the parent's composition pro rata, the parent's QC approval time, and the earliest harvest date.	P
GEN-08	Merge: allowed only for pallets with the same crop, product/pack spec, grade, customer (or none) and storage group. The merged pallet takes the union of compositions, the earliest QC approval time and the earliest harvest date (most conservative), and both parents close.	P
GEN-09	Rework / regrade: the source pallet(s) are consumed by a Rework JC; new FG Pallets inherit the source lots and harvest dates. Waste from regrade after long storage is coded 'long-storage waste' (SOP-WS).	C P
GEN-10	A Load contains 1..n FG Pallets; each FG Pallet appears in at most one departed Load. Each loaded pallet is linked to an order line.	P
GEN-11	Holds, QC/QA records, temperature exposures and attachments link to the specific IDs they concern, so a trace report returns them with the genealogy.	P
GEN-12	No ID is reused or deleted. A voided record keeps its ID with status 'Void' and a reason.	P

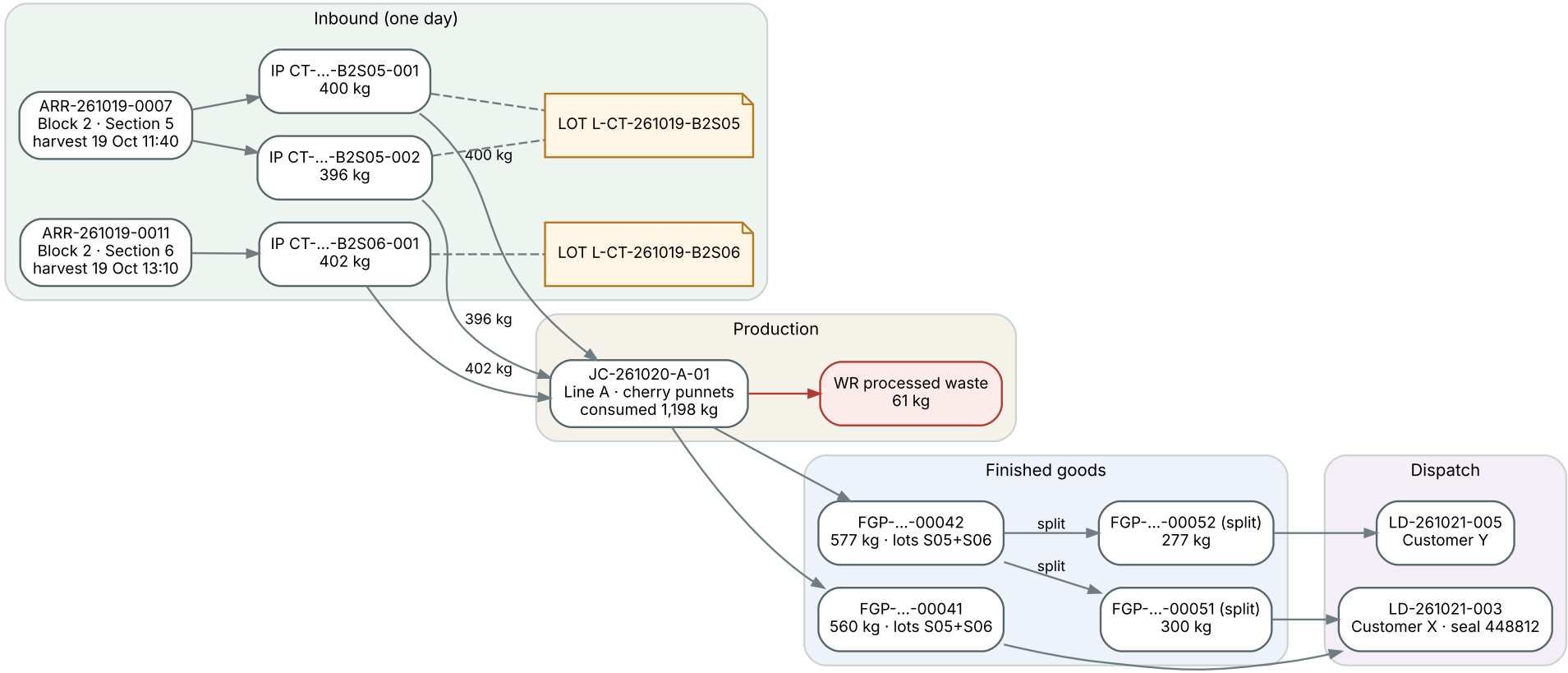


Figure 7.1 — Illustrative genealogy (invented IDs and weights): two arrivals, three incoming pallets, two lots, one Job Card, waste, a split, two loads. Mass balance: 400 + 396 + 402 = 1,198 kg in; 560 + 577 + 61 = 1,198 kg out.

7.3 Labels by stage

Stage	Label	Fields	Label status	Source
Receiving (incoming pallet)	Receiving label	Section and block number; date and time; weight; assigned pallet number. Proposed additions: crop, variety, harvest date (from tag), barcode of the IP ID, status (Registered / Accepted).	C P	SOP-RC; SOP-PL
In process	Production / lot label	Grade, harvest date and lot number (SOP flow diagram step 6). From receiving to dispatch every pallet shows grade and lot number (SOP-PC note).	C	SOP-PF; SOP-PC
Finished goods	FG label	Weight; lot number; supplier name; product description (and grade if not packed for a specific customer); customer name and address if packed for a customer; customer-specific requirements; product origin for certain orders. Traceability label adds crop, date, grade, customer and section number. Proposed: FG Pallet ID barcode; box count; harvest date.	C P	SOP-PL; SOP-PC
Finished goods	QC approval label	QC signature and stamp, not older than 24 h at dispatch. Proposed electronic equivalent: approval date/time, inspector ID, validity-until time printed or shown on scan.	C P	SOP-GR; SOP-DS
Waste	QC waste approval label	Attached before scaling and loading. Content not specified.	C (content N)	SOP-WS
Dispatch	Packing list	Corresponds to the sales order; truck seal number; logger pallet/box position. Proposed: generated from scanned FG Pallet IDs with lots and weights.	C P	SOP-DS; SOP-TR

7.4 Trace queries

ID	Query	Returns	Label
TQ-01	Backward from FG Pallet / Load / lot / box code	Load → order → FG pallets → compositions → lots → IPs → arrivals → source block/section or supplier, harvest date, tag supervisor → RMA, arrival check, QC-1/QC-2, dispatch QC records → temperature exposure per location → holds and overrides.	P
TQ-02	Forward from source (block/section or supplier) and harvest date, or from a lot or IP	Lots → IPs → JCs → FG pallets (incl. split/merge/rework descendants) → loads → customers, with quantity per customer and current location of any undispatched stock.	P
TQ-03	Recall scope	Union of forward results for the selected lots, with in-house stock auto-placed on hold on confirmation (QAM) and a customer contact list from orders.	P
TQ-04	Mass balance	Per JC, per lot and per day: inputs vs outputs + waste + weight loss + residue (formulas below).	P

7.5 Mass balance

ID	Balance	Formula	Notes
MB-01	Job Card balance	Input kg (Σ consumed IP kg) = Σ FG pallet net kg + Σ waste kg + residue returned kg + unexplained variance	Variance % = unexplained variance ÷ input kg × 100. Tolerance per crop: OD-28.
MB-02	Lot balance	Accepted lot kg = Σ consumed by JCs + in-stock kg + raw waste kg + weight-loss adjustment	Closes when in-stock = 0.
MB-03	Daily receiving balance (requires CP-01/CP-02)	Dispatched by farm (DN) = received registered kg + transit difference; Registered kg = accepted kg + rejected-to-waste kg + returned-to-farm kg	Without CP-01, rejected kg is not recorded and this balance cannot close (EV-03).
MB-04	Weight loss	Weekly adjustment per crop = harvest kg × agreed weight-loss % (SOP-WS), recorded separately from processed waste	% values and approver: OD-28.

8 Workflow rules

Each record type has an explicit state model. Transitions show the trigger and actor; line style shows the label: solid = Confirmed, dashed blue = Proposed, dotted amber = Not Yet Verified. Terminal states have a double border; red = rejection/waste/hold, amber = waiting or timed states.

8.1 State models

SM-IP — Incoming Pallet

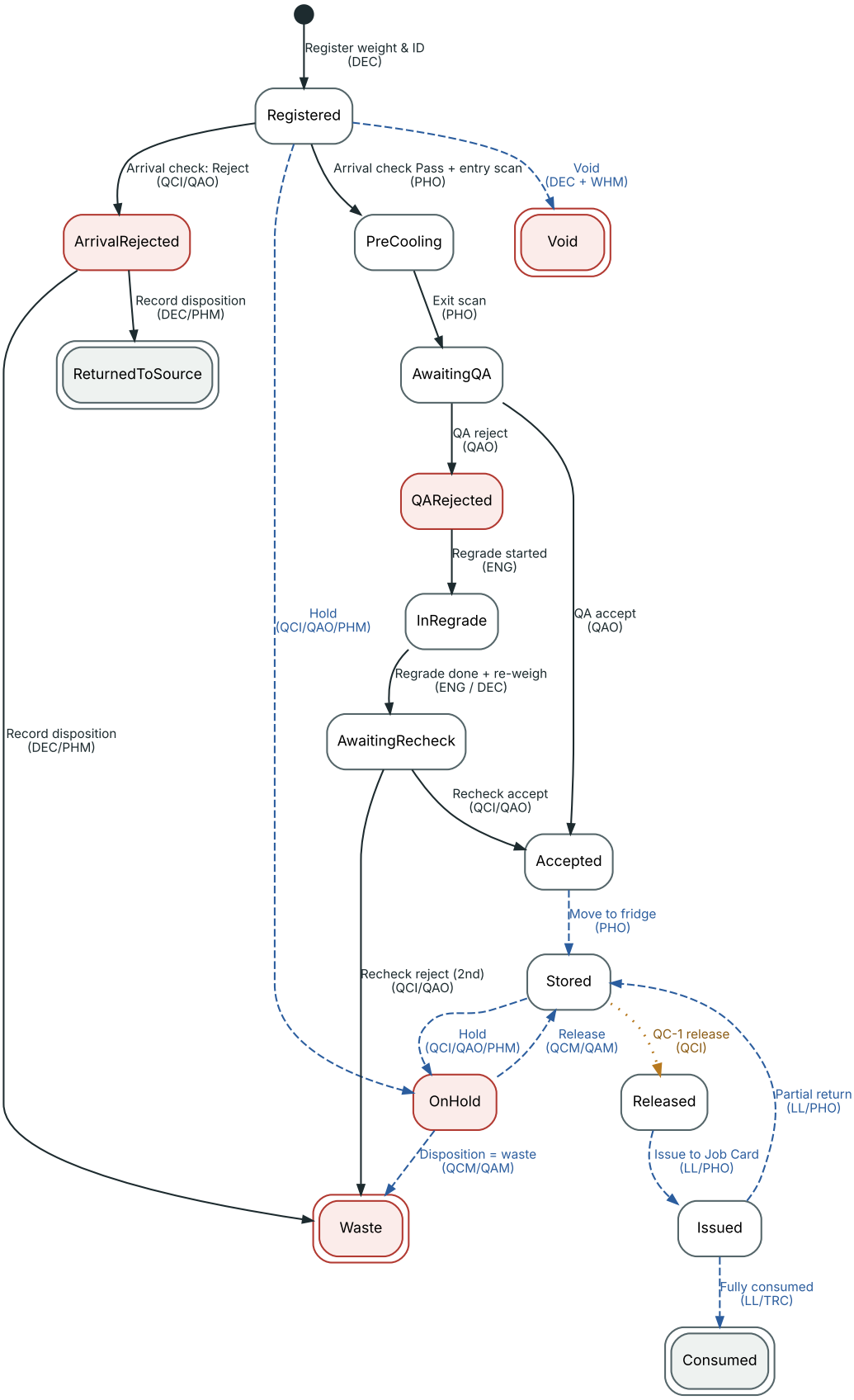


Figure 8.1 — Incoming Pallet state model (SM-IP; stages ST-02 → ST-08).

From	To	Trigger	Actor	Guard / rule	Label
[*]	Registered	Register weight & ID	DEC	Arrival exists; tag fields complete (V-01)	C P
Registered	ArrivalRejected	Arrival check: Reject	QCI/QAO	Excessive visible waste	C
ArrivalRejected	ReturnedToSource	Record disposition	DEC/PHM	Destination = farm/supplier	C
ArrivalRejected	Waste	Record disposition	DEC/PHM	Destination = waste area	C
Registered	PreCooling	Arrival check Pass + entry scan	PHO	G2 Pass	C
PreCooling	AwaitingQA	Exit scan	PHO	Dwell recorded (V-04)	C
AwaitingQA	Accepted	QA accept	QAO	RMA submitted with photos (V-23)	C
AwaitingQA	QARejected	QA reject	QAO	Reason mandatory	C
QARejected	InRegrade	Regrade started	ENG	Engineer notified automatically	C
InRegrade	AwaitingRecheck	Regrade done + re-weigh	ENG / DEC	New weight recorded	C
AwaitingRecheck	Accepted	Recheck accept	QCI/QAO	Recheck by a different person than ENG (SoD-02)	C
AwaitingRecheck	Waste	Recheck reject (2nd)	QCI/QAO	No third check (V-26)	C
Accepted	Stored	Move to fridge	PHO	Group and capacity check (V-07, V-08)	P
Stored	Released	QC-1 release	QCI	G4	NYV
Released	Issued	Issue to Job Card	LL/PHO	JC is Ready or InProgress (V-05)	P
Issued	Stored	Partial return	LL/PHO	Remaining kg > 0 recorded	P
Issued	Consumed	Fully consumed	LL/TRC	Remaining kg = 0	P
Registered	OnHold	Hold	QCI/QAO/PHM	Hold record created	P
Stored	OnHold	Hold	QCI/QAO/PHM	Hold record created	P
OnHold	Stored	Release	QCM/QAM	Disposition = release (SoD-05)	P
OnHold	Waste	Disposition = waste	QCM/QAM	Waste record created	P
Registered	Void	Void	DEC + WHM	Before any check; reason	P

SM-JC — Job Card (production order)

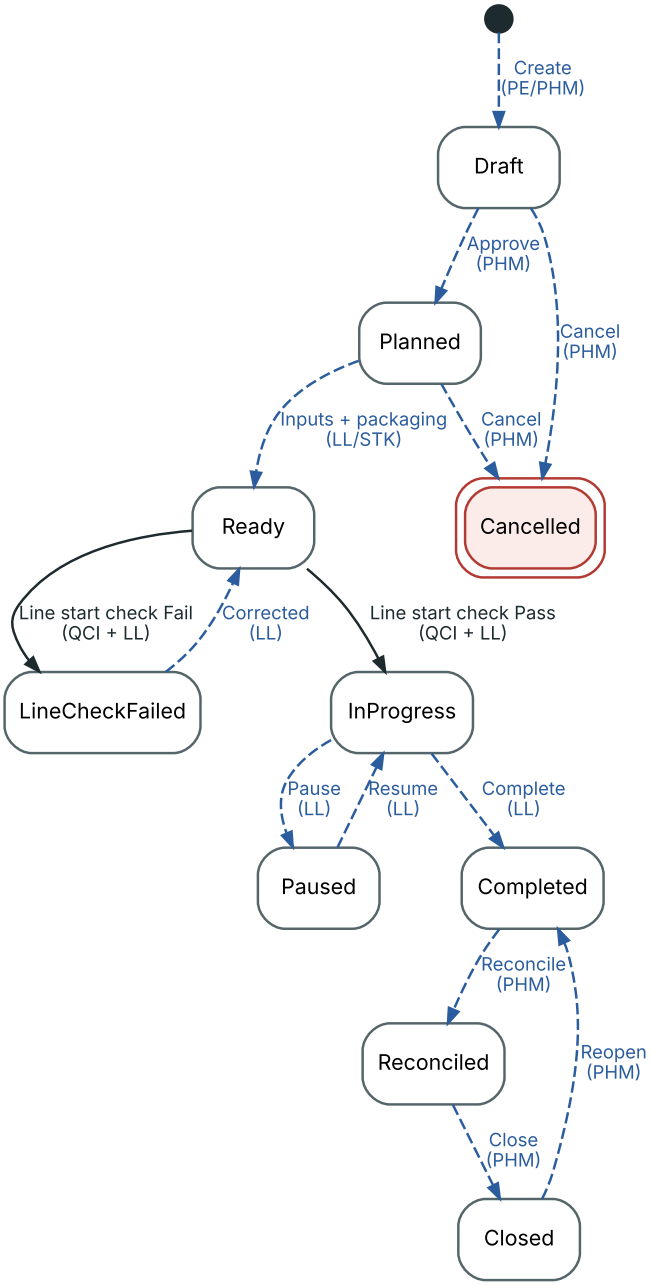


Figure 8.2 — Job Card (production order) state model (SM-JC; stages ST-08 / ST-15).

From	To	Trigger	Actor	Guard / rule	Label
[*]	Draft	Create	PE/PHM	—	P
Draft	Planned	Approve	PHM	Line, product, order line or grade target set	P
Planned	Ready	Inputs + packaging	LL/STK	≥1 released IP; packaging issued	P
Ready	InProgress	Line start check Pass	QCI + LL	G5 (SOP-PL)	C
Ready	LineCheckFailed	Line start check Fail	QCI + LL	Reason	C
LineCheckFailed	Ready	Corrected	LL	Re-check required	P
InProgress	Paused	Pause	LL	Reason code	P
Paused	InProgress	Resume	LL	—	P

From	To	Trigger	Actor	Guard / rule	Label
InProgress	Completed	Complete	LL	All inputs closed (consumed or returned)	P
Completed	Reconciled	Reconcile	PHM	Variance ≤ tolerance or explained (MB-01)	P
Reconciled	Closed	Close	PHM	Waste records submitted	P
Closed	Completed	Reopen	PHM	Reason; within reopen window (OD-15)	P
Draft	Cancelled	Cancel	PHM	No inputs issued	P
Planned	Cancelled	Cancel	PHM	No inputs issued	P

SM-FGP — Finished-Goods Pallet

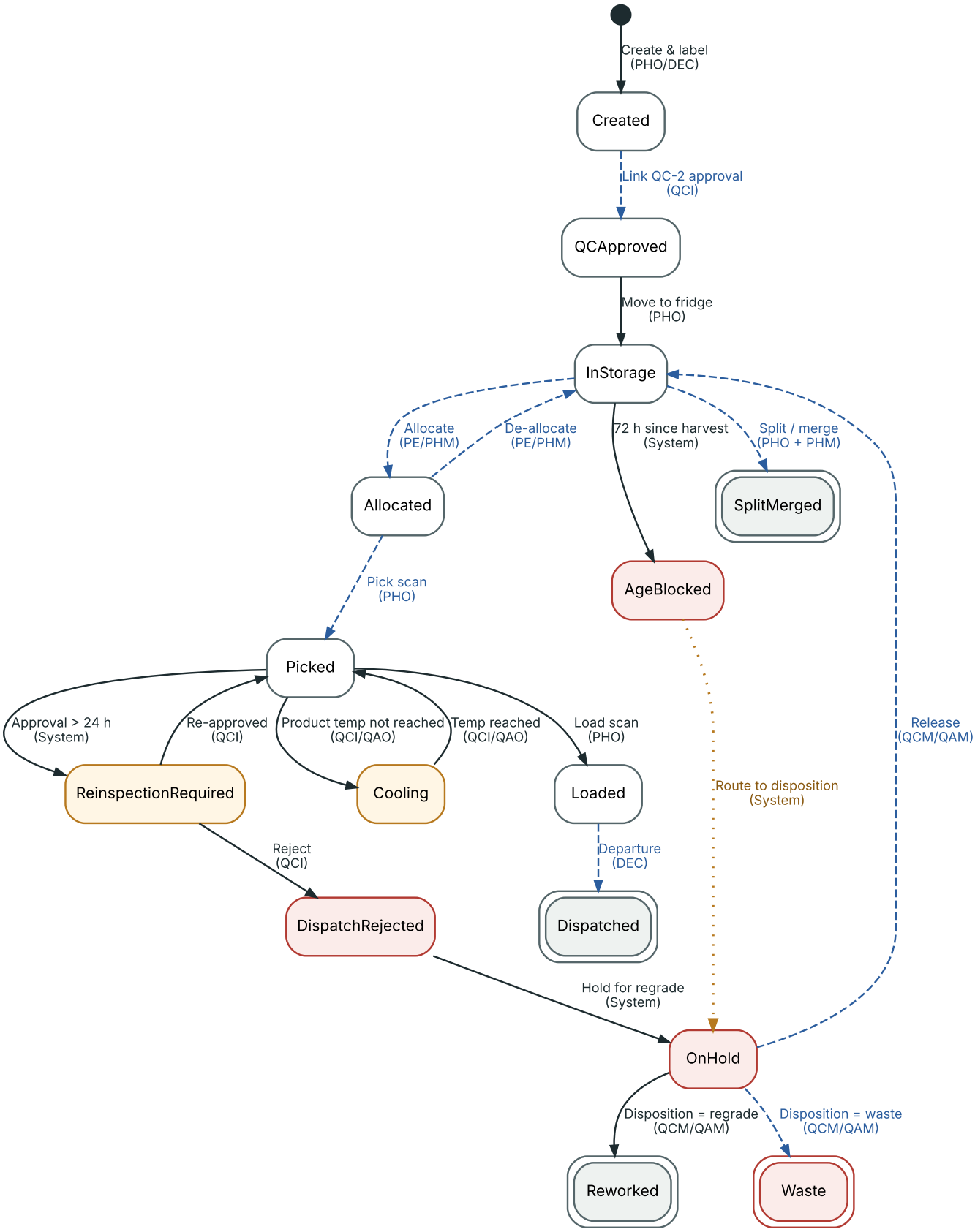


Figure 8.3 — Finished-Goods Pallet state model (SM-FGP; stages ST-10 → ST-14).

From	To	Trigger	Actor	Guard / rule	Label
[*]	Created	Create & label	PHO/DEC	QC-2 Pass for the output (G6)	C P

From	To	Trigger	Actor	Guard / rule	Label
Created	QCApproved	Link QC-2 approval	QCI	Approval time recorded (CP-06)	P
QCApproved	InStorage	Move to fridge	PHO	V-07, V-08	C
InStorage	Allocated	Allocate	PE/PHM	Spec matches order line	P
Allocated	InStorage	De-allocate	PE/PHM	—	P
Allocated	Picked	Pick scan	PHO	FIFO suggestion followed or reason (V-29)	P
Picked	Loaded	Load scan	PHO	QC approval ≤ 24 h; product temp OK; vehicle passed	C
Picked	ReinspectionRequired	Approval > 24 h	System	V-11	C
ReinspectionRequired	Picked	Re-approved	QCI	Dispatch QC Pass	C
ReinspectionRequired	DispatchRejected	Reject	QCI	Reason; photos	C
Picked	Cooling	Product temp not reached	QCI/QAO	Temperature recorded	C
Cooling	Picked	Temp reached	QCI/QAO	Re-measured	C
DispatchRejected	OnHold	Hold for regrade	System	Replacement = next FIFO pallet	C
OnHold	Reworked	Disposition = regrade	QCM/QAM	Rework JC consumes pallet	C
OnHold	InStorage	Release	QCM/QAM	SoD-05	P
OnHold	Waste	Disposition = waste	QCM/QAM	Waste record	P
InStorage	AgeBlocked	72 h since harvest	System	V-12	C
AgeBlocked	OnHold	Route to disposition	System	OD-10	NYV
InStorage	SplitMerged	Split / merge	PHO + PHM	V-22	P
Loaded	Dispatched	Departure	DEC	Load departed	P

SM-WR — Waste Record

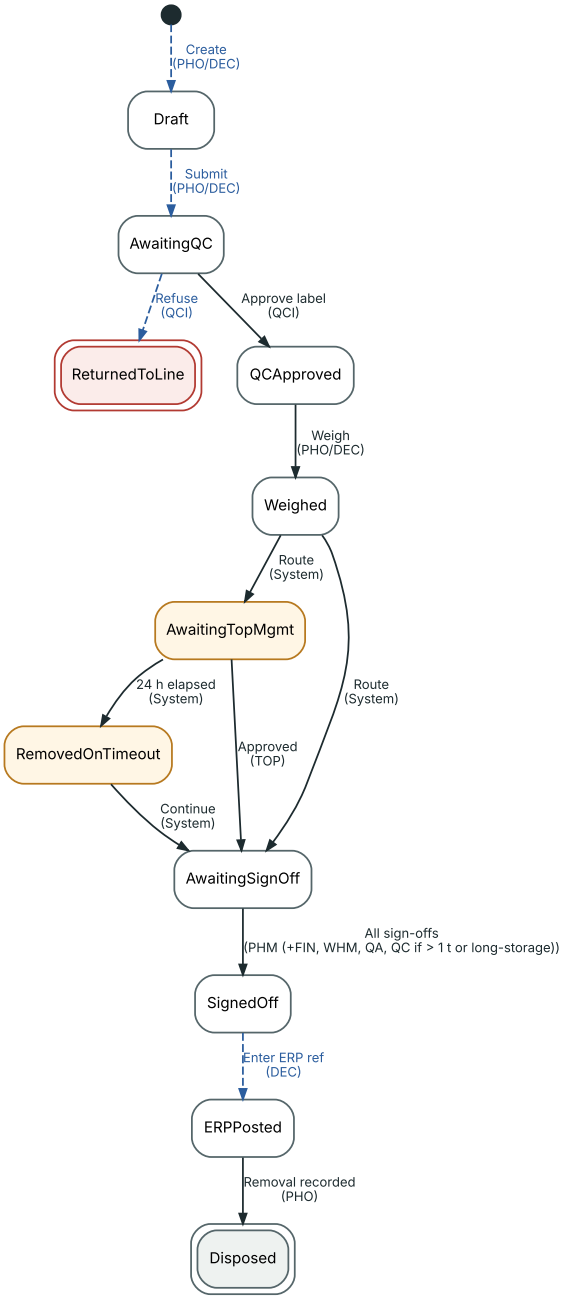


Figure 8.4 — Waste Record state model (SM-WR; stages ST-11).

From	To	Trigger	Actor	Guard / rule	Label
[*]	Draft	Create	PHO/DEC	Source record linked	P
Draft	AwaitingQC	Submit	PHO/DEC	Category; photos for long-storage	P
AwaitingQC	QCApproved	Approve label	QCI	Before scaling (SOP-WS)	C
AwaitingQC	ReturnedToLine	Refuse	QCI	Reason	P
QCApproved	Weighed	Weigh	PHO/DEC	Weight > 0 (V-16)	C
Weighed	AwaitingSignOff	Route	System	Processed or packaging waste; long-storage ≤ 1 t	C
Weighed	AwaitingTopMgmt	Route	System	Long-storage > 1 t (V-17)	C
AwaitingTopMgmt	AwaitingSignOff	Approved	TOP	—	C

From	To	Trigger	Actor	Guard / rule	Label
AwaitingTopMgmt	RemovedOnTimeout	24 h elapsed	System	Report + photos e-mailed (SOP-WS)	C
RemovedOnTimeout	AwaitingSignOff	Continue	System	Flag retained	C
AwaitingSignOff	SignedOff	All sign-offs	PHM (+FIN, WHM, QA, QC if > 1 t or long-storage)	V-17	C
SignedOff	ERPPosted	Enter ERP ref	DEC	Reference format check	P
ERPPosted	Disposed	Removal recorded	PHO	Start of next day shift (SOP-WS)	C

SM-NC — Hold / Nonconformity

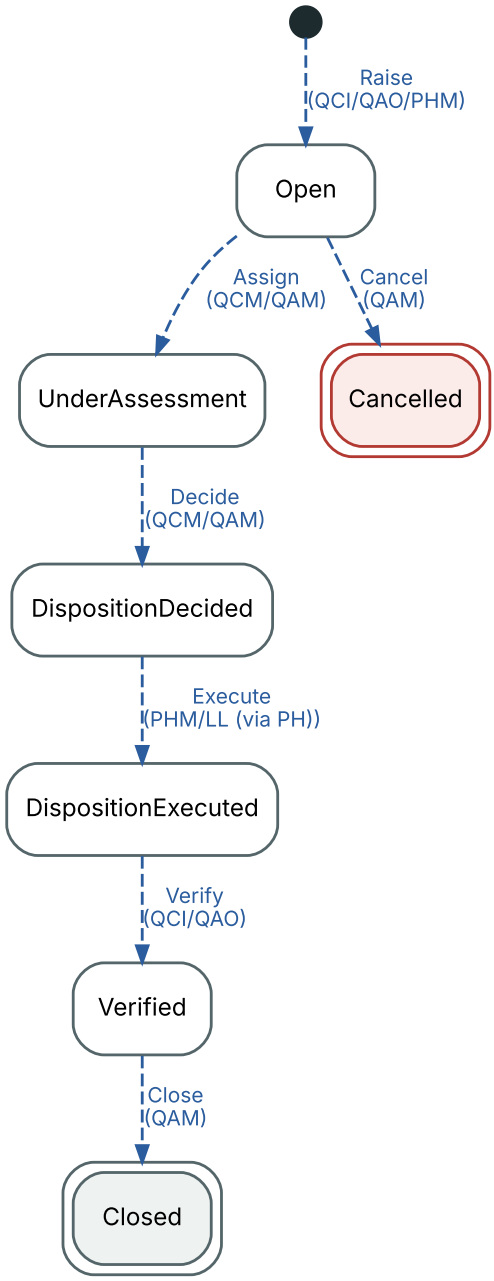


Figure 8.5 — Hold / Nonconformity state model (SM-NC; stages X-01).

From	To	Trigger	Actor	Guard / rule	Label
[*]	Open	Raise	QCI/QAO/PHM	≥1 item; reason	P
Open	UnderAssessment	Assign	QCM/QAM	—	P
UnderAssessment	DispositionDecided	Decide	QCM/QAM	Food-safety holds: QAM only	P
DispositionDecided	DispositionExecuted	Execute	PHM/LL (via PH)	Linked JC / waste / release	P
DispositionExecuted	Verified	Verify	QCI/QAO	Evidence attached	P
Verified	Closed	Close	QAM	CAPA linked if required	P
Open	Cancelled	Cancel	QAM	Reason	P

SM-LD — Dispatch Load

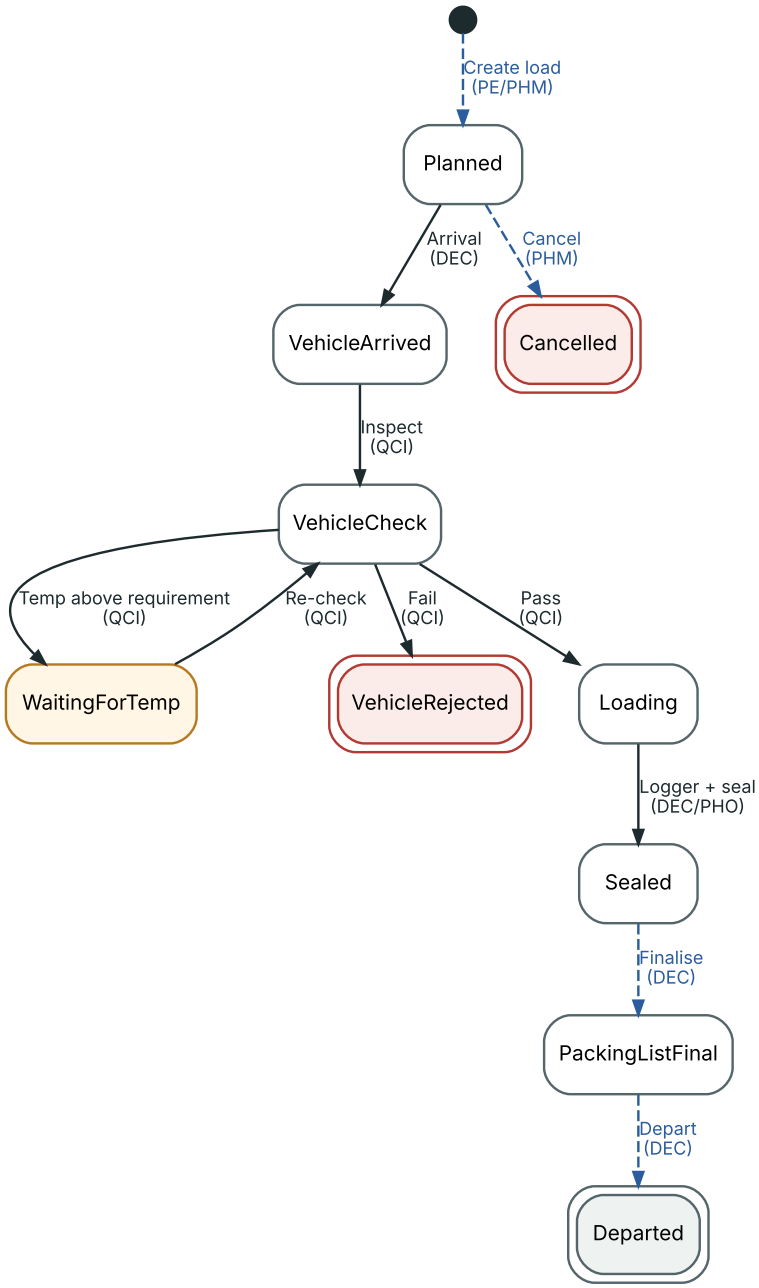


Figure 8.6 — Dispatch Load state model (SM-LD; stages ST-13 / ST-14).

From	To	Trigger	Actor	Guard / rule	Label
[*]	Planned	Create load	PE/PHM	Accepted order lines	P
Planned	VehicleArrived	Arrival	DEC	Time within window or override (V-15)	C P
VehicleArrived	VehicleCheck	Inspect	QCI	—	C
VehicleCheck	WaitingForTemp	Temp above requirement	QCI	V-14	C
WaitingForTemp	VehicleCheck	Re-check	QCI	—	C
VehicleCheck	VehicleRejected	Fail	QCI	Not lockable / dirty / smell / contamination	C
VehicleCheck	Loading	Pass	QCI	All checks recorded	C
Loading	Sealed	Logger + seal	DEC/PHO	Logger ID + position; seal no. (V-13)	C

From	To	Trigger	Actor	Guard / rule	Label
Sealed	PackingListFinal	Finalise	DEC	Scanned pallets = allocated or variance approved	P
PackingListFinal	Departed	Depart	DEC	Shelf-life sample recorded or waived by QAO (V-24)	P
Planned	Cancelled	Cancel	PHM	Reason	P

SM-CMP — Customer Complaint

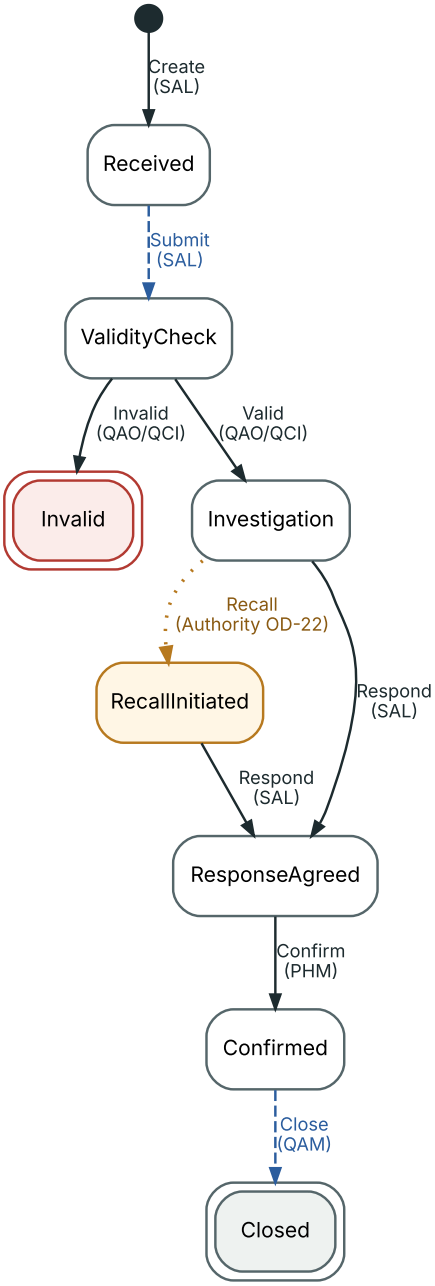


Figure 8.7 — Customer Complaint state model (SM-CMP; stages ST-16).

From	To	Trigger	Actor	Guard / rule	Label
[*]	Received	Create	SAL	Delivery date, lot, box codes, photos	C
Received	ValidityCheck	Submit	SAL	—	P
ValidityCheck	Invalid	Invalid	QAO/QCI	Missing info or > 24 h after delivery (V-18)	C
ValidityCheck	Investigation	Valid	QAO/QCI	—	C
Investigation	RecallInitiated	Recall	Authority OD-22	Health risk; ≤ 6 h from complaint	C NYV
Investigation	ResponseAgreed	Respond	SAL	≤ 24 h from complaint (V-18)	C
RecallInitiated	ResponseAgreed	Respond	SAL	—	C
ResponseAgreed	Confirmed	Confirm	PHM	—	C

From	To	Trigger	Actor	Guard / rule	Label
Confirmed	Closed	Close	QAM	CAPA linked if valid	P

8.2 Gates (QC/QA and operational)

Gate	Name	Where	Owner	Checks	Pass	Fail	Block	Label	Source
G1	Arrival gate	Gate / scales	Receiving (DEC); vehicle inspector NYV	Harvesting tag complete; vehicle condition (SOP-TR).	Registration allowed.	Vehicle exception: PHM + QA + Top mgmt approve or refuse.	Hard	C	SOP-RC; SOP-TR
G2	Arrival visual check	Receiving	QC/QA	Covered, palletised, crates, insects, disease, temperature, appearance, foreign matter, freshness, spec, GR3/waste.	To pre-cooling.	Rejected → farm or waste area.	Hard	C	SOP-RC; SOP-GR
G3	QA raw-material acceptance	After pre-cooling	QA	Sample per section: temperature, disease, deformity, shape, size, foreign matter, over-ripeness; photos/videos; tolerances (OD-05).	Accepted into raw stock.	Engineer regrade → recheck → 2nd reject = waste.	Hard	C	SOP-RC; SOP-GR
G4	QC-1 release	Sorting Area	QC	Quality and grading check (scope OD-17).	Released to production.	Hold.	Hard	NYV	SUM-1; SUM-3
G5	Line start check	Line	QC + Line Leader	Product compliance to requirements before packing commences.	JC InProgress.	JC cannot start.	Hard	C	SOP-PL
G6	QC-2 online & final	Line / finished product	QC	Random line checks; final inspection before weighing; grading to customer spec.	QC approval; FG pallet allowed.	Re-sort via PHM/PE instruction.	Hard	C	SOP-GR; SUM-3
G7	Waste QC approval	Waste point	QC	Product is waste (not saleable).	Weighing allowed.	Returned to line.	Hard	C	SOP-WS
G8	Dispatch release	Loading bay	QC (QA records)	Vehicle temp/cleanliness/smell/contamination; lockable; product temperature; packaging; logger; QC approval ≤ 24 h or re-inspection; photos/videos.	Load, seal, depart.	Wait for temperature / cool product / reject pallet (replace FIFO) / reject vehicle.	Hard	C	SOP-DS; SOP-GR; SOP-TR
G9	Hold release	Any	QC/QA (QAM for food safety)	Disposition evidence.	Item returns to prior state.	Disposition executed.	Hard	P	Proposed

All gates are hard blocks in the system: an item cannot take the next transition until the gate result is recorded. The physical product can still be moved; the system then shows it as out of sequence and raises an exception.

8.3 Validation rules

ID	Rule	Applies to	Type	Label	Source / OD
V-01	Harvest-tag fields mandatory: field (block/section), variety, harvest time, harvest date, supervisor. Supplier loads: supplier name and supplier reference.	ST-01/02	Block	C	SOP-RC; OD-31
V-02	Net weight must be > 0 and recorded with the scale ID. Weighing method (gross/tare/net): OD-29.	ST-02	Block	C NYV	SUM-1
V-03	Receipt after 17:00 is flagged 'late harvest receipt' and needs a reason.	ST-01	Flag	P	SOP-RC (17:00)
V-04	Pre-cooling dwell = exit − entry. Above 30 min raises an alert to PHM; the 15–30 min rule by receiving temperature applies when the rule table exists.	ST-04	Warn	C NYV	SOP-PC; OD-07
V-05	An IP can be issued to a JC only if its status is Released (QC-1) and it is not on hold.	ST-08	Block	P	SUM-3
V-06	Grading timer: IP not consumed 24 h after receiving = 'Grading overdue' (alert lead time OD-09).	ST-06	Flag	C	SOP-PC
V-07	Placement only into a fridge whose current storage group matches the crop's group; a mismatch needs a PHM override with reason.	ST-06/12	Block + override	C P	SOP-PC
V-08	Placement beyond the location's pallet capacity (pre-cooling 16; Harvest Fridge 64; Fridges 1–7 60 each) is warned.	ST-04/06/12	Warn	C	FCT; SUM-2
V-09	FG label mandatory fields per Section 7.3; label prints only when complete.	ST-10	Block	C	SOP-PL
V-10	Customer-specific pallet must reference an accepted order line; non-order pallet must carry a grade (Premium, GR1, GR2).	ST-10	Block	C	SOP-PL
V-11	At load scan: if (now − QC approval time) > 24 h the pallet goes to ReinspectionRequired; if ≤ 24 h dispatch rejection on product-quality grounds is not allowed.	ST-14	Block	C	SOP-GR; SOP-DS
V-12	Age since earliest harvest date: > 48 h needs a reason (specific order); > 72 h blocks loading.	ST-12/14	Warn / Block	C NYV	SOP-PC; OD-09; OD-10
V-13	Load cannot depart without seal number and, for loads in logger scope (OD-18), logger ID with pallet and box position.	ST-14	Block	C	SOP-DS; SOP-TR
V-14	Vehicle temperature above the crop's documented requirement blocks 'Loading' until a re-check passes. Tolerance: OD-08.	ST-14	Block	C NYV	SOP-DS
V-15	Loading or offloading event time between 01:00 and 09:00 needs a PHM override. LSS slots 11:00 and 19:00 are shown on the schedule.	ST-01/14	Block + override	C P	SOP-DS; OD-39
V-16	Waste weight can be entered only after QC approval.	ST-11	Block	C	SOP-WS
V-17	Waste sign-off routing: processed waste < 1 t → PHM; processed > 1 t → PHM, FIN, WHM, QA, QC; long-storage waste (any weight) → PHM, FIN/WHM, QC/QA; long-storage > 1 t → plus top-management approval (24 h rule). Threshold basis OD-11.	ST-11	Route	C NYV	SOP-WS
V-18	Complaint: mandatory fields per SOP-CP; created more than 24 h after delivery → proposed Invalid; resolution timer 24 h; recall timer 6 h.	ST-16	Flag + timers	C	SOP-CP
V-19	All IDs are unique system-wide and never reused.	All	Block	P	GEN-12
V-20	JC mass-balance variance above the crop tolerance blocks 'Reconciled' unless a variance reason is recorded.	ST-08	Block + reason	P	OD-28
V-21	Weight plausibility vs nominal kg per pallet (non-blocking). Tolerance OD-29.	ST-02/10	Warn	P	FCT
V-22	Merge only for the same crop, product/pack spec, grade, customer (or none) and storage group.	ST-10	Block	P	GEN-08
V-23	Photo/video attachment mandatory on QA raw-material acceptance and QC dispatch inspection; optional elsewhere.	ST-05/14	Block	C	SOP-GR; SOP-DS
V-24	Shelf-life sample per product, per customer, per load recorded before departure, or a QAO waiver with reason.	ST-14	Warn	C P	SOP-GR

ID	Rule	Applies to	Type	Label	Source / OD
V-25	Fridge temperature reading overdue per interval (OD-32) raises a task; reading outside the documented range raises an alert.	X-07	Flag	C NYV	SOP-PC
V-26	Incoming product gets at most one recheck after regrade; a second rejection goes to waste.	ST-05	Block	C	SOP-RC
V-27	Event time is server time. Records entered after the event store both times and a 'late entry' flag when the gap exceeds a threshold (OD-16).	All	Flag	P	ISO 27002 §8.17
V-28	No physical deletion. Errors are voided or corrected with reason; the original stays in history.	All	Block	P	ISO 9001 §7.5.3
V-29	Picking a pallet other than the FIFO suggestion needs a reason code (e.g. customer spec, colour, location).	ST-13	Reason	P	SOP-DS (FIFO)
V-30	A QC/QA instruction requiring PH staff action is routed to the PHM/PE task queue, not to individual PH staff.	All	Route	C	SOP-ST

8.4 Exception handling and escalation

ID	Exception	Detection	Immediate action	Owner	Escalation	Label
E-01	Incomplete or missing harvesting tag	V-01 at registration	Arrival held; DEC contacts farm supervisor; tag completed or load held	DEC	PHM if unresolved within a set time (OD-16)	C P
E-02	Incoming vehicle non-compliant	Vehicle inspection fail	Product not accepted until approved	PHM + QA + Top mgmt	Refusal → load returned	C
E-03	Arrival visual rejection	G2 fail	Pallets rejected; destination recorded (farm/waste)	QCI/QAO	Notify farm and PHM	C
E-04	QA rejection of incoming product	G3 fail	Engineer notified; regrade; re-weigh; recheck	QAO / ENG	2nd rejection → waste; notify PHM and farm	C
E-05	Pre-cooling dwell exceeded	V-04	Alert to PHM; move to QA queue	PHM	Repeated breaches → capacity review	C P
E-06	Grading overdue (> 24 h from receiving)	V-06	Alert PHM; prioritise in plan	PHM	QA informed; long-storage regrade rules apply	C P
E-07	Storage-group conflict or fridge full	V-07, V-08	Placement blocked/warned; PHM reallocates fridge or overrides	PHM	Override reviewed (X-06)	C P
E-08	Fridge temperature outside documented range	V-25	Alert PHM + QA; affected pallets listed	PHM / QAM	QAM decides hold (OD-08)	C NYV
E-09	Line start check fails	G5	JC blocked; correction; re-check	LL / QCI	Instruction via PHM/PE (V-30)	C
E-10	QC-2 nonconformance on line	G6 fail	Re-sort; affected output on hold	QCI → PHM/PE	Repeated → CAPA	C
E-11	Packaging shortage	Issue request unfilled	Notify WHM and PHM; JC paused	STK / WHM	Plan change by PHM	C P
E-12	Need to deviate from customer spec	Planner or QC	Temporary deviation request to customer in writing; produce only after written approval	Requester not stated (X-03)	Sales	C
E-13	Mass-balance variance above tolerance	V-20	JC cannot be reconciled without a variance reason	PHM	Repeated → investigation	P
E-14	FG age > 48 h without order / > 72 h	V-12	> 48 h: DC dispatch decision (SOP-PL); > 72 h: loading blocked	PHM / SAL	OD-10	C NYV
E-15	Dispatch vehicle above required temperature	V-14	Loading on hold until reached	QCI	LOG informed	C
E-16	Product temperature not reached at dispatch	Dispatch check	Pallet back to fridge for further cooling	QCI/QAO	—	C
E-17	Dispatch QC rejects a pallet (approval > 24 h)	V-11	Back to fridge; next FIFO pallet replaces; regrade via rework JC	QCI + PHM	Customer concession possible (X-03)	C
E-18	Vehicle cannot be locked	Vehicle check	No loading	QCI	LOG arranges another vehicle	C
E-19	Loading/offloading outside 09:00–01:00	V-15	Blocked unless PHM override	PHM	Override review	C P

ID	Exception	Detection	Immediate action	Owner	Escalation	Label
E-20	Long-storage waste > 1 t without top-management approval in 24 h	Timer	Waste removed due to contamination risk; report and photos e-mailed	PHM	Recorded as RemovedOnTimeout	C
E-21	Health-risk complaint	Complaint triage	Recall initiated within 6 h; trace report; in-house stock on hold	QAM / recall authority (OD-22)	Top management	C NYV
E-22	System unavailable	Monitoring / user report	Pre-numbered paper forms; back-entry with late-entry flag	PHM	IT	P
E-23	Label printer or scanner failure	User report	Spare printer / pre-printed ID labels; manual ID entry with double check	PHM	IT	P
E-24	Customer crates dirty	QC crate inspection	Wash by PH staff or return to supplier	QCI decides; PHM executes	—	C
E-25	Shelf-life sample shows mould before 14 days	QAO check	Sample removed; recorded; communicated to relevant parties	QAO	Investigation if pattern	C
E-26	Transit time or truck temperature change	LOG / SAL notice	PH informed ≥ 2 days ahead (transit); QA/QC and LOG 24 h ahead (temperature)	LOG	PHM	C

8.5 Audit trail

ID	Audit-trail requirement	Label	Basis
AT-01	Every create, edit, submit, verify, approve, reject, release, hold, cancel, reopen and override is logged with user, role used, server timestamp, record ID, field-level before/after values and reason (where required).	P	ISO 27002 §8.15; ALCOA+
AT-02	Audit entries are append-only; no role, including administrators, can edit or delete them.	P	ISO 27002 §5.33, §8.15
AT-03	Approval and verification actions are electronic signatures: user re-authenticates, and the meaning (approved / verified / rejected) is stored with the action.	P	Recognised e-signature practice
AT-04	Event time and entry time are both stored; late entries are flagged (V-27).	P	ALCOA+ (contemporaneous)
AT-05	Attachments (photos, videos, signed documents, customer approvals) are stored with hash, uploader and time and cannot be replaced, only superseded.	P	ISO 9001 §7.5.3
AT-06	Master-data versions (tolerances, temperatures, specs) are effective-dated; each quality record stores the version it was assessed against.	P	ISO 9001 §7.5.3; ISO 22000 §7.5
AT-07	Label prints and reprints are logged with the label content printed.	P	ISO 9001 §8.5.2
AT-08	Records and audit history are retained for the period decided in OD-21 (ISO 22000 §8.3 sets at least the product shelf life for traceability evidence).	NYV	ISO 22000 §8.3; OD-21
AT-09	Audit trail is viewable per record and searchable by user, date and action for the Auditor role.	P	ISO 9001 §9.2
AT-10	Administrative actions (role grants, configuration changes) are logged and reported weekly to the system owner.	P	ISO 27002 §8.2, §8.32

9 Data model and integration

9.1 Entities

Entity	Kind	Description	Data owner	Label
Crop	Master	Crop with storage group, documented temperature range, boxes per pallet, nominal kg per pallet, washer route.	PHM (operational) / QAM (temperature)	C NYV
Variety	Master	Variety under a crop (from harvesting tag).	Farm Operations	C
StorageGroup	Master	Tomato & cherry; mini cucumber; pepper & sweet bite; eggplant; mango (+ onion).	QAM	C
Source	Master	Own-farm block + section, or external supplier. Operations states 3 blocks × 8 sections (not in the supplied documents).	Farm Operations / Procurement	NYV
Grade	Master	Premium (export), GR1, GR2, waste (and GR3 at receiving).	QCM	C
Product	Master	Pack specification (crop, pack type, net weight, boxes per pallet, label template).	Sales + QCM	P
Customer	Master	Export UK/Abay/GU/EU, local, retail, LSS, DC.	Sales	C
CustomerSpec	Master (versioned)	Customer product specification and tolerances.	Sales (content) / QCM (approval)	NYV
ToleranceSet	Master (versioned)	Incoming raw-material grading schedule per crop.	QAM	NYV
Location	Master	Pre-cooling, Harvest Fridge, Fridges 1–7, sorting area, lines, waste area, loading bays; pallet capacity; current storage group.	PHM	C
Line	Master	Washer-dryer, Line A ELIFAB, Line B Aweta, Line C manual; crops allowed; nominal rates.	PHM	C
Equipment	Master	Scales, thermometers, printers; calibration due date.	PHM / QAM	P
User / Role	Master	Named users, roles, department, approver.	ADM (on department-head approval)	P
Order / OrderLine	Transaction	Customer order and lines; acceptance decision.	Sales / PHM	C P
Arrival	Transaction	Vehicle load at the gate; tag data; dispatch note; vehicle inspection.	Warehouse Mgmt (DEC)	P
IncomingPallet	Transaction	Field pallet as received.	Warehouse Mgmt (DEC)	C
Lot	Transaction	Raw-material lot (definition OD-02).	QAM / PHM	NYV
InspectionRecord	Transaction	Arrival check, RMA, recheck, QC-1, line start, QC-2, dispatch QC, vehicle check — one structure with type.	QCM / QAM	C P
Sample / Defect	Transaction	Sample measurements and defect counts within an inspection.	QCM / QAM	P
StockMove	Transaction	Location change of an IP or FG pallet.	PHM (custody OD-14)	P
TemperatureReading	Transaction	Location / vehicle / product temperature with time.	QAM	C P
JobCard	Transaction	Production order / line run.	PHM	P
JobCardInput	Transaction	IP consumed by a JC (kg, boxes).	PHM	P
FGPallet	Transaction	Finished pallet.	PHM	C
FGComposition	Transaction	Lots/IPs and kg in an FG pallet.	PHM	P
PackagingIssue	Transaction	Packaging issued to a JC (reference to ERP item).	Warehouse	C P
WasteRecord	Transaction	Waste by category, approvals and ERP reference.	PHM	C
Hold	Transaction	Hold / nonconformity and disposition.	QAM	P
CAPA	Transaction	Corrective action.	QAM	P

Entity	Kind	Description	Data owner	Label
Deviation	Transaction	Customer concession / temporary deviation.	Sales / QCM	C P
Load	Transaction	Outbound load: vehicle, logger, seal, packing list.	PHM / LOG	C P
LoadPallet	Transaction	FG pallet on a load, with order line.	PHM	P
ShelfLifeSample	Transaction	Sample per product/customer/load.	QAM	C
Complaint	Transaction	Customer complaint and investigation.	QAM	C
Override	Transaction	Emergency override and review.	Gate owner	P
Attachment	Transaction	Photo / video / document with hash.	Record owner	C P
AuditLog	System	Append-only audit trail.	System (read: AUD)	P
ExternalRef	System	ERP or other system document number linked to a record.	DEC / WHM	P

9.2 Entity-relationship model

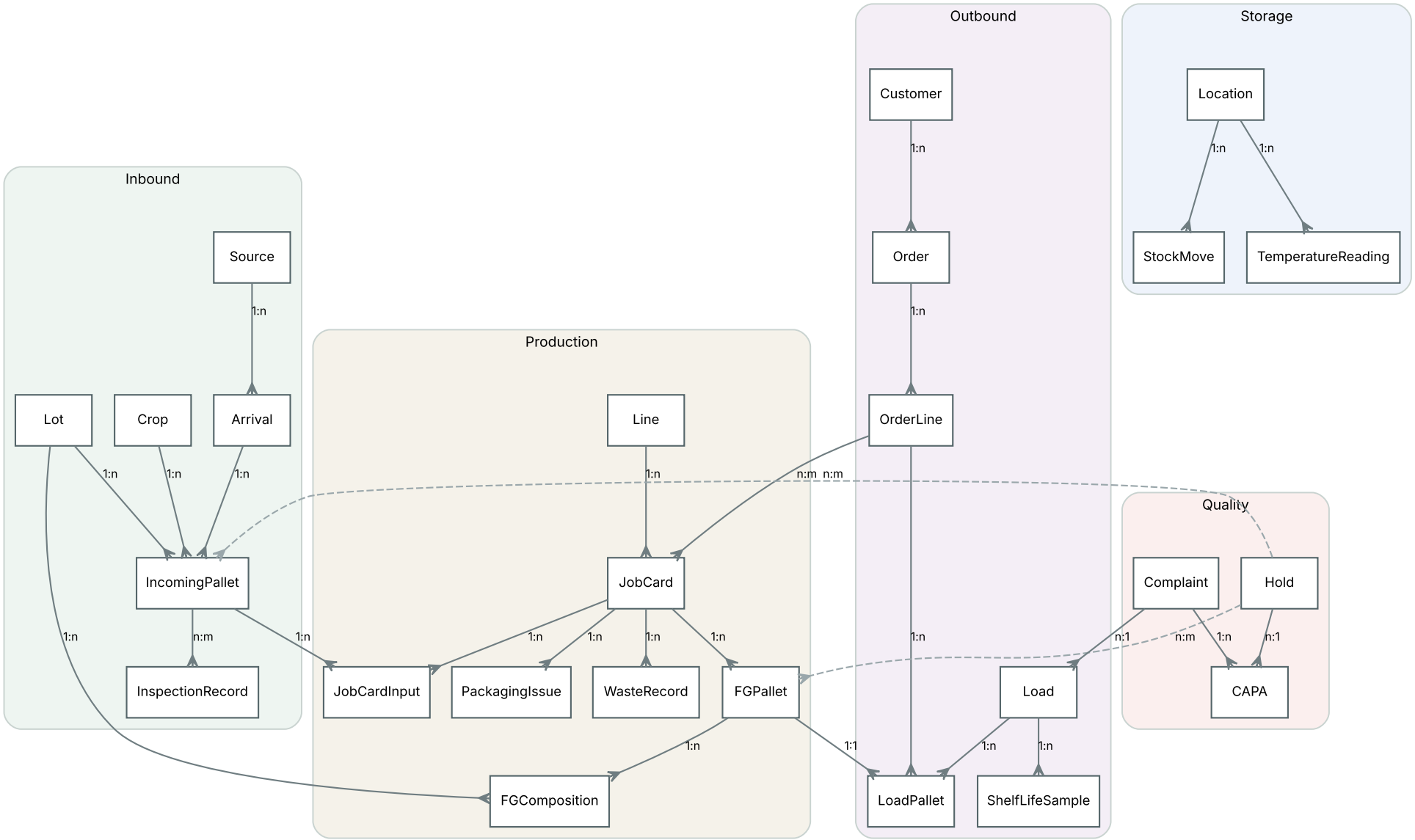


Figure 9.1 — Entity-relationship model (logical). Crow's foot marks the 'many' side. Dashed: Hold applies n:m to incoming and finished pallets.

9.3 Data dictionary (key entities)

Mandatory fields, their authoritative source and validation. Master-data entities are summarised in 9.1; their attributes follow the same pattern.

Arrival

Field	Type	Req.	Authoritative source	Validation	Label
arrival_id	ID	Y	System	Unique; ARR-YYMMDD-NNNN	P
arrived_at	Timestamp	Y	Server clock	—	P
source_type	Enum (own farm / supplier)	Y	DEC from tag / supplier docs	—	C
block_id, section_id	Ref → Source	Y if own farm	Harvesting tag	Valid block/section	C
supplier_id	Ref → Source	Y if supplier	Supplier documents	Approved supplier	C NYV
variety_id	Ref → Variety	Y	Harvesting tag	Belongs to crop	C
harvest_date, harvest_time	Date, time	Y	Harvesting tag	Not in future	C
harvest_supervisor	Text / Ref	Y	Harvesting tag	—	C
dispatch_note_no, dispatched_pallets, dispatched_boxes	Text, int, int	If CP-02	Farm / supplier dispatch note	—	P
vehicle_plate, driver	Text	Y	DEC	—	P
vehicle_check_result	Enum	Y	Inspector	Pass / Fail	C
vehicle_exception_id	Ref	If Fail	Approval record	3 approvals	C

IncomingPallet

Field	Type	Req.	Authoritative source	Validation	Label
ip_id	ID	Y	System	Unique, never reused	P
print_no	Text	Y	System	Per crop/section/block/day sequence (SOP-RC)	C
arrival_id	Ref	Y	System	—	P
crop_id	Ref	Y	Tag	—	C
box_count	Int	Y	DEC count	Default from crop (35; cherry 70)	C
net_kg	Decimal	Y	Scale	> 0 (V-02); plausibility V-21	C
scale_id	Ref → Equipment	Y	DEC	Calibration not expired (warn)	P
registered_at, registered_by	Timestamp, user	Y	System	—	P
status	Enum (SM-IP)	Y	System	Transitions per SM-IP	P
lot_id	Ref → Lot	Per OD-02	System	—	NYV
receiving_temp_c	Decimal	Y	QC/QA arrival check	—	C
precool_in, precool_out	Timestamp	Y	Scans	V-04	C P
location_id	Ref → Location	Y	Last stock move	V-07, V-08	P
remaining_kg	Decimal	Y	System	≥ 0	P

InspectionRecord

Field	Type	Req.	Authoritative source	Validation	Label
insp_id	ID	Y	System	—	P
type	Enum	Y	System	ARRIVAL, RMA, RECHECK, QC1, LINE_START, QC2_ONLINE, QC2_FINAL, DISPATCH_VEHICLE, DISPATCH_PRODUCT, WASTE_LABEL, CRATE	P
subject_refs	Refs	Y	User selection / scan	IP, section-day, JC, FG pallet, load	P
checklist_results	Structured	Y	Inspector	Items per type (SOP-RC, SOP-GR, SOP-DS)	C
tolerance_set_version / spec_version	Ref	Y for RMA, QC2	Master data	Effective on inspection date	NYV
product_temp_c	Decimal	Type-dependent	Inspector	—	C
result	Enum	Y	Inspector	Pass / Fail / Accept / Reject / Hold	C
reason	Text	If Fail	Inspector	—	P
attachments	Refs	Y for RMA, DISPATCH	Device camera	V-23	C
inspector, co_signer, signed_at	User, user, timestamp	Y	E-signature	SoD rules	C P

JobCard

Field	Type	Req.	Authoritative source	Validation	Label
jc_id	ID	Y	System	—	P
type	Enum	Y	Planner	PRODUCTION / REWORK	P
line_id	Ref	Y	Planner	Crop allowed on line	C
washer_route	Bool	Y	Crop route master	Per Figure 4.2	C
order_line_ids / grade_target	Refs / enum	One of	Planner	V-10	C
customer_spec_version	Ref	If order	Master data	—	NYV
planned_start, actual_start, actual_end	Timestamps	Y	Planner / LL	—	P
pauses	List (reason, from, to)	N	LL	—	P
crew_count	Int	N	LL	Needed for labour KPI (OD-34)	P
lot_attribution_method	Enum	Y	Planner	EXACT / TIME_WINDOW / PROPORTIONAL (GEN-06)	P
mass_balance_variance_kg / pct / reason	Decimal, decimal, text	Y at reconcile	System / PHM	V-20	P
status	Enum (SM-JC)	Y	System	—	P

FGPallet

Field	Type	Req.	Authoritative source	Validation	Label
fgp_id	ID	Y	System	FGP-... or SSCC (OD-20)	C P
jc_id	Ref	Y	System	—	P

Field	Type	Req.	Authoritative source	Validation	Label
product_id	Ref	Y	JC	—	P
grade / order_line_id	Enum / ref	One of	JC	V-10	C
box_count, net_kg	Int, decimal	Y	Count / scale	V-21	C
composition	List (lot, ip, kg)	Y	System (GEN-05/06)	Σ kg = net kg ± tolerance	P
earliest_harvest_at	Timestamp	Y	Derived	Used for V-12	P
qc_approval_id, qc_approved_at	Ref, timestamp	Y before storage	QC-2	V-11	C P
supplier_name, origin	Text	Per label rules	Source / order	SOP-PL	C
location_id	Ref	Y	Stock move	—	P
status	Enum (SM-FGP)	Y	System	—	P

WasteRecord

Field	Type	Req.	Authoritative source	Validation	Label
wr_id	ID	Y	System	—	P
category	Enum	Y	Creator	PROCESSED / LONG_STORAGE / PACKAGING / WEIGHT_LOSS (SOP-WS)	C
source_ref	Ref	Y	Creator	JC, IP, FG pallet or location	P
crop_id, bags, net_kg	Ref, int, decimal	Y	Creator / scale	V-16	C
qc_approval_id	Ref	Y	QCI	Before weight	C
signoffs	List (role, user, time)	Y	Approvers	V-17	C
topmgmt_requested_at / approved_at / timeout_flag	Timestamps, bool	If long-storage > 1 t	System / TOP	24 h rule	C
erp_ref	Text	Y before Disposed	DEC	Format check (IB-01)	P
disposed_at	Timestamp	Y	PHO	—	C

Load

Field	Type	Req.	Authoritative source	Validation	Label
load_id	ID	Y	System	= packing list no. (proposal)	P
customer / order_ids	Refs	Y	Planner	Accepted lines	C
vehicle_plate, transporter	Text	Y	DEC / LOG	—	P
vehicle_temp_c, cleanliness, smell, contamination, lockable	Decimal, enums, bool	Y	QCI	V-14; lockable = true	C
logger_id, logger_pallet_id, logger_box_position	Text, ref, text	Per OD-18	Loader	V-13	C
seal_no	Text	Y	Loader	V-13	C

Field	Type	Req.	Authoritative source	Validation	Label
pallets	List (fgp_id, order_line, scan time, dispatch QC ref)	Y	Scans	V-11	P
photos	Refs	Y	QCI	Per shipment (SOP-DS)	C
shelf_life_sample_ids	Refs	Y or waiver	QAO	V-24	C
departed_at	Timestamp	Y	DEC	—	P

Hold

Field	Type	Req.	Authoritative source	Validation	Label
hold_id	ID	Y	System	—	P
type	Enum	Y	Raiser	QUALITY / FOOD_SAFETY / OPERATIONAL (OD-23)	P
items	Refs	Y	Raiser	IP / JC output / FG pallet / load	P
reason, evidence	Text, refs	Y	Raiser	—	P
disposition	Enum	Y to close	QCM / QAM	RELEASE / REGRADE / REWORK / WASTE / RETURN / CONCESSION	C P
verified_by, closed_by	Users	Y	QC/QA	SoD-05	P
capa_id	Ref	N	QAM	—	P

Complaint

Field	Type	Req.	Authoritative source	Validation	Label
cmp_id	ID	Y	System	—	P
received_at, delivery_date	Timestamp, date	Y	SAL	V-18 (≤ 24 h)	C
nature, pct_out_of_spec	Enum/text, decimal	Y	Customer via SAL	SOP-CP	C
box_codes, lot_numbers, load_id	Text, refs	Y	Customer via SAL	Resolve to IDs	C NYV
photos	Refs	Y	Customer via SAL	SOP-CP rule on sufficiency	C
qty_supplied, qty_sampled, section	Decimal, decimal, ref	Y if valid	Investigation	SOP-CP	C
health_risk, recall_initiated_at	Bool, timestamp	Y	Investigator	≤ 6 h	C
response, agreed_action	Text	Y	SAL	≤ 24 h	C

9.4 Data ownership and system of record

Data domain	Owner	System of record today	System of record TO-BE	Label
Harvest tag data (source, variety, harvest time)	Farm Operations	Paper tag	New system (captured at gate)	C P
Receiving weights and Pallet IDs	Warehouse Mgmt (Data Entry)	'Pack house system' / ERP — identity not verified (IB-02)	New system, subject to OD-03	NYV
QA/QC inspection results and photos	QA / QC managements	Paper reports with photos/videos (format not stated)	New system	C P
Tolerances, temperatures, product specs	QAM / QCM (+ Sales for customer specs)	Grading schedule & product spec documents (not supplied)	New system master data (versioned)	NYV

Data domain	Owner	System of record today	System of record TO-BE	Label
Production records (JC, consumption, outputs)	PHM	Traceability team records (format not stated)	New system	NYV P
FG pallet IDs and labels	PHM	Not stated	New system, subject to OD-03	NYV
Fridge temperatures	QAM / PHM	Manual log + computerised record	Manual readings in new system; computerised record referenced (IB-03)	C P
Waste quantities (financial posting)	PHM / Finance	ERP + printed signed report	Operational record in new system; posting stays in ERP	C P
Packaging stock and consumption	Warehouse	ERP	ERP (unchanged)	C
Orders and customer master	Sales	Not stated (white board in PH)	Sales source unchanged; accepted lines in new system (OD-37)	NYV
Truck schedules	Logistics	Not stated	Unchanged; schedule visible in new system (IB-09)	NYV
Users and roles	Department heads (approval) / IT (administration)	—	New system	P

9.5 Integration boundaries

No API, file interface or integration is assumed for any existing system. Phase 1 avoids building interfaces and minimises double entry by making the new system the capture point and keeping ERP postings as summary postings with the ERP reference stored back. The cost of that remaining double entry is measured (H-10) before any integration is proposed.

ID	Existing system	Evidence	Data crossing	Phase-1 approach	Later option	Label	OD
IB-01	ERP (named in SOP001; product and version not stated in the supplied documents)	SOP-WS (waste entered in ERP); SOP-PL (packaging removed from ERP stock by Warehouse on dispatch); SOP-ST (data entry clerks handle ERP transactions).	Waste quantities by category; packaging consumed per load; possibly receiving quantities.	No interface. The system produces daily ERP-ready summaries; DEC keys them into the ERP as today and records the ERP document number on the source record. Missing references are flagged daily.	File export or API — only after ERP import capability, owner and cost are confirmed and the Phase 1–3 re-keying time is measured.	C (ERP use) / P (approach)	OD-04
IB-02	'Pack house system' (SOP-RC: product 'entered into the pack house system in order to receive a lot number and label')	SOP-RC only. Not identified: may be an ERP module, a label program or a spreadsheet.	Pallet numbers, lot numbers, labels.	Phase 0 identifies it. If it issues IDs today, decide whether the new system replaces it for the packing house or reads/keys its IDs. No replacement without an explicit decision.	Depends on identity.	NYV	OD-03
IB-03	Computerised fridge temperature record	SOP-PC.	Temperature series per fridge.	Manual readings in the system; the computerised record stays as is and is referenced by location and time.	Import only if the system offers an export (to be checked).	C (exists) / N (capability)	OD-32
IB-04	Scales (3 at receiving; FG and waste scales)	SUM-1 (3 scales).	Weights.	Manual entry with scale ID and plausibility check (V-21).	Direct reading only if indicators provide an output (to be checked on site).	C (exist) / N (capability)	OD-24
IB-05	Sales order source	SOP-FC; SOP-DS (white board).	Orders, customer specs.	Sales (or PH planner) enters accepted lines in the system; order template upload as an option.	Interface only if Sales holds orders in a system with export.	NYV	OD-37
IB-06	Grading machines (Aweta, ELIFAB)	SUM-2 (manufacturer descriptions only).	Possibly grading statistics.	Out of scope. Not assumed.	Evaluate only on a demonstrated need.	NYV	OD-40
IB-07	E-mail	SOP-WS (top-management approval via e-mail); SOP-FC (pack house group).	Notifications and approval requests.	System sends e-mail notifications; approvals are made in the system, with the e-mail as notice only.	—	C P	OD-11
IB-08	Existing QC/QA paper or electronic records	SOP-GR (reports with photos/videos).	Inspection records.	Replaced by system records for the pilot crop after parallel run; legacy records archived by QA.	—	NYV	OD-05
IB-09	Logistics truck schedule	SOP-FC (truck forecast, schedule one day before).	Truck ETA, slot.	LOG enters or uploads the schedule; no interface.	—	NYV	OD-37

10 Functional requirements

90 functional requirements in 16 modules, plus 12 non-functional requirements. Each states the user action, the system response, the governing business rule, the permitted roles, error handling and a testable acceptance criterion. Permissions refer to Section 6; rules refer to Section 8.

Code	Module	Requirements
MD	Master data & configuration	7
AC	Users, roles & access control	6
PL	Orders & daily planning (interface)	4
RC	Receiving & incoming pallet registration	6
QG	Quality gates (QC / QA)	7
ST	Storage, location & temperature	7
PR	Production / Job Cards	8
FG	Finished-goods pallets & labels	6
WS	Waste	6
DS	Dispatch	10
NC	Hold, nonconformity, CAPA & override	4
TR	Complaints, trace & recall	5
RP	Reporting & KPIs	4
AU	Audit trail & records	4
NT	Tasks, notifications & escalation	3
IN	External references & data exchange	3

10.1 Master data & configuration (FR-MD)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-MD-01	Crop master	ADM enters crop; PHM approves (QAM approves temperature fields).	Stores crop with storage group, documented temperature range, boxes per pallet, nominal kg per pallet, washer route and allowed lines; versioned on approval.	SoD-07 maker-checker; crop identity per OD-30.	ADM (maker); PHM/QAM (checker)	Reject approval by the maker; missing storage group blocks save.	A crop cannot be used in receiving until approved; changing boxes per pallet creates a new version and earlier records keep the old version ID.	C P	SOP-PC; SUM-2; FCT
FR-MD-02	Location master	ADM defines locations; PHM approves; PHM sets the current storage group.	Stores pallet capacity and current storage group per location; shows occupancy.	Capacities: pre-cooling 16, Harvest Fridge 64, Fridges 1–7 60 each.	ADM, PHM	Capacity below current occupancy cannot be saved.	Occupancy equals the count of pallets whose last move is into the location; changing a fridge's group is logged.	C	FCT; SUM-2; SOP-PC
FR-MD-03	Quality parameter sets	QAO/QCI drafts tolerance sets and temperature ranges; QAM/QCM approves with an effective date.	Version becomes active on its effective date; inspections store the version used.	AT-06; SoD-07.	QAO, QCI (maker); QAM, QCM (checker)	Overlapping effective dates for the same crop are blocked.	An RMA started before a new version's effective date shows the old version; audit trail shows the approver.	P	EV-08; OD-05; OD-08
FR-MD-04	Customers & customer specifications	SAL maintains customers and attaches spec content; QCM approves spec versions.	Spec versions selectable on order lines and JCs; grade-to-customer schedule visible to the line (SOP-PL).	Spec approval by QCM.	SAL, QCM	An order line referencing an unapproved spec cannot be accepted.	A JC shows the spec version of its order line; the QC-2 record stores the same version.	C P	SOP-PL; OD-06
FR-MD-05	Sources	ADM maintains blocks, sections and suppliers.	Valid sources for tag capture; supplier flag for separate reporting.	Own-farm structure given by Operations as 3 blocks × 8 sections (user-provided; not in the supplied documents) — confirm for the packhouse source list.	ADM; PHM approves	Inactive source cannot be selected.	Daily stock report separates own vs supplier crop by source (SOP-SR).	C NYV	SOP-SR; OD-31
FR-MD-06	Equipment & calibration	ADM/QAO records scales, thermometers and printers with calibration due date.	Warns when a weighing or temperature record uses equipment past its due date.	Warn only; calibration programme owner QAM.	ADM, QAO	—	A weight entered with an overdue scale shows a warning and the record is flagged for QA review.	P	ISO 9001 §7.1.5
FR-MD-07	Rule parameters	PHM (operational) or QAM (quality) proposes values; the other approves where SoD applies.	Time limits (24 h grading, 48/72 h, 30 min dwell, 24 h QC validity), weight thresholds (1 t), alert lead times stored as effective-dated parameters, not code.	Values documented in SOP001 are pre-loaded and labelled 'SOP-documented'.	PHM, QAM	Out-of-range entry (e.g. negative hours) blocked.	Changing a parameter changes rule behaviour from its effective date without a software release.	P	SOP-PC; SOP-GR; SOP-WS

10.2 Users, roles & access control (FR-AC)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-AC-01	Named user accounts	ADM creates a user from an approved access request.	User gets roles; department recorded; no shared accounts.	Request approved by the user's department head.	ADM	Request without approval cannot be fulfilled.	Every audit entry shows a named user; shared-account creation is impossible.	P	ISO 27002 §5.16, §5.18
FR-AC-02	Permission enforcement	User attempts any action.	Server checks role × object × action per Section 6 before executing.	Section 6 permission matrix.	All	Denied action shows 'not permitted for your role' and is logged.	Negative tests: each role attempting each non-granted action is denied (100 % of matrix cells tested).	P	Section 6
FR-AC-03	Segregation-of-duties enforcement	User attempts an action that conflicts with an earlier action on the same record.	Blocks the action, or (SoD-10) allows with a conflict flag where configured.	SoD-01 … SoD-10.	All	Clear message naming the conflicting earlier action.	The same user cannot create and approve a waste sign-off; the attempt is logged.	P	ISO 27002 §5.3
FR-AC-04	Electronic signature	User approves, verifies, rejects or releases.	Prompts PIN re-entry; stores meaning, user, role, time.	AT-03.	Approving roles	Three wrong PINs lock approvals for the session.	Every approval record shows signature meaning and time; approval without PIN is impossible.	P	AT-03
FR-AC-05	Access review	Department head opens the review task.	Lists own staff with roles and last activity; head confirms or revokes.	Frequency OD-41.	PHM, WHM, QCM, QAM	Unreviewed accounts escalate to system owner.	Review results stored with date and reviewer.	P	ISO 27002 §5.18
FR-AC-06	Fast user switching on shared devices	Worker badges in or enters PIN on a shared tablet.	Session switches to that user within seconds; idle sessions lock.	Idle timeout OD-41.	All floor users	—	Two users on one tablet produce two distinct audit identities.	P	ISO 27002 §8.5

10.3 Orders & daily planning (interface) (FR-PL)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-PL-01	Record order lines	SAL (or PH planner on Sales' behalf, OD-37) enters order lines or uploads a template.	Lines created as 'Pending acceptance'.	Mandatory: customer, product/spec, grade, quantity, required date, market type.	SAL	Template rows with errors are rejected individually with reasons.	An uploaded template of N valid rows creates N pending lines.	C NYV	SOP-FC
FR-PL-02	Accept / reject order lines	PHM reviews lines against forecast, stock and grading %.	Line Accepted or Rejected with reason; SAL notified.	PH management accepts or rejects (SOP-FC).	PHM	Reject without reason blocked.	SAL sees the decision within one minute of the PHM action.	C	SOP-FC
FR-PL-03	Daily plan view	PE opens plan for a date.	Shows accepted lines, FG stock by grade/customer, raw stock by crop with age, line availability (cleaning hour per shift).	—	PE, PHM	—	Plan shows raw stock older than 20 h first (alert lead time per OD-09).	P	SOP-FC; SOP-CL
FR-PL-04	Shortage notification	PHM flags a shortage on an accepted line.	Notifies SAL and LOG; records the notice time vs the 24 h (local) / 48 h (export) requirement.	SOP-FC confirmation windows.	PHM	—	The notice record shows whether it met the window.	C	SOP-FC

10.4 Receiving & incoming pallet registration (FR-RC)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-RC-01	Create arrival	DEC selects source and enters tag data (or scans a dispatch-note barcode if CP-02).	Creates Arrival with server time; flags arrivals after 17:00 (V-03).	V-01, V-03.	DEC	Missing tag field blocks submission and opens E-01.	An arrival cannot be submitted without all five tag fields.	C	SOP-RC
FR-RC-02	Incoming vehicle inspection & exception	Inspector records SOP-TR checks; on Fail, the system requests approval from PHM, QA and Top management.	Arrival 'On Hold — vehicle' until all three approve, or any one refuses.	SOP-TR three-party approval.	Inspector (OD-38); PHM, QAM, TOP	Partial approvals expire after a set time (OD-16).	A failed vehicle's pallets cannot be registered as accepted until three approvals exist; one refusal closes the arrival as 'Not accepted'.	C	SOP-TR
FR-RC-03	Register incoming pallet	DEC weighs, enters box count and net kg, prints label.	Creates IP with ID and per-crop/section/block/day print number; prints receiving label with barcode.	V-02, V-19, V-21. Entry point per OD-01.	DEC	Plausibility deviation → warning requiring confirmation; printer failure → E-23.	Registration of one pallet takes ≤ 60 s of keying in pilot timing (target to be confirmed in Phase 0); label shows SOP-RC fields.	C P	SOP-RC; SUM-3
FR-RC-04	Arrival visual check	QCI/QAO records the checklist per arrival with per-pallet Pass/Reject.	Updates IP status; rejected pallets require destination.	G2.	QCI, QAO	Reject without destination blocked.	All pallets of an arrival have a result before any enters pre-cooling.	C	SOP-RC; SOP-GR
FR-RC-05	Pre-cooling tracking	PHO scans pallet in and out.	Computes dwell; alerts PHM above 30 min.	V-04.	PHO, DEC	Exit without entry → prompt for entry time with late-entry flag.	Dwell is shown on the IP and on the pre-cooling board; alert fires at 30 min.	C P	SOP-PC
FR-RC-06	Void registration	DEC requests void with reason; WHM approves.	IP status Void; ID retained.	GEN-12; only before any check.	DEC, WHM	Void after a check blocked.	Voided IP appears in history and is excluded from stock.	P	V-28

10.5 Quality gates (QC / QA) (FR-QG)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-QG-01	Raw-material acceptance	QAO selects section/source and day, records samples, defects, temperature, photos; QCI co-signs.	Accept → all linked IPs Accepted (lot assigned per OD-02). Reject → QARejected and regrade task.	G3; V-23; tolerance version stored.	QAO; QCI co-sign	No photo → cannot submit; no active tolerance set → can only submit as 'Accepted — no tolerance set' with QAM flag until OD-05 closes.	Accepting an RMA moves every linked IP to Accepted in one action; the RMA shows the tolerance version.	C	SOP-RC; SOP-GR
FR-QG-02	Rejection loop	ENG marks regrade done; DEC re-weighs; QCI/QAO rechecks.	Recheck Accept → Accepted; Reject → Waste with a waste record; no third check.	V-26; SoD-02.	ENG, DEC, QCI, QAO	Recheck by the regrading engineer blocked.	A second rejection creates a waste record automatically linked to the IP.	C	SOP-RC
FR-QG-03	QC-1 release	QCI scans pallets in the sorting area and releases.	IP status Released.	G4 (scope OD-17).	QCI	Pallet on hold cannot be released.	Only Released pallets can be issued to a JC.	NYV	SUM-3
FR-QG-04	Line start check	QCI and LL complete the check together.	JC moves to InProgress only with both signatures.	G5.	QCI + LL	Single signature leaves JC in Ready.	A JC cannot record outputs before a passed line start check.	C	SOP-PL

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-QG-05	QC-2 online and final inspection	QCI records random line checks and the final inspection.	Final Pass creates the QC approval used by FG pallets (time-stamped).	G6; CP-06.	QCI	Fail → hold on affected output and PHM/PE task (V-30).	FG pallets of a JC show the approval time of the covering final inspection.	C P	SOP-GR
FR-QG-06	Instruction routing	QCI records a finding needing staff action.	Creates a task for PHM/PE; PH staff never receive QC tasks directly.	V-30.	QCI → PHM/PE	—	No task assignment from a QC user to a PH operator is possible.	C	SOP-ST
FR-QG-07	Inspection templates	QCM/QAM maintains checklist items per inspection type.	Templates versioned; inspections store the version.	SOP checklists pre-loaded.	QCM, QAM	—	Arrival checklist contains all 10 SOP-RC items on day one.	C P	SOP-RC; SOP-GR; SOP-DS

10.6 Storage, location & temperature (FR-ST)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-ST-01	Stock move by scan	PHO scans pallet and destination.	Records move; checks group (V-07) and capacity (V-08).	PHM override for group mismatch.	PHO, DEC	Unknown barcode → error, no move.	Location history of any pallet is complete from registration to dispatch in pilot sampling (100 % of 50 sampled pallets).	C P	SOP-PC
FR-ST-02	FIFO suggestion	PHO/PE requests pallets for a JC or order line.	Lists eligible pallets oldest first (receiving time for IPs; earliest harvest for FG).	V-29 for deviations.	PHO, PE	—	Picking a non-first pallet requires a reason code.	C P	SOP-PC; SOP-DS
FR-ST-03	Ageing monitor	—	Flags IPs at 24 h from receiving; FG at 48 h and 72 h from harvest; alerts per lead time.	V-06, V-12.	PHM view	—	A pallet reaching 72 h cannot be loaded (test with a back-dated harvest time in UAT).	C NYV	SOP-PC
FR-ST-04	Temperature readings	QAO/PHO enters reading per fridge.	Stores reading; compares with documented range of the fridge's group; alerts on excursion; overdue-reading tasks.	V-25; OD-08; OD-32.	QAO, PHO (OD-32)	Out-of-range reading requires a remark.	Excursion alert reaches PHM and QA within one minute of entry.	C NYV	SOP-PC
FR-ST-05	Location board	User opens a fridge.	Shows pallets, crop, grade/customer, age, holds, QC validity.	—	All ops roles (view)	—	Board counts match a physical count in the pilot stock check (variance target set in Phase 0).	P	CP-09
FR-ST-06	Fridge group assignment	PHM changes a fridge's storage group.	Allowed only when empty, or with override listing pallets present.	SOP-PC allocations may change; groups stay separate.	PHM	—	Group change history visible per fridge.	C P	SOP-PC
FR-ST-07	Customer crates & LSS boxes	QCI records crate inspection; DEC records LSS box movements.	Wash/return decision stored; LSS box balance per customer.	SOP-CC.	QCI, DEC	—	LSS box balance can be printed per day.	C	SOP-CC

10.7 Production / Job Cards (FR-PR)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-PR-01	Create & approve Job Card	PE creates JC (line, product, order lines or grade target); PHM approves.	JC Planned; visible to LL and STK.	Line must allow the crop (Figure 4.2).	PE, PHM	Disallowed crop/line combination blocked.	A cherry tomato JC cannot be planned on Line B.	C P	SUM-1; OD-15
FR-PR-02	Issue inputs	LL/PHO scans IPs into the JC.	Status Issued; kg reserved; partial return supported.	V-05.	LL, PHO	Non-released or held IP rejected with reason.	Sum of consumed + returned kg equals issued kg for every IP.	P	GEN-04
FR-PR-03	Packaging issue reference	STK records packaging issued to the JC (item, quantity) as today's store issue.	Stored on the JC for consumption reporting; no ERP posting from the system.	IB-01.	STK	—	Per-load packaging summary can be exported for WHM's ERP entry.	C P	SOP-PL
FR-PR-04	Run control	LL starts, pauses (reason), resumes, completes.	Records times; cleaning pause type tracks the 1 h/shift.	—	LL	Complete with open inputs blocked.	Run time and pause reasons available per JC.	P	SOP-CL
FR-PR-05	Record outputs	PHO/DEC creates FG pallets; waste records; residue return.	Outputs linked to JC; composition derived per GEN-06.	V-09, V-10.	PHO, DEC	—	Every FG pallet created in the pilot has ≥ 1 lot in its composition.	P	GEN-05
FR-PR-06	Mass balance	LL completes; PHM reconciles.	Computes MB-01; requires a reason above tolerance.	V-20; OD-28.	PHM	—	Reconciled JC shows input, output, waste, residue and variance.	P	MB-01
FR-PR-07	Rework Job Card	QCM/QAM disposition = regrade/rework.	Creates a Rework JC with source pallets; lots and harvest dates inherited.	GEN-09.	PE/PHM after disposition	—	New pallets from rework show the source pallets in their trace.	C P	SOP-GR; SOP-DS
FR-PR-08	Customer deviation	Requester records the deviation and attaches the customer's written approval.	JC can use the deviation only when approval is attached.	X-03.	Requester not stated (X-03); SAL proposed	No attachment → cannot link.	A JC produced under deviation shows the deviation and approval document.	C	SOP-PL

10.8 Finished-goods pallets & labels (FR-FG)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-FG-01	Create FG pallet	PHO/DEC selects JC, enters box count and net kg.	Creates pallet with composition, earliest harvest time and QC-2 reference.	QC-2 Pass required (G6).	PHO, DEC	No covering QC-2 Pass → blocked.	Pallet cannot be created for a JC without a passed final inspection.	C P	SOP-GR; SUM-3
FR-FG-02	Print FG label	User prints.	Template per product/customer; prints only when V-09 passes; logs print.	Section 7.3 fields.	PHO, DEC	Missing field listed on screen.	Printed label contains all SOP-PL fields for the case (customer vs grade).	C	SOP-PL
FR-FG-03	QC validity	—	Shows remaining validity (24 h − age of approval) on scan and boards.	V-11.	All (view)	—	Scanning a pallet approved 25 h ago shows 'Re-inspection required'.	C	SOP-GR; SOP-DS

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-FG-04	Split pallet	PHO splits into n children; PHM approves.	Parent closed; children inherit composition pro rata, QC time and earliest harvest.	GEN-07.	PHO, PHM	Child quantities not summing to parent → blocked.	Trace of each child returns the parent's lots.	P	GEN-07
FR-FG-05	Merge pallets	PHO merges; PHM approves.	New pallet; parents closed; union composition; earliest QC time and harvest.	V-22; GEN-08.	PHO, PHM	Incompatible pallets blocked.	A merged pallet takes the older QC approval time.	P	GEN-08
FR-FG-06	Relabel / void label	DEC relabels with reason.	Old label voided; new print logged.	AT-07.	DEC	—	Label history shows both prints.	P	AT-07

10.9 Waste (FR-WS)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-WS-01	Create waste record	PHO/DEC selects category and source.	Draft waste record; long-storage requires photos.	SOP-WS categories.	PHO, DEC	—	Waste can be reported by category per day and per crop.	C	SOP-WS
FR-WS-02	QC waste approval	QCI approves or refuses.	Approval enables weighing; refusal returns product.	V-16; G7.	QCI	—	Weight field is disabled until approval exists.	C	SOP-WS
FR-WS-03	Sign-off routing	System routes after weighing.	Requests sign-offs by threshold; long-storage > 1 t e-mails TOP and starts a 24 h timer.	V-17; E-20.	PHM, FIN, WHM, QA, QC, TOP	Timer expiry → RemovedOnTimeout with report and photos e-mailed.	A 1.2 t long-storage record requests 6 parties; after 24 h without TOP approval it shows RemovedOnTimeout.	C NYV	SOP-WS; OD-11
FR-WS-04	ERP reference capture	DEC enters the ERP document number after posting.	Stores reference; daily report lists signed-off waste without reference.	IB-01.	DEC	Duplicate reference warns.	Missing-reference report is empty when all ERP postings are done.	P	SOP-WS; IB-01
FR-WS-05	Weight-loss adjustment	Authorised user records weekly weight loss per crop using the agreed %.	Recorded separately from processed waste.	MB-04; OD-28.	PHM (OD-28)	—	Stock report shows weight loss as its own line.	C	SOP-WS
FR-WS-06	Packaging waste count	WHM records counted boxes/pallets; sign-offs collected.	Record for ERP removal by Warehouse.	SOP-WS signatories.	WHM, FIN, PHM, QA/QC	—	Record shows all signatories before 'Ready for ERP'.	C	SOP-WS

10.10 Dispatch (FR-DS)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-DS-01	Create load & allocate	PE/PHM creates load from accepted lines.	Proposes FIFO pallets matching spec, with QC validity and age.	V-12, V-29.	PE, PHM	Insufficient stock → shortage proposal including issue from Harvest Fridge for grading (SOP-DS).	Allocation proposal never includes held or age-blocked pallets.	C P	SOP-DS

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-DS-02	Load by scan	PHO scans each pallet onto the vehicle.	Checks allocation, QC validity, hold, age; prompts re-inspection when > 24 h.	V-11, V-12.	PHO	Non-allocated pallet → warning; held pallet → block.	Scanned list equals packing list (100 % in UAT).	C P	SOP-DS
FR-DS-03	Vehicle inspection	QCI records temperature, cleanliness, smell, contamination, lockable.	Pass → Loading; temperature fail → WaitingForTemp; other fail → VehicleRejected.	V-14; E-18.	QCI	—	A non-lockable vehicle cannot reach Loading.	C	SOP-DS; SOP-TR
FR-DS-04	Product temperature at dispatch	QCI/QAO records product temperature.	Below requirement → pallet to Cooling and back to fridge.	E-16; OD-08.	QCI, QAO	—	Pallet in Cooling cannot be loaded until a passing re-measure.	C NYV	SOP-DS
FR-DS-05	Dispatch rejection & replacement	QCI rejects a pallet (approval > 24 h only).	Pallet → DispatchRejected → hold; system proposes the next FIFO pallet.	V-11; E-17.	QCI; PHM consulted	Reject of a < 24 h approval blocked with SOP reference.	Replacement proposal appears immediately and is FIFO-oldest eligible.	C	SOP-DS; SOP-GR
FR-DS-06	Logger & seal	Loader records logger ID, pallet and box position, seal number.	Stored on the load; logger box marked instruction shown.	V-13; OD-18.	DEC/PHO (OD-18)	Departure without seal blocked.	Every departed load in UAT has a seal number.	C	SOP-DS; SOP-TR
FR-DS-07	Packing list	DEC finalises.	Generates packing list from scans: pallets, lots, weights, seal, logger.	Variance vs allocation needs approval.	DEC	—	Printed packing list matches scanned pallets one-for-one.	C P	SOP-DS
FR-DS-08	Loading window	User records loading start between 01:00 and 09:00.	Blocks unless PHM override.	V-15.	PHM override	—	Override recorded with reason.	C P	SOP-DS
FR-DS-09	Shelf-life samples	QAO records sample per product per customer per load.	Creates disposal task at day 14; mould removal recorded.	V-24; X-04.	QAO	—	Each departed load shows samples or a waiver.	C	SOP-GR
FR-DS-10	Departure	DEC records departure.	Load Departed; pallets Dispatched; ERP packaging summary available.	SM-LD.	DEC	—	Pallets on a departed load cannot be moved or relabelled.	P	SOP-DS

10.11 Hold, nonconformity, CAPA & override (FR-NC)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-NC-01	Raise hold	QCI/QAO/PHM selects items and reason.	Items blocked from any onward transition.	X-01.	QCI, QAO, PHM	—	A held pallet scanned at loading is refused.	P	X-01
FR-NC-02	Disposition & release	QCM/QAM records disposition; PH executes; QC/QA verifies.	Disposition creates linked records (rework JC, waste, release).	SoD-05.	QCM, QAM	PHM release attempt denied.	Hold closes only after verification.	P	ISO 9001 §8.7
FR-NC-03	CAPA	QAM opens CAPA from hold, complaint or audit.	Tracks root cause, actions, owner, due date, effectiveness check.	X-02.	QAM, QCM	Overdue actions escalate.	CAPA cannot close without an effectiveness note.	P	ISO 9001 §10.2

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-NC-04	Emergency override	Authorised role invokes override with reason and duration.	Bypasses the named rule for named records; notifies gate owner; schedules review.	X-06; OD-16.	PHM, QCM, QAM	Non-overridable rules (X-06) cannot be selected.	Every override has a review result within the review window, or escalates.	P	X-06

10.12 Complaints, trace & recall (FR-TR)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-TR-01	Complaint intake	SAL enters complaint details and attachments.	Checks completeness and the 24 h window; starts 6 h recall and 24 h resolution timers.	V-18.	SAL (OD-33)	Incomplete → 'Invalid — incomplete' proposal for QAO decision.	Timers visible on the complaint; alerts before expiry.	C	SOP-CP
FR-TR-02	Backward trace	User enters any ID (FG pallet, load, lot, box code if defined).	Returns TQ-01 report as screen and PDF.	—	QAO, QAM, QCM, PHM	Unknown ID → 'not found' with search suggestions.	Mock recall: backward trace of 10 random dispatched pallets complete (all links present) in ≤ 30 min total (target to be confirmed by QAM).	P	CP-12
FR-TR-03	Forward trace & recall scope	QAM selects lots or sources and dates.	Returns TQ-02/TQ-03 with customers, quantities and in-house stock; confirm → bulk hold.	—	QAM	—	Bulk hold places all in-house descendants on hold in one action.	P	SOP-CP
FR-TR-04	Mock recall mode	QAM starts a mock recall.	Times the exercise; reports completeness (traced kg ÷ produced kg).	ISO 22000 §8.3 verify/test.	QAM	—	Report shows elapsed time and % reconciled.	P	ISO 22000 §8.3
FR-TR-05	Investigation & response	QAO/QCI investigate; PHM confirms; SAL records response.	Links trace report, samples, CAPA; farm notification letter for residue/MRL cases.	SOP-CP.	QAO, QCI, PHM, SAL	—	Closed complaint has investigation, PHM confirmation and response.	C	SOP-CP

10.13 Reporting & KPIs (FR-RP)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-RP-01	Daily stock report	PHM generates and issues.	Per crop: daily harvest, stock by customer and grade, waste, season-to-date harvest and waste, crop colour, own vs supplier.	SOP-SR.	PHM; view many	—	Report reproduces the SOP-SR content and reconciles to transactions.	C	SOP-SR
FR-RP-02	Exception board	User opens.	Lists overdue grading, age alerts, holds, open overrides, missing ERP refs, temperature excursions.	—	PHM, QCM, QAM	—	Each item links to its record.	P	—
FR-RP-03	KPI reports	Manager selects period.	Computes K1–K13 with definitions shown.	Section 12.4.	Managers	—	Each KPI value can be drilled down to its records.	P	Section 12.4

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-RP-04	Export	User exports a list.	CSV/XLSX export with the filters applied.	Permission-filtered.	By role	—	Exported rows equal on-screen rows.	P	—

10.14 Audit trail & records (FR-AU)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-AU-01	Audit capture	—	Logs every action per AT-01.	AT-01, AT-02.	System	—	Audit sample of 100 actions shows before/after for each changed field.	P	AT-01
FR-AU-02	Audit viewer	AUD/QAM searches.	Filter by record, user, date, action.	Read-only.	AUD, QAM, ADM	—	No edit control exists in the viewer.	P	AT-09
FR-AU-03	Corrections	Author requests correction after verification.	Correction record with reason; approver per record type.	SoD-09; V-28.	Record-dependent	—	Original values remain visible.	P	V-28
FR-AU-04	Downtime back-entry	User enters a paper-form record after downtime.	Stores event time from the form, entry time from server, form number and late-entry flag.	V-27; E-22.	Original role	Duplicate form number blocked.	Back-entered records are reportable as a list per downtime event.	P	E-22

10.15 Tasks, notifications & escalation (FR-NT)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-NT-01	Tasks & notifications	—	Creates tasks for the role/queue that must act (e.g. RMA due, regrade, sign-off, re-inspection).	V-30 routing.	System	—	Each gate in Section 8.2 produces a task for its owner when work is waiting.	P	—
FR-NT-02	E-mail notifications	—	Sends e-mail for approvals outside the floor (TOP, FIN) with a link; approval itself happens in the system.	IB-07.	System	Bounce → in-app alert to PHM.	TOP approval request e-mail sent within one minute of routing.	C P	SOP-WS
FR-NT-03	Escalation timers	—	Escalates unacted tasks after configured times to the next level.	Times per OD-16.	System	—	Escalation log shows who was notified and when.	P	—

10.16 External references & data exchange (FR-IN)

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-IN-01	External references	DEC/WHM records an external document number.	Links ERP or other references to records.	IB-01.	DEC, WHM	Format validation per system.	Any waste or load shows its ERP reference(s).	P	IB-01

ID	Function	User action	System response	Business rule	Permission	Error handling	Acceptance criteria	Label	Source
FR-IN-02	ERP-ready summaries	DEC/WHM exports daily summaries.	CSV of waste by category/crop and packaging per load, in the column order WHM specifies.	No API assumed.	DEC, WHM	—	Summary totals equal system totals for the day.	P	IB-01
FR-IN-03	Import templates	ADM/SAL uploads master data or orders.	Validates row by row; reports errors.	—	ADM, SAL	Row errors listed; valid rows imported only on confirmation.	Re-importing the same file creates no duplicates.	P	—

10.17 Non-functional requirements

ID	Area	Requirement	Label	Source / OD
NFR-01	Availability	System available during all receiving, production and loading hours (receiving until after 17:00; loading 09:00–01:00). Planned maintenance only between 01:00 and 09:00.	P	SOP-DS
NFR-02	Floor performance	Scan-to-confirmation ≤ 2 s on site Wi-Fi for 95 % of scans (target to confirm after the network survey).	P	OD-24
NFR-03	Devices	Works on rugged handhelds/tablets with a camera and on desktop browsers; label printing to the printers selected in Phase 0.	P	OD-24
NFR-04	Connectivity resilience	Short Wi-Fi loss in cold rooms or loading bays does not lose data: captured actions queue on the device and sync with original event time.	P	OD-24
NFR-05	Language	User interface in the language(s) decided in OD-26; data fields support Arabic and English text.	NYV	OD-26
NFR-06	Security	Encrypted transport; named accounts; PIN re-authentication for approvals; role-based access enforced server-side.	P	ISO 27002 §5.15, §8.5
NFR-07	Backup & recovery	Daily backup with tested restore; recovery point and time objectives to be set by IT and management.	NYV	OD-25; ISO 27002 §8.13
NFR-08	Retention	Records, attachments and audit trail retained for the period set in OD-21; not less than product shelf life (ISO 22000 §8.3).	NYV	OD-21
NFR-09	Time	All server clocks synchronised to a reliable time source; time zone Asia/Amman.	P	ISO 27002 §8.17
NFR-10	Photo storage	Photo/video capture compressed on device; storage sized from Phase 0 counts (photos per report × reports per day).	P	SOP-GR
NFR-11	Configurability	Time limits, thresholds, checklists, label templates and tolerance sets are configurable data, not code (FR-MD-07).	P	—
NFR-12	Usability on the floor	Each floor action completes in ≤ 3 screens with gloves-friendly controls; barcode scan preferred over typing.	P	—

11 Operational bottleneck analysis

Confirmed issues are documented in the sources. Hypotheses are plausible from the documented facts but unmeasured; none should drive investment until measured. Every hypothesis has a metric, a method and a proposed owner.

11.1 Confirmed issues (documented)

ID	Area	Issue	Evidence	Effect	Measure / validate
B-01	Data	Farm dispatch quantity is not recorded; receiving cannot be reconciled with the farm.	EV-01 (SUM-3 key point)	Loss or miscount between farm and packhouse is invisible.	Dispatched vs received pallet/box counts for 2 weeks after CP-02 pilot.
B-02	Data	Rejected incoming product is not entered in the system.	EV-03 (SOP-RC)	Waste and rejection rates by section are unknown; mass balance cannot close.	Rejected pallets and kg per section per week (manual tally in Phase 0).
B-03	Data	Three conflicting lot-assignment points; two conflicting system-entry points.	EV-04, EV-05	Ambiguous traceability key; risk of different practices per shift.	Shadow the receiving and production desks for 3 days: record when lot numbers are actually written.
B-04	Data / documents	Acceptance tolerances and the organogram referenced by the SOP are missing from the supplied file; revision status inconsistent.	EV-08, EV-17	Gates cannot be configured; authority cannot be confirmed from documents.	Obtain the controlled documents from QA document control.
B-05	Handoff	Orders reach the floor via a white board.	EV-32	No record of what was planned vs loaded; changes not traceable.	Count same-day order changes and the time between change and floor awareness.
B-06	Approval	Waste above 1 t needs five signatures; long-storage above 1 t needs top-management e-mail approval.	EV-15	Approval latency; waste may sit in the waste area (contamination risk is why the SOP adds the 24 h rule).	Time from weighing to final signature, per waste report, for 4 weeks.
B-07	Rule definition	Pre-cooling dwell depends on receiving temperature, but no rule table exists; fridge-reading interval undefined.	EV-06, EV-20	Rules cannot be enforced or audited.	QA to define; then measure compliance.
B-08	Rule definition	48 h clock start ambiguous; 72 h maximum vs long-storage regrade conflict; logger scope conflict.	EV-14, EV-23	Inconsistent decisions at dispatch.	Decision only (OD-09, OD-10, OD-18).
B-09	Master data	ELIFAB punnet-filling rate unknown; 'Blocky' crop identity ambiguous; nominal kg per pallet inconsistent in Option 3.	SUM-3 item 1; EV-29; EV-31	Capacity planning and storage rules may be wrong.	Rate from the line; crop list from PHM; weigh 20 pallets per crop.

11.2 Hypotheses and measurement plan

ID	Area	Hypothesis	Why plausible	Metric & method	Period	Owner (proposed)
H-01	Receiving	One data-entry employee for 3 gates/3 scales creates a queue at peak arrival before 17:00.	EV-02; last harvest by 17:00 concentrates arrivals.	Arrival time per pallet, registration start/end, queue length every 15 min.	2 weeks at peak season	WHM

ID	Area	Hypothesis	Why plausible	Metric & method	Period	Owner (proposed)
H-02	QA gate	Pallets wait after pre-cooling for the QA sample decision.	QA samples per section; QA is outside PH (EV-18).	Pre-cooling exit → RMA decision time per section.	2 weeks	QAM
H-03	Line capacity	On peak days, receipts exceed what the lines can grade within 24 h, pushing product to Fridges 5–7 and into long storage.	Arithmetic (Section 11.3): a full Harvest Fridge of cherry (25.6 t) needs ≈ 21.3 ELIFAB running hours at the nominal 1.2 t/h.	Daily received kg by crop vs graded kg by line; hours run; % IPs graded within 24 h.	One full season month	PHM
H-04	Labour (Line C)	Manual line output is limited by crew size for slow categories (e.g. mini cucumber 40 kg/h/worker ≈ 11.25 worker-hours per 450 kg pallet).	SUM-2 labour targets; FCT nominal weights.	Crew count and kg per JC on Line C; actual kg/h/worker vs target.	4 weeks	PHM
H-05	Approval	Multi-party waste sign-offs delay closure and ERP posting.	EV-15.	Weighing → final sign-off; sign-off → ERP posting.	4 weeks	PHM + FIN
H-06	Dispatch	Trucks wait for temperature, re-inspection (> 24 h approvals) or replacements.	EV-13, EV-23.	Truck arrival → loading start → departure; re-inspection count; replacements.	4 weeks	PHM + LOG
H-07	Dispatch QC load	A large share of pallets need re-inspection because they wait > 24 h after approval.	EV-13; 48 h waiting rule for unsold product (SOP-PL).	% loaded pallets with approval age > 24 h.	4 weeks	QCM
H-08	Instruction routing	Routing QC findings through PHM/PE delays corrective action on the line.	EV-16.	QC finding → PHM/PE acknowledgement → action time.	Pilot	PHM + QCM
H-09	Packaging	Line stoppages occur from packaging not collected or not issued in time (daily collection; 24 h stock limit inside PH).	EV-33.	Line pause minutes by reason code.	Pilot	PHM + WHM
H-10	Data entry	Phase 1 adds keying (system + ERP) and slows the data-entry role.	IB-01 (no interface in Phase 1).	Minutes per pallet and per day of keying, before vs during pilot.	Pilot	WHM
H-11	Labelling	FG label printing cannot keep up with line output at peak.	CP-13.	Pallets per hour per line vs printer cycle; queue at printer.	Pilot	PHM + IT
H-12	Storage	Storage-group separation reduces usable cold capacity on mixed-crop days.	EV-21.	Daily occupancy per fridge vs capacity, by group.	4 weeks	PHM
H-13	Infrastructure	Wi-Fi does not reach cold rooms and loading bays reliably.	No evidence either way.	Site survey: signal and scan success per location.	Phase 0 (1 day)	IT

11.3 Capacity arithmetic (from documented figures)

Item	Figure	Derivation	Status
Nominal kg per pallet (35 boxes; cherry 70)	Pepper/Sweetbite 350; Mini/Eggplant 450; Tomato 700; Cherry 400; Onion 600	Location total ÷ pallet positions in FCT (e.g. 44,800 ÷ 64 = 700)	C (recomputed)
ELIFAB grading (cherry)	≈ 3.0 pallets/h	1,200 kg/h ÷ 400 kg/pallet	C (inputs) / derived
Aweta grading (tomato)	≈ 3.1 pallets/h	2,200 kg/h ÷ 700 kg/pallet; 'per product' meaning to confirm	C (inputs) / N (meaning)
Full Harvest Fridge of cherry	25.6 t ≈ 21.3 ELIFAB running hours	25,600 ÷ 1,200	Derived
Full Harvest Fridge of tomato	44.8 t ≈ 20.4 Aweta running hours	44,800 ÷ 2,200	Derived
Theoretical 24 h ceiling, cherry	≈ 25–26 t/day at 100 % uptime	1.2 t/h × (24 h − 1 h cleaning per shift × 2 or 3 shifts). Shift pattern not documented.	Assumption-dependent
Pre-cooling theoretical turnover	32–64 pallets/h	16 positions × 60 ÷ dwell (30 or 15 min); ignores handling time	Derived
Line C worker-hours per pallet	Mini cucumber 11.25; Cherry punnet 4.4; Sweetbite 4.1; Eggplant 2.1	kg per pallet ÷ kg/h/worker target	Derived

Reading the arithmetic

- At the nominal ELIFAB rate, a full Harvest Fridge of cherry tomato is about 21 running hours of grading. If cherry receipts on a peak day approach Harvest Fridge capacity, the 24-hour grading rule (SOP-PC) cannot be met by Line A alone, whatever software is used.
- Pre-cooling (16 positions, 15–30 min) turns over 32–64 pallets per hour in theory, an order of magnitude above line grading rates (about 3 pallets per hour per machine line). Pre-cooling is unlikely to be the constraint; receiving registration and QA sampling are more likely (H-01, H-02).
- Line C is labour-bound: mini cucumber at 40 kg/h/worker needs about 11 worker-hours per pallet.
- These are ceilings from nameplate-style rates, not observed performance. Measure first (H-03, H-04).

12 Implementation plan

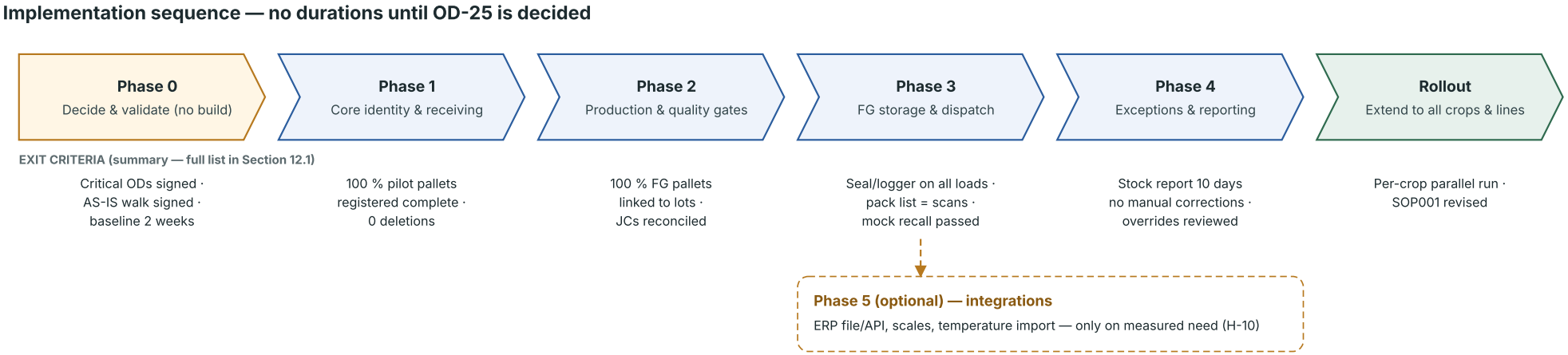


Figure 12.1 — Implementation sequence. Durations are not estimated: they depend on the build approach (OD-25) and Phase 0 findings.

12.1 Phases, dependencies and exit criteria

Phase	Name	Scope	Dependencies	Testing	Exit criteria (measurable)
Phase 0	Decide & validate (no build)	Close critical open decisions; walk the AS-IS process with PHM, QC, QA and Warehouse and sign it off; obtain missing documents (grading schedule, specifications, organogram, current SOP revision); identify the 'pack house system'; Wi-Fi/device/printer survey; baseline measurements B-01…B-09, H-01…H-13 (manual where needed); choose pilot crop and period (OD-27).	Management sponsorship; availability of PHM, QCM, QAM, WHM.	Document review; site walk sign-off.	All OD items marked 'Critical' decided and signed by their owners; AS-IS maps signed by PHM; tolerance set for the pilot crop approved by QAM; network survey report; baseline data for at least 2 weeks.
Phase 1	Core identity & receiving	Master data (FR-MD), access (FR-AC), arrival, registration, arrival check, pre-cooling, RMA and rejection loop (FR-RC, FR-QG-01/02), storage moves and temperature (FR-ST-01…06), labels, audit (FR-AU), basic backward trace to arrival.	Phase 0 exit; devices and printers installed.	Unit and workflow tests; permission negative tests for Phase 1 objects; UAT T-02…T-07; 2–4 week parallel run with paper.	100 % of pilot-crop pallets received during the parallel run registered with all mandatory fields (K1); 0 deletions in audit; RMA linked to 100 % of accepted pallets; physical-vs-system location check ≥ 98 % on a 50-pallet sample (threshold to be confirmed in Phase 0).
Phase 2	Production & quality gates	Job Cards, inputs, line start check, QC-1, QC-2, FG pallets, FG labels, split/merge, waste capture and approval routing, mass balance (FR-PR, FR-FG, FR-WS, FR-QG-03…07).	Phase 1 stable; OD-02, OD-13, OD-15, OD-28 decided.	UAT T-08…T-11, T-16; mass-balance test on 10 JCs per line.	100 % of FG pallets have ≥ 1 lot in composition; 100 % of JCs reconciled or variance-explained; QC approval recorded for 100 % of FG pallets before storage.

Phase	Name	Scope	Dependencies	Testing	Exit criteria (measurable)
Phase 3	Finished-goods storage & dispatch	FG storage, FIFO suggestions, ageing, load planning, vehicle checks, dispatch QC, logger/seal, packing list, shelf-life samples (FR-DS, FR-ST-02/03).	Phase 2; OD-09, OD-10, OD-18, OD-20, OD-37 decided.	UAT T-12…T-15; mock recall T-22.	100 % of loads in the pilot have seal and (per OD-18) logger recorded; packing list equals scans on 100 % of loads; mock recall (backward and forward) passes within the target agreed by QAM.
Phase 4	Exceptions, complaints & reporting	Holds/NC, CAPA, overrides, complaints and recall scope, daily stock report, exception board, KPIs (FR-NC, FR-TR, FR-RP, FR-NT).	Phase 3.	UAT T-17…T-21.	Daily stock report issued from the system for 10 consecutive working days with zero manual corrections; every override reviewed within the window; complaint timers verified in UAT.
Rollout	Extend to all crops & lines	Crop by crop, using the Phase 1–4 configuration.	Each crop's tolerance set, specs and labels approved.	Per-crop smoke test and one-week parallel run.	Same exit criteria as Phases 1–3 per crop; SOP001 revised by QA document control to reference system records.
Phase 5 (optional)	Integrations	ERP summary file or API, scale reading, temperature-record import — each only if justified.	Measured re-keying time (H-10); confirmed interface capability; cost approved.	Interface reconciliation tests.	For each integration: measured reduction in keying time, and daily reconciliation differences = 0 for 2 weeks.

12.2 Acceptance test scenarios

ID	Scenario	Steps	Expected result
T-01	End-to-end happy path (pilot crop)	Arrival → registration → checks → pre-cooling → RMA accept → storage → QC-1 → JC → QC-2 → FG pallet → storage → load → departure; trace backward and forward.	All states reached in order; trace returns every record.
T-02	Arrival rejection	G2 reject with destination farm.	IP ReturnedToSource; arrival shows rejected count.
T-03	QA reject → regrade → accept	G3 reject; ENG regrade; re-weigh; recheck by a different user.	IP Accepted; new weight stored; SoD-02 enforced.
T-04	QA reject → regrade → reject	Second rejection.	IP Waste; waste record auto-created; no third recheck possible.
T-05	Non-compliant incoming vehicle	Vehicle fail; approvals by PHM, QA, TOP; then one refusal variant.	Accepted only with 3 approvals; refusal closes the arrival.
T-06	Grading overdue	Back-date receiving by 25 h.	Flag raised; appears on exception board.
T-07	Storage-group conflict	Place pepper into a tomato fridge.	Blocked; PHM override with reason succeeds and is logged.
T-08	Partial issue	Issue 1 IP to 2 JCs.	Consumed + returned = issued; composition correct.
T-09	Line start fail	QC + LL fail; correct; pass.	JC cannot start until pass.
T-10	QC-2 fail	Final inspection fail.	Output on hold; PHM/PE task created, not an operator task.
T-11	Split & merge	Split one pallet; merge two compatible pallets; attempt incompatible merge.	Rules GEN-07/08 applied; incompatible merge blocked.
T-12	Dispatch re-inspection & rejection	Pallet approved 25 h ago; QC rejects; replacement proposed; rework JC.	FIFO replacement; rework genealogy intact; < 24 h rejection attempt blocked.
T-13	Vehicle temperature	Vehicle above requirement; re-check passes.	Loading blocked until pass.
T-14	Non-lockable vehicle	Lockable = false.	VehicleRejected; no loading.
T-15	72 h age	Harvest time back-dated 73 h.	AgeBlocked; cannot be allocated or loaded.
T-16	Waste thresholds	0.8 t processed; 1.2 t processed; 1.2 t long-storage with no TOP approval for 24 h.	Correct sign-off sets; RemovedOnTimeout after 24 h.
T-17	Valid complaint & recall scope	Complaint with lot; forward trace; bulk hold; CAPA.	In-house stock held; timers shown; CAPA linked.
T-18	Late complaint	Complaint 30 h after delivery.	Proposed Invalid with reason.
T-19	Override	Invoke, expire, review; attempt self-review; attempt a non-overridable rule.	Self-review and non-overridable rule blocked.
T-20	Permission matrix	Every role × every non-granted action.	All denied and logged.
T-21	Downtime	Paper forms during simulated outage; back-entry.	Late-entry flags; duplicate form numbers blocked.
T-22	Mock recall	Random 10 dispatched pallets backward; 2 lots forward.	Complete and within the agreed time; % reconciled reported.

12.3 Governance and change control

Item	Arrangement
System owner	Proposed: Operations (transformation owner), accountable for scope and change control.
Process owner	PHM for Packing House stages; QCM and QAM for their gates and master data; WHM for receiving registration and ERP references.
Steering review	Proposed: PHM, QCM, QAM, WHM, Operations and IT, at each phase exit.

Item	Arrangement
Change control	Any change to a Confirmed behaviour needs the owner's written approval and an SOP001 revision by QA document control. Proposed behaviours are approved through the Open Decisions Register.
Document control	This specification is versioned. Decisions update it; each update lists changed IDs.

12.4 Process and system KPIs

Targets are not invented. Where SOP001 sets a limit (24 h, 48/72 h, 6 h, 24 h), the KPI measures compliance with it; the % target is set after the Phase 1–3 baseline.

ID	KPI	Formula	Source	Frequency	Owner	Target basis
K1	Registration completeness	Pallets registered with all mandatory fields ÷ pallets received	System + gate count	Daily	WHM	Baseline in pilot; target set by WHM
K2	Graded within 24 h of receiving	IPs consumed ≤ 24 h after receiving ÷ IPs consumed	System	Daily	PHM	SOP limit 24 h; % target after baseline
K3	Pre-cooling dwell compliance	IPs with dwell ≤ limit ÷ IPs pre-cooled	System	Daily	PHM	SOP 15–30 min; rule table OD-07
K4	QA first-pass acceptance	IPs accepted at first RMA ÷ IPs assessed, by crop/section	System	Weekly	QAM	Baseline only (informs farm)
K5	Loaded within 48 h of harvest	Loaded pallets ≤ 48 h ÷ loaded pallets; count > 72 h (should be 0 per SOP)	System	Weekly	PHM	SOP 48 h / 72 h; clock per OD-09
K6	Mass-balance variance	|Unexplained variance| ÷ input kg per JC	System	Per JC / weekly	PHM	Tolerance OD-28
K7	Waste rate by category	Waste kg (processed / long-storage / weight loss) ÷ accepted kg, by crop	System	Weekly	PHM	Baseline then target by management
K8	Dispatch re-inspection and rejection	Pallets re-inspected ÷ loaded; pallets rejected ÷ loaded	System	Weekly	QCM	Baseline
K9	Traceability test	Mock-recall elapsed time; traced kg ÷ produced kg	Mock recall	Per test (frequency OD-21)	QAM	Target agreed by QAM; recall must start ≤ 6 h (SOP-CP)
K10	Complaint resolution in 24 h	Complaints resolved ≤ 24 h ÷ valid complaints	System	Monthly	QAM	SOP 24 h
K11	Overrides	Overrides per gate per week; % reviewed within window	System	Weekly	PHM / QCM / QAM	Review 100 % (proposed)
K12	Labour productivity (optional)	Packed kg ÷ (crew × hours) per category	JC crew & time	Weekly	PHM	SUM-2 targets 40–210 kg/h/worker (OD-34, OD-35)
K13	Truck turnaround	Departure − vehicle arrival, per load	System	Weekly	PHM / LOG	Baseline

13 Open decisions register

Each decision has an owner from the department that holds the authority. Recommendations are proposals only. A decision closes when the owner records the choice and the date; the specification is then re-issued with the affected IDs updated.

Critical — needed before any build (Phase 0)

- OD-01 — Point of system entry; registration of rejected pallets; farm dispatch note. (owner: PHM + WHM + QAM; Farm Operations for CP-02)
- OD-02 — Lot definition and assignment point. (owner: QAM + PHM)
- OD-03 — Identity of the 'pack house system' and who issues Pallet IDs, lot numbers and labels. (owner: PHM + WHM + IT)
- OD-05 — Incoming raw-material grading schedule (tolerances) per crop. (owner: QAM)
- OD-24 — Hardware and infrastructure: devices, scanners, label printers, Wi-Fi coverage, scale outputs. (owner: IT + PHM)
- OD-25 — Build and hosting approach (in-house build with Claude Code, low-code platform, or vendor); hosting, backup, RPO/RTO. (owner: Top management + IT + Operations)

ID	Decision required	Why it matters	Options	Recommendation (proposed)	Owner	Priority	Needed by	Source
OD-01	Point of system entry; registration of rejected pallets; farm dispatch note.	Defines what 'received' means and whether quantities reconcile (EV-01, EV-03, EV-04).	A: register after QA acceptance (current SOP). B: register every pallet at the scale with status (CP-01) + dispatch note (CP-02).	B — it is the only option that closes MB-03.	PHM + WHM + QAM; Farm Operations for CP-02	Critical	Phase 0	SOP-RC; SUM-3 #6
OD-02	Lot definition and assignment point.	The lot is the traceability key on labels and complaints (EV-05, EV-25).	A: lot at QA acceptance (crop × source × harvest date). B: lot at production (per JC). C: both — raw lot printed on labels, JC as batch.	C.	QAM + PHM	Critical	Phase 0	SOP-RC; SOP-PF; SUM-3 #8
OD-03	Identity of the 'pack house system' and who issues Pallet IDs, lot numbers and labels.	Avoids two systems issuing IDs; respects 'no replacement without explicit requirement'.	A: new system issues IDs for the packing house. B: existing system issues; new system reads/keys them.	Decide after identification in Phase 0.	PHM + WHM + IT	Critical	Phase 0	SOP-RC
OD-04	ERP boundary and integration.	Duplicate keying vs interface cost (IB-01, H-10).	A: no interface; ERP references captured (Phase 1). B: file export. C: API.	A now; reassess with measured keying time.	WHM + FIN + IT	High	Phase 0 (boundary) / Phase 5	SOP-WS; SOP-PL; SOP-ST
OD-05	Incoming raw-material grading schedule (tolerances) per crop.	G3 cannot be configured without it (EV-08).	Supply the controlled schedule; or QA drafts and approves one.	Supply the controlled document.	QAM	Critical	Phase 0 (pilot crop)	SOP-RC; SOP-GR
OD-06	Product / customer specifications: owner, format, approval.	QC-2 and labels depend on them.	Sales maintains, QCM approves (proposed).	As proposed.	SAL + QCM	High	Phase 2	SOP-PL; SOP-GR

ID	Decision required	Why it matters	Options	Recommendation (proposed)	Owner	Priority	Needed by	Source
OD-07	Pre-cooling dwell rule by receiving temperature.	V-04 enforcement.	Rule table (temperature band → minutes); or a single 30 min maximum.	Rule table if QA can define it; else 30 min max.	QAM + PHM	High	Phase 1	SOP-PC
OD-08	Temperature limits: confirm SOP storage ranges as operational limits; alarm tolerances; vehicle and product temperature requirements at dispatch.	Alerts and dispatch blocks (V-14, V-25).	Adopt SOP table with tolerances set by QA.	QA to set; this specification invents none.	QAM	High	Phase 1	SOP-PC; SOP-DS
OD-09	Start of the 48 h clock and link to 'Order Received — Max. 48 Hours'; alert lead times.	V-12 and K5 depend on it (EV-14).	A: from harvest (SOP). B: from order receipt (manager's note). C: both, as separate limits.	C if both rules are intended; confirm.	PHM + SAL	High	Phase 3	SOP-PC; SUM-3 #7
OD-10	Can regraded product be loaded beyond 72 h after harvest?	SOP says > 72 h is strictly not allowed but also describes regrading after long storage.	A: never beyond 72 h (waste). B: allowed after regrade with QA approval.	Owner decision; system blocks until decided.	PHM + QAM	High	Phase 3	SOP-PC; SOP-GR
OD-11	Waste thresholds: basis of '1 t' (per report, shift, day, event); QA/QC signing level; whether electronic sign-off replaces wet signatures.	Routing V-17.	Per waste report per shift (proposed).	Per report per shift; e-signature accepted.	PHM + FIN + WHM + QAM + QCM	High	Phase 2	SOP-WS
OD-12	Which engineer regrades rejected incoming product (department, person)?	Task routing and SoD-02.	Farm engineer / Production Engineer / other.	—	PHM + Farm Operations	Medium	Phase 1	SOP-RC
OD-13	Who is the 'traceability team', and who records JC inputs and FG pallets?	System user assignment for ST-08/ST-10.	Data-entry clerks (Warehouse Mgmt) / PH staff / dedicated team.	—	PHM + WHM	High	Phase 2	SOP-PC; SOP-ST
OD-14	Custody of fridge stock: who records moves in and out (PH vs Warehouse store keeper)?	Stock-move accountability.	PH operators / store keeper / data entry.	—	PHM + WHM	High	Phase 1	SOP-PL; SOP-PC
OD-15	Who creates and approves Job Cards; JC granularity (per line × product × shift?); reopen window.	ST-08 design.	PE creates, PHM approves; one JC per line per product per shift.	As proposed.	PHM	High	Phase 2	Proposed (CP-04)
OD-16	Emergency override authority, non-overridable rules, maximum duration, review window; escalation times for tasks and partial approvals.	X-06 safety and control.	As listed in Section 6.5 (proposal).	Approve list in Section 6.5.	PHM + QCM + QAM + Top management	High	Phase 1	Proposed
OD-17	Scope of QC-1 versus arrival check and QA acceptance.	Avoid duplicate inspection (EV-11).	A: keep QC-1 as a separate gate. B: merge with QA acceptance. C: QC-1 only on issue from storage > X h.	—	QCM + PHM	Medium	Phase 2	SUM-3 #4
OD-18	Temperature logger: all loads or export loads only?	V-13 (EV-23).	All loads (SOP-DS) / export only (SOP-GR).	—	QAM + LOG	Medium	Phase 3	SOP-DS; SOP-GR
OD-19	Definition and format of 'box code'.	Complaints must quote it (EV-25).	Box code = lot + date; or a printed box serial.	—	PHM + SAL	Medium	Phase 2	SOP-CP

ID	Decision required	Why it matters	Options	Recommendation (proposed)	Owner	Priority	Needed by	Source
OD-20	Label formats, barcode type; GS1 SSCC adoption; physical vs electronic QC approval label.	Customer acceptance of labels.	Internal barcode / GS1 SSCC; stamped label kept or replaced.	Internal barcode unless customers require SSCC; keep stamped label until customers confirm.	SAL + QAM + PHM	Medium	Phase 1	SOP-PL; SOP-GR
OD-21	Record retention period; mock-recall frequency.	Storage sizing and evidence for audits.	≥ shelf life (ISO 22000 §8.3) vs customer/legal periods.	QA to set from customer and legal requirements.	QAM (+ legal)	Medium	Phase 1	ISO 22000 §8.3
OD-22	Applicable certifications, customer standards and regulatory requirements (Jordan and export markets); recall authority.	Determines which standard clauses are mandatory requirements vs good practice.	—	QA to list; this specification claims no compliance.	QAM + Top management	High	Phase 0	SUM-3 (QA role)
OD-23	HACCP study: are any packhouse steps CCPs or OPRPs (e.g. washer-dryer water, cold chain)? Which holds count as food-safety holds?	The system must record monitoring for any CCP/OPRP but must not define them (EV-36).	—	QA / food-safety team to confirm from the existing HACCP plan.	QAM	High	Phase 1	ISO 22000 §8.5; CXC 1-1969 Ch.2
OD-24	Hardware and infrastructure: devices, scanners, label printers, Wi-Fi coverage, scale outputs.	Phase 1 cannot start without it.	—	Survey in Phase 0.	IT + PHM	Critical	Phase 0	CP-13
OD-25	Build and hosting approach (in-house build with Claude Code, low-code platform, or vendor); hosting, backup, RPO/RTO.	Cost, timeline and support model.	—	Decide after this specification is approved.	Top management + IT + Operations	Critical	Phase 0	—
OD-26	User-interface language(s).	Usability on the floor.	Arabic / English / both.	Both, selectable per user.	PHM	Medium	Phase 1	—
OD-27	Pilot crop and pilot period.	Phase 1–3 scope.	e.g. a single-line crop to limit variables.	—	PHM	Medium	Phase 0	—
OD-28	Mass-balance tolerance per crop; weekly weight-loss % per crop and its approver.	V-20, MB-04.	—	Set from pilot data.	PHM + FIN	Medium	Phase 2	SOP-WS
OD-29	Weighing method (gross/tare/net); nominal kg per pallet; plausibility tolerance; status of Option 3 (49 boxes).	V-02, V-21 (EV-31).	—	Weigh 20 pallets per crop in Phase 0.	PHM + WHM	Medium	Phase 1	FCT
OD-30	Crop master: identity of 'Blocky'; storage group per crop; washer-dryer crop list.	Storage rules and routes (EV-29).	—	—	PHM (+ QAM for storage groups)	High	Phase 0	SUM-1; SOP-PC; SUM-3 #5
OD-31	External supplier produce: same flow? supplier lot reference? approval of suppliers?	Receiving design (EV-27).	—	—	QAM + PHM (+ Procurement)	Medium	Phase 1	SOP-SR; SOP-TR
OD-32	Fridge temperature reading interval; who records; identity and export capability of the computerised record.	V-25; IB-03.	—	—	QAM + PHM	Medium	Phase 1	SOP-PC

ID	Decision required	Why it matters	Options	Recommendation (proposed)	Owner	Priority	Needed by	Source
OD-33	Who logs complaints in the system (Sales or QA)?	FR-TR-01.	SAL (proposed).	SAL.	SAL + QAM	Medium	Phase 4	SOP-CP
OD-34	Is labour productivity capture in scope?	K12; crew recording effort.	In scope from Phase 2 / later / never.	Later, after core traceability works.	PHM (+ HR)	Low	Phase 4	SUM-2
OD-35	Assign the unlabelled labour values (230; 30).	Only if labour module is in scope.	—	—	Owner of the productivity table	Low	Phase 4	SUM-3 #3
OD-36	Current controlled revision of SOP001; who updates it to reference system records.	Document control (EV-17).	—	QA document control issues one revision.	QAM + PHM	High	Phase 0	SOP
OD-37	Order source and format; who enters accepted lines; truck schedule input.	IB-05, IB-09; ST-00.	Sales enters / PH planner enters / template upload.	Sales enters; template upload as fallback.	SAL + PHM + LOG	High	Phase 3	SOP-FC; SOP-DS
OD-38	Who inspects incoming vehicles and which 'top management' role approves vehicle exceptions.	FR-RC-02.	—	—	Top management + PHM + QAM	Medium	Phase 1	SOP-TR
OD-39	Loading-window enforcement: hard block or override.	V-15.	Block / PHM override.	PHM override with reason.	PHM	Low	Phase 3	SOP-DS
OD-40	Installed machine models (Ingro, ELIFAB, Aweta) and ELIFAB punnet-filling rate.	Equipment master and capacity (B-09).	—	Record from nameplates.	PHM	Low	Phase 2	SUM-3 #1, #2
OD-41	Access governance: review frequency, joiner/mover/leaver timing, idle timeout, dual-role (SoD-10) approvals.	Section 6.6.	—	Quarterly review (proposed).	IT + department heads	Medium	Phase 1	ISO 27002 §5.18

14 Standards-to-requirements traceability matrix

Status describes the specification only: 'Specified' means a requirement exists in this document; it does not mean implemented, compliant or certified. 'Type' distinguishes formal standard requirements (binding only if the company adopts or is certified to the standard) from codes of practice, security controls and recommended practices.

ID	Reference	Principle (paraphrased)	Type	Design requirement	Evidence / control needed	Status
STD-01	ISO 9001:2015 §4.4	Processes defined with inputs, outputs, owners, criteria and measures.	Standard requirement (if the organisation adopts ISO 9001)	Stage cards ST-00…ST-17; KPIs K1–K13.	Signed AS-IS and TO-BE maps; KPI reports.	Specified
STD-02	ISO 9001:2015 §5.3	Responsibilities and authorities assigned and communicated.	Standard requirement (conditional)	Section 6 responsibility and permission matrices.	Approved role matrix; access review records.	Specified — pending OD-12, OD-13, OD-14, OD-15
STD-03	ISO 9001:2015 §7.1.5	Measuring resources fit for purpose; calibrated where traceability of measurement is required.	Standard requirement (conditional)	FR-MD-06 equipment calibration status; scale ID on every weight.	Calibration certificates (outside the system).	Specified
STD-04	ISO 9001:2015 §7.5; ISO 22000:2018 §7.5	Documented information controlled, protected and retained.	Standard requirement (conditional)	AT-01…AT-10; V-28; versioned master data.	Audit-trail tests; retention decision.	Specified — pending OD-21
STD-05	ISO 9001:2015 §8.2.3	Requirements reviewed before committing to supply.	Standard requirement (conditional)	FR-PL-02 order acceptance.	Acceptance records.	Specified
STD-06	ISO 9001:2015 §8.4	Externally provided products verified.	Standard requirement (conditional)	G2/G3 for own-farm and supplier loads; supplier source master.	RMA records per supplier.	Specified — pending OD-31
STD-07	ISO 9001:2015 §8.5.2	Outputs identified, status identified, traceability controlled.	Standard requirement (conditional)	Identifiers (Section 7.1); status per state model; labels (Section 7.3).	Label samples; trace tests T-22.	Specified — pending OD-02, OD-19, OD-20
STD-08	ISO 9001:2015 §8.5.3	Customer property identified, protected, reported if lost or damaged.	Standard requirement (conditional)	FR-ST-07 customer crates and LSS boxes.	Crate inspection and LSS box balance records.	Specified
STD-09	ISO 9001:2015 §8.5.4	Outputs preserved (storage, handling).	Standard requirement (conditional)	V-07, V-08, FIFO, ageing, temperature records.	Temperature and location history.	Specified — pending OD-08
STD-10	ISO 9001:2015 §8.6	Release only after planned arrangements are met; release authority recorded.	Standard requirement (conditional)	G6, G8; QC approval with e-signature.	Release records with signer.	Specified
STD-11	ISO 9001:2015 §8.7	Nonconforming outputs identified and controlled; concession recorded.	Standard requirement (conditional)	X-01 holds; X-03 concessions; SM-NC.	Hold records with disposition and verification.	Specified
STD-12	ISO 9001:2015 §9.1.3; §10.2	Analysis of data; nonconformity and corrective action.	Standard requirement (conditional)	KPIs; CAPA (X-02); complaints (ST-16).	CAPA effectiveness evidence.	Specified
STD-13	ISO 22000:2018 §8.3	Traceability system identifies incoming material and first distribution step; relates received lots to end products; covers rework and distribution; evidence retained at least for shelf life; effectiveness verified and tested (reconciliation of quantities is a recognised way).	Standard requirement (conditional)	GEN-01…GEN-12; MB-01…MB-04; TQ-01…TQ-04; FR-TR-04 mock recall.	Mock-recall reports; mass-balance reports.	Specified — pending OD-01, OD-02, OD-21

ID	Reference	Principle (paraphrased)	Type	Design requirement	Evidence / control needed	Status
STD-14	ISO 22000:2018 §8.5 (HACCP / hazard control plan); Codex CXC 1-1969 Ch.2	Monitoring, corrective action, verification and records for CCPs/OPRPs defined by the food-safety team.	Standard requirement (conditional)	Configurable inspection templates and temperature records able to carry CCP/OPRP monitoring; the system does not define CCPs.	Existing HACCP plan.	Not started — pending OD-23
STD-15	ISO 22000:2018 §8.7	Monitoring and measuring methods and equipment controlled.	Standard requirement (conditional)	FR-MD-06.	Calibration records.	Specified
STD-16	ISO 22000:2018 §8.9.4–8.9.5	Potentially unsafe product held until evaluated; withdrawal/recall procedures.	Standard requirement (conditional)	X-01 food-safety holds (QAM only); FR-TR-03 recall scope and bulk hold.	Recall test records.	Specified — pending OD-22, OD-23
STD-17	Codex CXC 1-1969 (Rev. 2022) Ch.1 §7.2	Time and temperature control.	Recognised code of practice	Pre-cooling dwell, ageing timers, temperature readings and alerts.	Temperature and dwell records.	Specified — pending OD-07, OD-08
STD-18	Codex CXC 1-1969 Ch.1 §7.4, §7.5, §8.1	Documentation and records; recall procedures; lot identification and traceability.	Recognised code of practice	Audit trail; recall scope; lot labels.	As STD-13.	Specified
STD-19	Codex CXC 1-1969 Ch.1 §9	Transportation: vehicles clean, suitable, temperature-capable.	Recognised code of practice	Incoming and dispatch vehicle inspections (FR-RC-02, FR-DS-03).	Vehicle inspection records.	Specified
STD-20	Codex CAC/RCP 53-2003 §5.2.2.3–5.2.2.4, §5.3, §5.5, §5.7, §5.8	Fresh fruit and vegetable packing: cooling and cold storage controlled and monitored, condensate must not drip on produce; incoming material requirements; water in the packing establishment; records kept for recall; lot identification for recall.	Recognised code of practice	Temperature records; storage rules; RMA; records retention; trace. Wash-water control is not in the SOP and is referred to QA.	Wash-water records if QA defines them.	Partly specified — wash water pending OD-23
STD-21	ISO/IEC 27001:2022 Annex A / ISO/IEC 27002:2022 §5.3	Segregation of duties.	Standard control (information security)	SoD-01…SoD-10; FR-AC-03.	SoD test results (T-20).	Specified
STD-22	ISO/IEC 27002:2022 §5.15, §5.16, §5.17, §5.18, §8.3, §8.5	Access control, identity, authentication, access rights and review, least privilege.	Standard control (information security)	FR-AC-01…06; Section 6.6.	Access review records.	Specified — pending OD-41
STD-23	ISO/IEC 27002:2022 §8.2	Privileged access restricted and controlled.	Standard control (information security)	SoD-06; AT-10.	Admin activity reports.	Specified
STD-24	ISO/IEC 27002:2022 §8.15, §8.17, §5.33	Logging; clock synchronisation; protection of records.	Standard control (information security)	AT-01…AT-10; V-27; NFR-09.	Log integrity tests.	Specified
STD-25	ISO/IEC 27002:2022 §8.13, §8.32	Backup; change management.	Standard control (information security)	NFR-07; versioned configuration (FR-MD-07).	Restore test; change log.	Specified — pending OD-25
STD-26	ALCOA+ data-integrity principles	Records attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring, available.	Recognised practice (not a legal requirement here)	Named users; server time; late-entry flags; no deletion; attachments hashed.	Audit-trail tests.	Specified
STD-27	GS1 SSCC / GS1-128 logistic labels	Globally unique pallet identification.	Optional practice; a requirement only if customers require it	CP-14; OD-20.	Customer requirements.	Not started — pending OD-20

ID	Reference	Principle (paraphrased)	Type	Design requirement	Evidence / control needed	Status
STD-28	Regulation (EC) No 178/2002 Art. 18 (EU) and the UK retained equivalent	One-step-back / one-step-forward traceability for food business operators in those markets.	Legal requirement for operators in the EU/UK; relevance to this company via export customers to be confirmed	Supplier and customer links (GEN-02, GEN-10).	QA confirmation of applicability.	Supported by design — applicability pending OD-22

Appendix A Glossary

Term	Meaning
ALCOA+	Data-integrity principles: attributable, legible, contemporaneous, original, accurate, plus complete, consistent, enduring, available.
ARR	Arrival ID — one vehicle load at the gate.
CAPA	Corrective and preventive action.
CCP / OPRP / PRP	Critical control point / operational prerequisite programme / prerequisite programme (ISO 22000, Codex HACCP).
CP	Change proposal — a TO-BE change to the AS-IS process (Section 5.5).
DC	Distribution centre.
DEC	Data Entry Clerk (reports to Warehouse Management).
FG / FGP	Finished goods / Finished-Goods Pallet.
FIFO	First in, first out.
G1 – G9	Workflow gates (Section 8.2).
GR1 / GR2 / Premium	Grades of finished product; Premium is used for export (SOP-PL).
IP	Incoming Pallet — a field pallet as received.
JC	Job Card — a production order for one line run.
LSS	A customer named in SOP001 (dedicated boxes and truck slots).
MRL	Maximum residue limit.
OD	Open decision (Section 13).
PHM / PE / LL	Packing House Manager / Production Engineer / Line Leader.
QCI / QCM / QAO / QAM	QC Inspector / QC Manager / QA Officer / QA Manager.
RMA	Raw-material acceptance (QA sample check per section).
SoD	Segregation of duties.
SSCC	GS1 Serial Shipping Container Code — an 18-digit pallet identifier.
ST-xx	TO-BE stage (Section 5.6).
Tolerance set	QA-owned acceptance criteria per crop (the 'incoming raw material grading schedule').

Appendix B Notes for implementation (Claude Code)

These notes apply only after Phase 0. They keep the build faithful to this specification.

- Do not start Phase 1 build until the Phase 0 exit criteria (Section 12.1) are met and this document is re-issued with the decisions applied.
- Use the IDs in this document (ST-, FR-, V-, G-, SM-, E-, OD-, CP-) as stable references in code, tests and commit messages.
- Implement each state model (SM-xx) as an explicit transition table with guards, enforced on the server. No status field may be written directly.
- Load the permission matrix and SoD rules from the 'Permissions' sheet of the companion workbook as data; generate the negative permission tests (T-20) from the same data.
- Every value labelled Not Yet Verified must be configuration with a 'pending decision' default that warns or blocks — never a hard-coded constant.
- Rule parameters (24 h, 30 min, 48/72 h, 1 t) are seeded from SOP001 values and marked 'SOP-documented' in the parameter table (FR-MD-07).
- The audit log is append-only from the first commit. No delete endpoints for business records.

- No integration code in Phases 1–4. ERP interaction is limited to storing external references and exporting summaries (FR-IN-01, FR-IN-02).
- Acceptance tests T-01 – T-22 (Section 12.2) are the definition of done for the phases that list them.

Appendix C Companion files

File	Content	Edit with
PH-OAS-SPEC-001.pdf	This specification (review copy)	—
PH-OAS-SPEC-001_Registers.xlsx	Every table in this document as filterable sheets: evidence, stages, permissions, rules, data dictionary, FRs, bottlenecks, plan, decisions, standards	Excel / Google Sheets
PH-OAS-SPEC-001.docx	Editable text of the specification (portrait; diagrams as images)	Word
diagrams/*.drawio	Swimlanes (Figures 4.1, 4.3, 5.3, 5.4)	diagrams.net (draw.io)
diagrams/*.dot	State models, ERD, genealogy, exception paths	Graphviz / any text editor
diagrams/*.svg	All figures as vector images	Any vector editor

End of document · PH-OAS-SPEC-001 v0.1 · DRAFT — for management review. Not approved. Not for build.